BME302 Previous-Year Question Bank: Solutions

Fluid Mechanics and Fluid Machines · AKTU examination questions, topic-wise and mapped to the syllabus

Dr. Abhishek Pandey, Department of Mechanical Engineering, G. L. Bajaj Institute of Technology & Management

About this solution book

This book gives a model answer to every question in the BME302 Previous-Year Question Bank, in the same order and with the same numbering. Each question carries a source marking:

AKTU 2023-24 | BME302 | Q2(c) | 7 Marks

This gives the university, the session of the examination, the paper code, the question number in that paper and the marks. A question asked in several papers shows one marking for each paper, so the number of markings shows how often a topic is asked.

Papers used. All papers were taken from the question-paper library of ABES Institute of Technology (ABESIT), Ghaziabad, which hosts the original AKTU papers.

Paper	Subject	Session (exam)	Semester	Max marks
BME302	Fluid Mechanics & Fluid Machines	2023-24	III	70
BME302	Fluid Mechanics & Fluid Machines	2025-26	III	70
KME302	Fluid Mechanics and Fluid Machines	2022-23	III	100
KME302	Fluid Mechanics & Fluid Machines	2021-22	III	100
KCE303	Fluid Mechanics (Civil Engg.)	2022-23	III	100
ME401	Fluid Machinery	2016-17	IV	100
RME601	Fluid Machinery	2018-19	VI	70

All seven papers are AKTU papers. UPTU, GBTU and MTU papers (before 2015) were not available in the ABESIT library, so none are included. The KME302 papers for 2019-20 and 2020-21 are listed there but could not be read (one is a scanned image), so they are not included either.

Wording and data. The questions keep the data exactly as printed. The wording has been lightly edited for clarity. Where a value is printed unclearly (a lost degree sign, a garbled unit), the value taken is stated in italics after the question. Real examination questions are kept even when their data are not realistic; the solution points this out in a **Note** so that students are not confused.

Mapping to the syllabus. Units I to V follow the BME302 syllabus (2023-24 onwards). Two groups in Unit II, stream function and velocity potential (II.5) and dimensional analysis (II.6), are asked in AKTU papers but are not named in the current unit list; dimensional analysis belongs to CO2 of the course. The last section, **Outside the current BME302 syllabus**, collects questions from the civil-engineering and fluid-machinery papers whose topics BME302 no longer covers.

Topic-wise frequency

Unit	Topic	Questions	Times asked	Papers
I	I.1 Definition of fluid and basic properties	5	5	4
I	I.2 Viscosity and Newton’s law of viscosity	4	7	4
I	I.3 Compressibility	1	1	1
I	I.4 Surface tension and capillarity	3	4	3
I	I.5 Euler’s and Bernoulli’s equations	2	2	2
I	I.6 Flow measurement: venturimeter, orifice meter, Pitot tube	5	6	5
II	II.1 Types of flow	7	10	4
II	II.2 Reynolds number and Froude number	2	5	4
II	II.3 Flow lines	1	1	1
II	II.4 Continuity equation	3	5	4
II	II.5 Stream function and velocity potential (asked in AKTU papers; not named in the current Unit II syllabus)	7	7	4
II	II.6 Dimensional analysis and similarity (CO2 of BME302; not listed in the unit topics)	5	5	2
III	III.1 Laminar flow through pipes	5	5	4

Unit	Topic	Questions	Times asked	Papers
	and between plates			
III	III.2 Resistance to flow, turbulence and losses	8	8	4
III	III.3 Boundary layer	4	6	5
III	III.4 Drag and lift	2	2	2
IV	IV.1 Impact of jets	6	6	4
IV	IV.2 Classification and general ideas	8	9	5
IV	IV.3 Pelton wheel	4	9	6
IV	IV.4 Reaction turbines (Francis)	5	5	4
IV	IV.5 Specific speed, unit quantities and performance	5	5	3
IV	IV.6 Cavitation in turbines	1	2	2
V	V.1 Centrifugal pump: basics and efficiencies	6	9	6
V	V.2 Centrifugal pump numericals	4	4	3
V	V.3 Cavitation in pumps	3	3	3
V	V.4 Reciprocating pump	8	13	5
V	V.5 Air vessels and comparison	3	4	3
Outside	X.1 Notches, weirs and orifice flow	3	3	3
Outside	X.2 Hydrostatics, buoyancy and vortex motion	4	4	1
Outside	X.3 Hydraulic devices	5	5	2

UNIT I. Fluid properties, Bernoulli's equation and flow measurement

I.1 Definition of fluid and basic properties

QB I.1 AKTU 2023-24 | BME302 | Q1(a) | 2 Marks Define specific weight and specific gravity.

SOLUTION

Specific weight w is the weight of a fluid per unit volume: $w = \dfrac{\text{Weight}}{\text{Volume}} = \rho g$. Unit: N/m^3. For water, $w = 1000 \times 9.81 = 9810\ \text{N/m}^3$.

Specific gravity (relative density) S is the ratio of the specific weight (or density) of a fluid to that of a standard fluid: water for liquids (at 4°C), air for gases.

$S = \dfrac{w_{\text{liquid}}}{w_{\text{water}}} = \dfrac{\rho_{\text{liquid}}}{\rho_{\text{water}}}$. It has no unit; for mercury $S = 13.6$.

QB I.2 AKTU 2022-23 | KME302 | Q1(b) | 2 Marks Differentiate between mass density and specific volume.

SOLUTION

Mass density ρ	**Specific volume** v
Mass per unit volume, $\rho = m/V$	Volume per unit mass, $v = V/m$
Unit kg/m^3	Unit m^3/kg
Water: $1000\ \text{kg/m}^3$	Water: $0.001\ \text{m}^3/\text{kg}$
Used mostly for liquids	Used mostly for gases

The two are reciprocals: $v = 1/\rho$.

QB I.3 AKTU 2021-22 | KME302 | Q1(a) | 2 Marks Define an ideal fluid.

SOLUTION

An **ideal fluid** is an imaginary fluid that is **incompressible** and has **zero viscosity** (no shear resistance, so no friction losses) and no surface tension. Its density is constant: $\rho = \text{constant}$, $\mu = 0$. No real fluid is ideal, but air and water far from walls behave nearly so, which is why Euler's and Bernoulli's equations are derived for an ideal fluid. A fluid with viscosity is called a **real fluid**.

QB I.4 AKTU 2022-23 | KCE303 | Q1(a) | 2 Marks Define the no-slip condition.

SOLUTION

The **no-slip condition** states that a real (viscous) fluid in contact with a solid surface has **zero velocity relative to that surface**: the fluid layer touching the wall sticks to it. For a fixed wall, $u = 0$ at $y = 0$; for a moving wall, the fluid moves with the wall's velocity. Because of it a velocity gradient du/dy forms near the wall, giving the shear stress $\tau = \mu\, du/dy$ and the boundary layer.

QB I.5 AKTU 2022-23 | KCE303 | Q1(b) | 2 Marks One litre of petrol weighs 7 N. Find its specific weight, mass density, specific volume and relative density.

SOLUTION

Given: $V = 1$ litre $= 0.001\ \mathrm{m}^3$, $W = 7\ \mathrm{N}$.

1. $w = \dfrac{W}{V} = \dfrac{7}{0.001} = 7000\ \mathrm{N/m^3}$
2. $\rho = \dfrac{w}{g} = \dfrac{7000}{9.81} = 713.6\ \mathrm{kg/m^3}$
3. $v = \dfrac{1}{\rho} = \dfrac{1}{713.6} = 1.401 \times 10^{-3}\ \mathrm{m^3/kg}$
4. $S = \dfrac{\rho}{\rho_w} = \dfrac{713.6}{1000} = 0.714$

I.2 Viscosity and Newton's law of viscosity

QB I.6 AKTU 2025-26 | BME302 | Q1(a) | 2 Marks AKTU 2022-23 | KME302 | Q1(a) | 2 Marks Discuss the effect of temperature on the viscosity of (i) liquids and (ii) gases.

SOLUTION

(i) Liquids: viscosity **decreases** as temperature rises. Viscosity of a liquid comes mainly from **intermolecular cohesion**; heating weakens cohesion, so the liquid flows more easily (hot oil is thinner than cold oil).

(ii) Gases: viscosity **increases** as temperature rises. In a gas, cohesion is negligible and viscosity comes from **molecular momentum exchange** between layers; heating makes molecules move faster, so the exchange, and hence the shear resistance, increases.

QB I.7 AKTU 2023-24 | BME302 | Q3(b) | 7 Marks AKTU 2021-22 | KME302 | Q3(a) | 10 Marks Explain the concept of viscosity. Discuss the effect of an increase in temperature on the viscosity of fluids, with the reason.

SOLUTION

1. Concept of viscosity. Viscosity is the property of a fluid by which it offers resistance to the movement of one layer of fluid over an adjacent layer. It is the "internal friction" of a fluid.

Sketch: fluid between a fixed lower plate and a moving upper plate; the velocity varies from 0 at the fixed plate to U at the moving plate, showing layers sliding over one another.

Consider two layers a distance dy apart, moving with velocities u and $u + du$. The upper (faster) layer drags the lower layer forward and the lower layer retards the upper one, so a **shear stress** τ acts between them. Experiments show that τ is proportional to the rate of shear strain (velocity gradient):

$$\tau \propto \frac{du}{dy} \qquad \Rightarrow \qquad \tau = \mu \frac{du}{dy}$$

The constant μ is the **coefficient of dynamic viscosity**:

$$\mu = \frac{\tau}{du/dy}$$

2. Units. SI: $\mathrm{N{\cdot}s/m^2} = \mathrm{Pa{\cdot}s}$; CGS: poise, with $1\ \text{poise} = 0.1\ \mathrm{N{\cdot}s/m^2}$. Dimensions $\left[ML^{-1}T^{-1}\right]$.

3. Kinematic viscosity. The ratio of dynamic viscosity to density, $\nu = \mu/\rho$, unit $\mathrm{m^2/s}$ (CGS: stoke, $1\ \text{stoke} = 10^{-4}\ \mathrm{m^2/s}$).

4. Causes of viscosity. Two mechanisms act together:

1. **Cohesion** between molecules: dominant in liquids, where molecules are close together.
2. **Molecular momentum exchange**: molecules moving randomly between layers carry momentum from the fast layer to the slow one; dominant in gases, where molecules are far apart.

5. Effect of increase in temperature.

Fluid	Effect of rise in temperature	Reason
Liquids	Viscosity **decreases**	Heating weakens intermolecular cohesive forces, which are the main cause of viscosity in liquids
Gases	Viscosity **increases**	Heating increases random molecular motion, so more momentum is exchanged between layers

For liquids (Bansal): $\mu = \mu_0 \dfrac{1}{1+\alpha t+\beta t^2}$; for gases: $\mu = \mu_0 + \alpha t - \beta t^2$, where μ_0 is the viscosity at $0^\circ\mathrm{C}$ and t is in $^\circ\mathrm{C}$. For water, μ falls from about 1.79×10^{-3} at $0^\circ\mathrm{C}$ to $0.28\times10^{-3}\ \mathrm{N\cdot s/m^2}$ at $100^\circ\mathrm{C}$; for air it rises from about 1.7×10^{-5} to $2.2\times10^{-5}\ \mathrm{N\cdot s/m^2}$ over the same range.

Significance. Engine oils thin out when hot, so multigrade oils are used to keep enough viscosity at running temperature; in gas bearings and hot gas lines friction rises with temperature.

QB I.8 AKTU 2025-26 | BME302 | Q2(a) | 7 Marks AKTU 2022-23 | KME302 | Q2(a)(i) | 10 Marks (part)

Explain (i) Newton's law of viscosity and (ii) absolute, gauge and vacuum pressures.

SOLUTION

(i) Newton's law of viscosity

Statement. The shear stress on a layer of fluid is directly proportional to the rate of shear strain (the velocity gradient normal to the layer).

Sketch: velocity profile $u\,(\,y\,)$ above a fixed plate; two layers dy apart with velocities u and $u+du$.

$$\tau \propto \frac{du}{dy} \qquad \Rightarrow \qquad \tau = \mu \frac{du}{dy}$$

where μ is the dynamic viscosity ($\mathrm{N\cdot s/m^2}$). Points to note:

1. Fluids obeying this law (constant μ at a given temperature and pressure) are **Newtonian fluids**: water, air, oils, petrol.
2. Fluids not obeying it are **non-Newtonian**: blood, paint, toothpaste, polymer solutions.
3. *Sketch (rheogram):* τ against du/dy is a straight line through the origin of slope μ for a Newtonian fluid; an ideal fluid lies along the du/dy axis ($\tau = 0$), an ideal solid along the τ axis.

(ii) Absolute, gauge and vacuum pressures

Sketch: a vertical pressure scale with absolute zero at the bottom and the atmospheric pressure line above it; a point A above the atmospheric line (gauge pressure measured upward from it) and a point B below it (vacuum measured downward from it), both with absolute pressure measured from absolute zero.

1. **Absolute pressure** p_{abs}: pressure measured above **absolute zero** (perfect vacuum).
2. **Gauge pressure** p_{gauge}: pressure measured above the **local atmospheric pressure** by a gauge that reads zero in the atmosphere.
3. **Vacuum (negative gauge) pressure** p_{vac}: amount by which a pressure is **below** atmospheric.

$$p_{\mathrm{abs}} = p_{\mathrm{atm}} + p_{\mathrm{gauge}}, \qquad p_{\mathrm{abs}} = p_{\mathrm{atm}} - p_{\mathrm{vac}}$$

Standard atmospheric pressure: $101.325\ \mathrm{kN/m^2} = 10.33\ \mathrm{m\ of\ water} = 760\ \mathrm{mm\ of\ mercury}$.

Example: a gauge reads $200\ \text{kPa}$, so $p_{\text{abs}} = 101.3 + 200 = 301.3\ \text{kPa}$; a vacuum of $40\ \text{kPa}$ means $p_{\text{abs}} = 61.3\ \text{kPa}$.

QB I.9 AKTU 2023-24 | BME302 | Q3(a) | 7 Marks A square plate 0.8 m × 0.8 m weighing 300 N slides down an inclined plane at 30° to the horizontal with a uniform velocity of 0.3 m/s. The gap between the plate and the plane is filled with a lubricating oil film 1.5 mm thick. Find the dynamic viscosity of the oil. *(The unit of weight is unclear in the printed paper; it is taken as 300 N.)*

SOLUTION

The unit of weight is unclear in the paper; as noted, $W = 300\ \text{N}$ is used.

Given: $A = 0.8 \times 0.8 = 0.64\ \text{m}^2$; $W = 300\ \text{N}$; $\theta = 30^\circ$; $u = 0.3\ \text{m/s}$; $dy = 1.5\ \text{mm} = 1.5 \times 10^{-3}\ \text{m}$.

To find: μ.

Sketch: plate on a 30° incline; the weight W resolved into $W\sin\theta$ along the plane and $W\cos\theta$ normal to it; the oil film shear force τA acts up the plane.

Assumption: the velocity varies linearly across the thin film, from 0 at the plane to $0.3\ \text{m/s}$ at the plate.

1. Uniform velocity, so the viscous force balances the weight component along the plane: $F = W\sin 30^\circ = 300 \times 0.5 = 150\ \text{N}$
2. Shear stress: $\tau = \dfrac{F}{A} = \dfrac{150}{0.64} = 234.4\ \text{N/m}^2$
3. Velocity gradient: $\dfrac{du}{dy} = \dfrac{0.3 - 0}{1.5 \times 10^{-3}} = 200\ \text{s}^{-1}$
4. Newton's law of viscosity: $\mu = \dfrac{\tau}{du/dy} = \dfrac{234.4}{200} = 1.172\ \text{N·s/m}^2 = 11.72\ \text{poise}$

Answer: dynamic viscosity of the oil $\mu = 1.172\ \text{N·s/m}^2$ (11.72 poise), a heavy lubricating oil.

I.3 Compressibility

QB I.10 AKTU 2025-26 | BME302 | Q1(b) | 2 Marks Define compressibility and bulk modulus.

SOLUTION

Bulk modulus K is the ratio of the increase in pressure to the resulting volumetric strain:

$$K = \frac{dp}{-dV/V}$$

Unit N/m^2; for water $K \approx 2.1 \times 10^9\ \text{N/m}^2$.

Compressibility is the reciprocal of bulk modulus, $\beta = 1/K$ (unit $\mathrm{m^2/N}$). It measures how much a fluid's volume decreases per unit rise in pressure. Liquids have very low compressibility and are usually treated as incompressible.

I.4 Surface tension and capillarity

QB I.11 AKTU 2021-22 | KME302 | Q1(b) | 2 Marks AKTU 2022-23 | KME302 | Q2(a)(ii) | 10 Marks (part)
Describe capillary rise (capillarity).

SOLUTION

Capillarity is the rise or fall of a liquid in a small-diameter tube dipped in it, caused by surface tension together with adhesion and cohesion.

- **Capillary rise:** when adhesion (liquid–glass) exceeds cohesion, the liquid wets the tube, the meniscus is concave and the liquid rises (water in glass, $\theta \approx 0^\circ$).
- **Capillary depression:** when cohesion exceeds adhesion, the meniscus is convex and the liquid falls (mercury in glass, $\theta \approx 130^\circ$).

Sketch: a glass tube in water with a concave meniscus raised by h; a tube in mercury with a convex meniscus depressed by h.

Expression. At equilibrium, the vertical surface tension force round the tube perimeter balances the weight of the raised column:

$$\sigma\,\pi d\cos\theta = \rho g \times \frac{\pi}{4} d^2 h \qquad \Rightarrow \qquad h = \frac{4\sigma\cos\theta}{\rho g d}$$

h is positive (rise) for $\theta < 90^\circ$ and negative (depression) for $\theta > 90^\circ$. Since $h \propto 1/d$, the effect is significant only in narrow tubes; manometer and piezometer tubes are therefore kept wider than about 6 mm.

QB I.12 AKTU 2025-26 | BME302 | Q3(b) | 7 Marks Describe surface tension, and state its unit and dimensional formula. Find the surface tension in a soap bubble of 40 mm diameter when the pressure inside it is 2.5 N/m² above atmospheric.

SOLUTION

1. Surface tension. Surface tension σ is the tensile force per unit length acting along the free surface of a liquid, at right angles to any line drawn on the surface. It arises because a molecule inside the liquid is pulled equally on all sides by cohesion, while a molecule at the surface is pulled only inward and sideways. The surface therefore behaves like a stretched membrane. It explains the spherical shape of droplets, capillarity and insects walking on water.

2. Unit and dimensions. $\sigma = \dfrac{\text{Force}}{\text{Length}}$: SI unit $\mathrm{N/m}$; dimensions $\dfrac{[MLT^{-2}]}{[L]} = [MT^{-2}]$.
For water–air at $20^\circ\mathrm{C}$, $\sigma \approx 0.073\ \mathrm{N/m}$.

3. Pressure inside a soap bubble. A soap bubble has **two** free surfaces (inner and outer). *Sketch:* half of the bubble as a free body; pressure excess p acts on the projected area $\frac{\pi}{4}d^2$; surface tension acts on two circumferences πd.

$$p \times \frac{\pi}{4} d^2 = 2 \times \sigma\, \pi d \qquad \Rightarrow \qquad p = \frac{8\sigma}{d}$$

(For a liquid droplet, with one surface, $p = 4\sigma/d$.)

4. Numerical.

Given: $d = 40\ \text{mm} = 0.04\ \text{m}$; $p = 2.5\ \text{N/m}^2$ (above atmospheric).

To find: σ.

1. $p = \dfrac{8\sigma}{d} \Rightarrow \sigma = \dfrac{p\,d}{8}$
2. $\sigma = \dfrac{2.5 \times 0.04}{8} = 0.0125\ \text{N/m}$

Answer: surface tension of the soap film $\sigma = 0.0125\ \text{N/m}$, a typical value for soap solution (much less than for pure water).

QB I.13 AKTU 2022-23 | KME302 | Q3(a) | 10 Marks Find the capillary effect (rise or depression) in a glass tube of 4 mm diameter immersed in (i) water and (ii) mercury at 20 °C. Surface tension in contact with air: water 0.073575 N/m, mercury 0.51 N/m. Contact angle: water 0°, mercury 130°. Density of water 998 kg/m³. *(The temperature and angle are printed unclearly; 20 °C and 130° are taken.)*

SOLUTION

The temperature and contact angle are unclear in the paper; as noted, 20°C and 130° are used. The density of mercury is not given; $\rho_{Hg} = 13\,600\ \text{kg/m}^3$ is assumed.

Given: $d = 4\ \text{mm} = 0.004\ \text{m}$; water: $\sigma = 0.073575\ \text{N/m}$, $\theta = 0^\circ$, $\rho = 998\ \text{kg/m}^3$; mercury: $\sigma = 0.51\ \text{N/m}$, $\theta = 130^\circ$, $\rho = 13\,600\ \text{kg/m}^3$.

To find: capillary height h in each case.

Formula. *Sketch:* tube of diameter d with the liquid column raised (or lowered) by h; surface tension σ acts along the meniscus at angle θ to the wall. Equating the vertical component of surface tension round the perimeter to the weight of the column:

$$\sigma\, \pi d \cos\theta = \rho g \frac{\pi}{4} d^2 h \qquad \Rightarrow \qquad h = \frac{4\sigma \cos\theta}{\rho g d}$$

(i) Water

1. $\cos 0^\circ = 1$
2. $h = \dfrac{4 \times 0.073575 \times 1}{998 \times 9.81 \times 0.004} = \dfrac{0.2943}{39.16} = 7.515 \times 10^{-3}\ \text{m}$
3. $h = 7.52\ \text{mm}$ (rise)

(ii) Mercury

1. $\cos 130^\circ = -0.6428$

2. $h = \dfrac{4 \times 0.51 \times (-0.6428)}{13\,600 \times 9.81 \times 0.004} = \dfrac{-1.311}{533.7} = -2.457 \times 10^{-3}\ \text{m}$
3. $h = -2.46\ \text{mm}$ (depression)

Answer: water rises by about 7.52 mm; mercury is depressed by about 2.46 mm. The negative sign for mercury shows a fall, because its contact angle exceeds 90° (cohesion exceeds adhesion, convex meniscus).

I.5 Euler's and Bernoulli's equations

QB I.14 AKTU 2022-23 | KCE303 | Q1(f) | 2 Marks State the assumptions made in deriving Euler's equation of motion.

SOLUTION

Euler's equation $\dfrac{dp}{\rho} + g\,dz + v\,dv = 0$ is derived with these assumptions:

1. The fluid is **ideal** (non-viscous): only pressure and gravity forces act; friction is zero.
2. The flow is **steady**.
3. The flow is along a **streamline**, and the element is small, so velocity and pressure are uniform over its cross-section.
4. (For integration to Bernoulli's equation) the fluid is **incompressible** and the flow is irrotational or along one streamline.

QB I.15 AKTU 2016-17 | ME401 | Q2(a) | 10 Marks Derive Euler's equation of motion along a streamline and obtain Bernoulli's equation from it.

SOLUTION

Euler's equation is Newton's second law applied to a small fluid element moving along a streamline. Integrating it for an incompressible fluid gives Bernoulli's equation.

Assumptions.

1. The fluid is ideal: viscous forces are zero; only pressure and gravity act.
2. The flow is steady.
3. The element is small, so pressure is uniform over its cross-section.
4. The element moves along a streamline; s is the distance along it.
5. For Bernoulli's equation: the fluid is incompressible (ρ constant).

Sketch: a small cylinder of cross-section dA and length ds on a streamline inclined at θ to the vertical; pressure $p\,dA$ on the upstream face, $\left(p + \frac{\partial p}{\partial s} ds\right) dA$ on the downstream face, weight $\rho g\,dA\,ds$ acting vertically down.

Part A: Euler's equation

Step 1. The element. Volume $dA\,ds$, mass $\rho\,dA\,ds$, weight $\rho g\,dA\,ds$. A step ds along the streamline raises it by $dz = ds\cos\theta$.

Step 2. Pressure forces along s.

$$F_{\text{up face}} = p\,dA, \qquad F_{\text{down face}} = \left(p + \frac{\partial p}{\partial s} ds\right) dA$$

The pressure on the curved sides is normal to s and has no component along it.

Step 3. Weight component along s (acts backward when the flow rises):

$$W_s = \rho g\, dA\, ds\, \cos\theta$$

Step 4. Newton's second law along s.

$$p\,dA - \left(p + \frac{\partial p}{\partial s} ds\right) dA - \rho g\, dA\, ds\, \cos\theta = \rho\, dA\, ds\, a_s$$

Step 5. Acceleration. In steady flow $v = v\,(\,s\,)$ only, so

$$a_s = \frac{dv}{dt} = \frac{dv}{ds}\frac{ds}{dt} = v\frac{dv}{ds}$$

Step 6. Simplify. Cancel $p\,dA$ and put $\cos\theta = dz/ds$:

$$-\frac{\partial p}{\partial s}\, ds\, dA - \rho g\, dA\, ds \frac{dz}{ds} = \rho\, dA\, ds\, v\frac{dv}{ds}$$

Step 7. Divide by $\rho\, dA\, ds$ and multiply by ds (along the streamline $\frac{\partial p}{\partial s} ds = dp$):

$$-\frac{dp}{\rho} - g\, dz = v\, dv$$

Result (Euler's equation):

$$\frac{dp}{\rho} + g\, dz + v\, dv = 0$$

Part B: Bernoulli's equation

Step 8. Integrate along the streamline with ρ constant (incompressible):

$$\frac{1}{\rho}\int dp + g\int dz + \int v\, dv = C \quad \Rightarrow \quad \frac{p}{\rho} + gz + \frac{v^2}{2} = C$$

Step 9. Divide by g and write V for the speed:

$$\frac{p}{\rho g} + \frac{V^2}{2g} + z = \text{constant}$$

Step 10. Between two points 1 and 2 on the same streamline:

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

Each term is energy per unit weight (m): pressure head $p/\,(\,\rho g\,)$, velocity head $V^2/\,(\,2g\,)$, datum head z.

Significance. The total head is constant along a streamline in ideal flow: where velocity rises, pressure falls. For a real fluid a loss term is added, $\frac{p_1}{\rho g}+\frac{V_1^2}{2g}+z_1=\frac{p_2}{\rho g}+\frac{V_2^2}{2g}+z_2+h_L$. It is the basis of the venturimeter, orifice meter and Pitot tube.

I.6 Flow measurement: venturimeter, orifice meter, Pitot tube

QB I.16 AKTU 2023-24 | BME302 | Q1(b) | 2 Marks Define the coefficient of contraction C_c and the coefficient of velocity C_v for an orifice meter.

SOLUTION

Coefficient of contraction C_c: ratio of the area of the jet at the vena contracta to the area of the orifice, $C_c=\frac{a_c}{a_0}$ (typically 0.61–0.69).

Coefficient of velocity C_v: ratio of the actual velocity of the jet at the vena contracta to the theoretical velocity, $C_v=\frac{V_{\text{actual}}}{\sqrt{2gH}}$ (typically 0.95–0.99).

Their product gives the coefficient of discharge: $C_d=C_c\times C_v$ (about 0.6–0.65).

QB I.17 AKTU 2023-24 | BME302 | Q2(a) | 7 Marks AKTU 2022-23 | KME302 | Q3(b) | 10 Marks Explain the use of a venturimeter and derive the expression for the discharge through it.

SOLUTION

1. Use. A venturimeter is a device for measuring the **rate of flow (discharge) through a pipe**. It works on Bernoulli's principle: when the flow area is reduced, the velocity increases and the pressure falls; the pressure drop measured by a differential manometer gives the discharge. Its gradual cones keep losses small, so C_d is high (0.95–0.99) and it is used for permanent metering in water mains, pump test rigs and process lines.

2. Construction. *Sketch:* a pipe of diameter d_1 (section 1), a short converging cone (included angle about 20–22°), a throat of diameter d_2 (section 2, usually $1/3$ to $3/4$ of d_1), and a long diverging cone (included angle about 5–7°); a U-tube mercury manometer connected between sections 1 and 2 shows a reading x.

1. **Converging part:** flow accelerates.
2. **Throat:** smallest area, maximum velocity, minimum pressure.
3. **Diverging part:** long and gradual so the flow slows without separation and most of the pressure is recovered.

3. Derivation of discharge.

Assumptions: steady, incompressible flow; ideal fluid between sections 1 and 2 (losses taken later by C_d); uniform velocity at each section. Let a_1, p_1, V_1 and a_2, p_2, V_2 be the area, pressure and velocity at the inlet and the throat.

Step 1. Bernoulli's equation between 1 and 2 (horizontal meter, $z_1=z_2$):

$$\frac{p_1}{\rho g}+\frac{V_1^2}{2g}=\frac{p_2}{\rho g}+\frac{V_2^2}{2g}$$

Step 2. Define the differential head $h=\dfrac{p_1-p_2}{\rho g}$:

$$h=\frac{V_2^2-V_1^2}{2g}$$

Step 3. Continuity:

$$a_1V_1=a_2V_2 \quad\Rightarrow\quad V_1=\frac{a_2}{a_1}V_2$$

Step 4. Eliminate V_1**:**

$$2gh=V_2^2-\frac{a_2^2}{a_1^2}V_2^2=V_2^2\,\frac{a_1^2-a_2^2}{a_1^2}$$

Step 5. Solve for V_2**:**

$$V_2=\frac{a_1}{\sqrt{a_1^2-a_2^2}}\sqrt{2gh}$$

Step 6. Theoretical discharge $Q_{th}=a_2V_2$:

$$Q_{th}=\frac{a_1a_2\sqrt{2gh}}{\sqrt{a_1^2-a_2^2}}$$

Step 7. Actual discharge. Friction makes the actual discharge smaller; $C_d=Q_{\text{act}}/Q_{th}$:

$$Q=C_d\,\frac{a_1a_2\sqrt{2gh}}{\sqrt{a_1^2-a_2^2}}$$

4. Value of h **from the manometer.** For a differential manometer reading x with a liquid heavier than the flowing liquid ($S_h>S_o$):

$$h=x\left(\frac{S_h}{S_o}-1\right)$$

For water with mercury, $h=x\,(\,13.6-1\,)\,=12.6\,x$. For a lighter manometer liquid (inverted U-tube), $h=x\left(1-\dfrac{S_l}{S_o}\right)$. For an inclined or vertical meter, $h=\left(\dfrac{p_1}{\rho g}+z_1\right)-\left(\dfrac{p_2}{\rho g}+z_2\right)$, and the same manometer formulas still give h directly.

Significance. $Q\propto\sqrt{h}$: doubling the flow needs four times the head difference. The venturimeter has a high C_d and low permanent loss, but it is long and costly compared with an orifice meter.

QB I.18 AKTU 2021-22 | KME302 | Q2(a) | 10 Marks A horizontal venturimeter with inlet diameter 200 mm and throat diameter 100 mm carries water. The differential mercury manometer reads 180 mm. Taking $C_d = 0.98$, find the rate of flow.

SOLUTION

Given: $d_1 = 200\ \text{mm} = 0.2\ \text{m}$; $d_2 = 100\ \text{mm} = 0.1\ \text{m}$; manometer reading $x = 180\ \text{mm} = 0.18\ \text{m}$ of mercury ($S_h = 13.6$); water $S_o = 1$; $C_d = 0.98$.

To find: discharge Q.

Sketch: horizontal venturimeter, inlet section 1 and throat section 2, with a mercury U-tube between them showing a 180 mm difference.

1. Areas: $a_1 = \dfrac{\pi}{4} \times 0.2^2 = 0.031\,42\ \text{m}^2$; $a_2 = \dfrac{\pi}{4} \times 0.1^2 = 0.007\,854\ \text{m}^2$
2. Differential head: $h = x\left(\dfrac{S_h}{S_o} - 1\right) = 0.18 \times (\,13.6 - 1\,) = 2.268\ \text{m of water}$
3. $\sqrt{2gh} = \sqrt{2 \times 9.81 \times 2.268} = 6.671\ \text{m/s}$
4. $\sqrt{a_1^2 - a_2^2} = \sqrt{0.031\,42^2 - 0.007\,854^2} = 0.030\,42\ \text{m}^2$
5. $Q = C_d \dfrac{a_1 a_2 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}} = 0.98 \times \dfrac{0.031\,42 \times 0.007\,854 \times 6.671}{0.030\,42}$
6. $Q = 0.98 \times 0.054\,11 = 0.053\,03\ \text{m}^3/\text{s} = 53.0\ \text{litres/s}$

Answer: rate of flow $Q \approx 0.0530\ \text{m}^3/\text{s}$ (53 L/s).

QB I.19 AKTU 2025-26 | BME302 | Q3(a) | 7 Marks A horizontal venturimeter with inlet diameter 30 cm and throat diameter 10 cm carries water. The pressure at the inlet is 13.734 N/cm² (gauge) and the vacuum at the throat is 37 cm of mercury. The head lost between the inlet and the throat is 4% of the differential head. Find (i) the rate of flow and (ii) the coefficient of discharge.

SOLUTION

Given: $d_1 = 30\ \text{cm} = 0.3\ \text{m}$; $d_2 = 10\ \text{cm} = 0.1\ \text{m}$; $p_1 = 13.734\ \text{N/cm}^2 = 137\,340\ \text{N/m}^2$ (gauge); $p_2 = 37\ \text{cm of Hg vacuum}$; $h_L = 0.04\,h$; water $\rho = 1000\ \text{kg/m}^3$; $S_{Hg} = 13.6$.

To find: (i) Q; (ii) C_d.

1. Areas: $a_1 = \dfrac{\pi}{4} \times 0.3^2 = 0.070\,69\ \text{m}^2$; $a_2 = \dfrac{\pi}{4} \times 0.1^2 = 0.007\,854\ \text{m}^2$
2. Inlet pressure head: $\dfrac{p_1}{\rho g} = \dfrac{137\,340}{1000 \times 9.81} = 14.0\ \text{m of water}$
3. Throat pressure head: $\dfrac{p_2}{\rho g} = -0.37 \times 13.6 = -5.032\ \text{m of water}$
4. Differential head: $h = 14.0 - (\,-5.032\,) = 19.032\ \text{m}$
5. Head loss: $h_L = 0.04 \times 19.032 = 0.7613\ \text{m}$
6. Bernoulli with loss (horizontal):
   $\dfrac{p_1}{\rho g} + \dfrac{V_1^2}{2g} = \dfrac{p_2}{\rho g} + \dfrac{V_2^2}{2g} + h_L \Rightarrow \dfrac{V_2^2 - V_1^2}{2g} = h - h_L = 18.27\ \text{m}$

7. Continuity: $V_1 = \dfrac{a_2}{a_1}V_2 = \dfrac{d_2^2}{d_1^2}V_2 = \dfrac{V_2}{9}$
8. $\dfrac{V_2^2}{2g}\left(1 - \dfrac{1}{81}\right) = 18.27 \Rightarrow V_2 = \sqrt{\dfrac{2 \times 9.81 \times 18.27 \times 81}{80}} = 19.05\ \text{m/s}$

(i) Rate of flow

9. $Q = a_2 V_2 = 0.007\,854 \times 19.05 = \ 0.1496\ \text{m}^3/\text{s}$

(ii) Coefficient of discharge

10. Theoretical discharge (no loss):
$$Q_{th} = \frac{a_1 a_2\sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}} = \frac{0.070\,69 \times 0.007\,854 \times \sqrt{2 \times 9.81 \times 19.032}}{\sqrt{0.070\,69^2 - 0.007\,854^2}} = 0.1527\ \text{m}^3/\text{s}$$
11. $C_d = \dfrac{Q}{Q_{th}} = \dfrac{0.1496}{0.1527} = \ 0.980$

(Check: $C_d = \sqrt{(h - h_L)/h} = \sqrt{0.96}\ = 0.980$.)

Answer: $Q \approx 0.1496\ \text{m}^3/\text{s}$ (about 150 L/s); $C_d = 0.98$.

QB I.20 AKTU 2022-23 | KCE303 | Q2(c) | 10 Marks An orifice meter with a 20 cm orifice is fitted in a 50 cm pipe carrying water at 30 °C ($\rho = 987$ kg/m³). A mercury manometer ($\rho = 13\,600$ kg/m³) connected across it reads 30 cm. Taking $C_d = 0.6$, find the rate of flow and the velocity in the pipe.

SOLUTION

Given: orifice $d_0 = 20\ \text{cm} = 0.2\ \text{m}$; pipe $d_1 = 50\ \text{cm} = 0.5\ \text{m}$; water $\rho_o = 987\ \text{kg/m}^3$; mercury $\rho_h = 13\,600\ \text{kg/m}^3$; $x = 30\ \text{cm} = 0.3\ \text{m}$; $C_d = 0.6$.

To find: Q and the pipe velocity V_1.

Sketch: a thin plate with a 20 cm hole in a 50 cm pipe; pressure tappings at section 1 (upstream, pipe) and section 2 (vena contracta) connected to a mercury U-tube reading 30 cm.

Formula (same form as the venturimeter, with C_d allowing for contraction and losses):

$$Q = C_d \frac{a_0 a_1\sqrt{2gh}}{\sqrt{a_1^2 - a_0^2}}$$

1. Areas: $a_0 = \dfrac{\pi}{4} \times 0.2^2 = 0.031\,42\ \text{m}^2$; $a_1 = \dfrac{\pi}{4} \times 0.5^2 = 0.1963\ \text{m}^2$
2. Differential head in water:
$$h = x\left(\frac{\rho_h}{\rho_o} - 1\right) = 0.3 \times \left(\frac{13\,600}{987} - 1\right) = 0.3 \times 12.78 = 3.834\ \text{m}$$
3. $\sqrt{2gh} = \sqrt{2 \times 9.81 \times 3.834}\ = 8.673\ \text{m/s}$
4. $\sqrt{a_1^2 - a_0^2} = \sqrt{0.1963^2 - 0.031\,42^2} = 0.1938\ \text{m}^2$
5. $Q = 0.6 \times \dfrac{0.031\,42 \times 0.1963 \times 8.673}{0.1938} = 0.6 \times 0.2760 = \ 0.1656\ \text{m}^3/\text{s}$

6. Velocity in the pipe: $V_1 = \dfrac{Q}{a_1} = \dfrac{0.1656}{0.1963} = 0.843\ \mathrm{m/s}$

Answer: $Q \approx 0.166\ \mathrm{m^3/s}$ (166 L/s); velocity in the pipe $\approx 0.84\ \mathrm{m/s}$.

Note. The printed density $987\ \mathrm{kg/m^3}$ is low for water at $30^\circ\mathrm{C}$ (the true value is about $996\ \mathrm{kg/m^3}$). Using 996 changes h to $3.797\ \mathrm{m}$ and Q by less than 1%, so the answer above stands.

UNIT II. Fluid kinematics

II.1 Types of flow

QB II.1 AKTU 2023-24 | BME302 | Q1(c) | 2 Marks AKTU 2022-23 | KME302 | Q1(c) | 2 Marks AKTU 2022-23 | KCE303 | Q1(g) | 2 Marks Differentiate between laminar and turbulent flow (explain turbulent flow).

SOLUTION

Laminar flow	Turbulent flow
Fluid moves in smooth parallel layers (laminae) that do not mix	Particles move in a zig-zag, random way; layers mix intensely through eddies
Viscous forces dominate; pipe $\mathrm{Re} < 2000$	Inertia forces dominate; pipe $\mathrm{Re} > 4000$
Velocity profile parabolic, $u_{\max} = 2\bar{V}$	Velocity profile flatter, $u_{\max} \approx 1.2\bar{V}$
Head loss $h_f \propto V$	Head loss $h_f \propto V^{1.75\text{–}2}$

Turbulent flow is flow in which the velocity at a point fluctuates randomly in magnitude and direction about a mean value, causing rapid mixing (e.g. water in city mains, rivers, flow past aircraft).

QB II.2 AKTU 2022-23 | KME302 | Q1(d) | 2 Marks Differentiate between steady and unsteady flow.

SOLUTION

Steady flow	Unsteady flow
Flow properties at a point do not change with time	Flow properties at a point change with time
$\left(\dfrac{\partial V}{\partial t}\right)_{x,y,z} = 0,\ \left(\dfrac{\partial p}{\partial t}\right)_{x,y,z} = 0$	$\left(\dfrac{\partial V}{\partial t}\right)_{x,y,z} \neq 0$
Example: constant discharge through a pipe	Example: flow while a valve is being closed; tank draining
Streamlines, pathlines and streaklines coincide	They are generally different

QB II.3 AKTU 2023-24 | BME302 | Q2(b) | 7 Marks Explain the principle of the continuity equation. Explain steady and unsteady flow.

SOLUTION

1. Principle of continuity. The continuity equation is the **law of conservation of mass** applied to a flowing fluid: mass can be neither created nor destroyed. For a fixed region (control volume) in steady flow, the mass entering per second equals the mass leaving per second.

Sketch: a stream tube (or pipe) of varying section; section 1 with area A_1, velocity V_1, density ρ_1; section 2 with A_2, V_2, ρ_2.

Step 1. Mass entering per second at section 1: $\dot{m}_1 = \rho_1 A_1 V_1$.

Step 2. Mass leaving per second at section 2: $\dot{m}_2 = \rho_2 A_2 V_2$.

Step 3. In steady flow nothing accumulates between 1 and 2, so $\dot{m}_1 = \dot{m}_2$:

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Step 4. For an incompressible fluid ($\rho_1 = \rho_2$):

$$Q = A_1 V_1 = A_2 V_2$$

So where a pipe narrows the velocity increases, and where it widens the velocity falls. In three dimensions the same principle gives $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$ for incompressible flow.

2. Steady flow. Flow in which the velocity, pressure, density, etc. at any point do **not change with time**:

$$\left(\frac{\partial V}{\partial t}\right)_{x_0, y_0, z_0} = 0, \quad \left(\frac{\partial p}{\partial t}\right)_{x_0, y_0, z_0} = 0, \quad \left(\frac{\partial \rho}{\partial t}\right)_{x_0, y_0, z_0} = 0$$

Example: water through a pipe from a constant-head tank with the valve set at one opening.

3. Unsteady flow. Flow in which these properties at a point **change with time**:

$$\left(\frac{\partial V}{\partial t}\right)_{x_0, y_0, z_0} \neq 0$$

Example: flow from a draining tank, flow while a valve is being opened or closed (water hammer), tidal flow.

Significance. Most pipe and machine problems are treated as steady, which allows the simple form $Q = AV$ and Bernoulli's equation to be used.

QB II.4 AKTU 2022-23 | KME302 | Q2(b) | 10 Marks Define (i) uniform and non-uniform flow, (ii) one-, two- and three-dimensional flow and (iii) compressible and incompressible flow.

SOLUTION

(i) Uniform and non-uniform flow

These describe how the velocity changes with **position** (space) at a given instant.

1. **Uniform flow:** the velocity does not change, in magnitude or direction, from point to point along the flow:

$$\left(\frac{\partial V}{\partial s}\right)_{t\,=\,\text{const}} = 0$$

Example: flow through a long straight pipe of constant diameter (mean velocity).

2. **Non-uniform flow:** the velocity changes from point to point:

$$\left(\frac{\partial V}{\partial s}\right)_{t\,=\,\text{const}} \neq 0$$

Example: flow through a tapering pipe, a bend, or a river of varying depth.

Combined with time, four types arise: steady uniform (constant-diameter pipe, constant flow), steady non-uniform (tapering pipe, constant flow), unsteady uniform (constant-diameter pipe while the flow rate changes), unsteady non-uniform (tapering pipe while the flow rate changes, e.g. flood wave).

(ii) One-, two- and three-dimensional flow

The number of space coordinates needed to describe the velocity field.

1. **One-dimensional:** velocity varies with one coordinate only (e.g. along the pipe axis), variations across the section are neglected: $u = f\,(\,x\,)$, $v = 0$, $w = 0$. Example: mean flow in a pipe.
2. **Two-dimensional:** velocity varies with two coordinates; it is the same in all planes parallel to one plane: $u = f_1\,(\,x, y\,)$, $v = f_2\,(\,x, y\,)$, $w = 0$. Example: flow between two wide parallel plates; flow over a long weir or a long cylinder.
3. **Three-dimensional:** velocity varies with all three coordinates: $u = f_1\,(\,x, y, z\,)$, $v = f_2\,(\,x, y, z\,)$, $w = f_3\,(\,x, y, z\,)$. Example: flow in a river bend, flow round a car or a pump impeller.

Sketch: a pipe with a single arrow along its axis (1D); parallel plates with a velocity profile in the x–y plane (2D); a converging duct with flow changing in x, y, z (3D).

(iii) Compressible and incompressible flow

1. **Compressible flow:** density changes from point to point, $\rho \neq \text{constant}$. Example: high-speed gas flow in nozzles, compressors, jet engines; gas flow at Mach number $M > 0.3$.
2. **Incompressible flow:** density is constant, $\rho = \text{constant}$. Example: flow of liquids; air flow at low speed ($M < 0.3$, density change below about 5%).

The Mach number $M = V/C$ (flow speed / speed of sound) decides which applies for gases.

Significance. These classifications decide which equations are used: e.g. $Q = AV$ and Bernoulli's equation in their simple form hold only for steady, incompressible flow.

QB II.5 AKTU 2023-24 | BME302 | Q1(d) | 2 Marks AKTU 2025-26 | BME302 | Q1(c) | 2 Marks Differentiate between rotational and irrotational flow.

SOLUTION

Rotational flow	Irrotational flow
Fluid particles rotate about their own mass centres while moving	Fluid particles do not rotate about their own centres
$\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) \neq 0$	$\omega_z = 0$, i.e. $\frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$
Velocity potential ϕ does not exist	Velocity potential ϕ exists
Example: forced vortex, flow in a boundary layer, pipe flow	Example: free vortex, flow outside the boundary layer

QB II.6 AKTU 2025-26 | BME302 | Q4(b) | 7 Marks Explain rotational and irrotational flow. State whether the flow $\vec{V} = 8x^3\,\hat{i} - 10x^2y\,\hat{j}$ is rotational or irrotational.

SOLUTION

1. Rotational flow. A flow in which the fluid particles, while moving along their paths, also **rotate about their own mass centres**. Example: forced vortex (liquid in a rotating tank), flow near a wall where the velocity gradient makes particles spin.

2. Irrotational flow. A flow in which the fluid particles **do not rotate** about their own centres (they may still move along curved paths). Example: free vortex (water draining from a basin, away from the core), flow outside the boundary layer.

Sketch: two small crosses carried along a curved path; in rotational flow the cross turns as it moves, in irrotational flow it keeps the same orientation.

3. Test. The rotation components of a fluid element are

$$\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right), \quad \omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right), \quad \omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)$$

The flow is irrotational if $\omega_x = \omega_y = \omega_z = 0$. For a 2D flow in the x–y plane only ω_z remains, and the condition is $\frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$. Vorticity $\zeta = 2\omega$.

4. Given flow.

Given: $u = 8x^3$, $v = -10x^2y$, $w = 0$.

1. $\frac{\partial v}{\partial x} = -20xy$; $\frac{\partial u}{\partial y} = 0$
2. $\omega_z = \frac{1}{2}(-20xy - 0) = -10xy$
3. $\omega_x = \omega_y = 0$ (no w, no z-dependence)

Since $\omega_z = -10xy \neq 0$ (except on the x and y axes), the flow is **rotational**.

Answer: the flow is **rotational**, with rotation $\omega_z = -10xy$ (clockwise where $xy > 0$) and vorticity $\zeta = -20xy$.

Note. This field does not satisfy incompressible continuity: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 24x^2 - 10x^2 = 14x^2 \neq 0$. As printed it cannot be a liquid flow; the question asks only about rotation, which is answered above.

QB II.7 AKTU 2022-23 | KCE303 | Q1(d) | 2 Marks Define reversible flow.

SOLUTION

A **reversible flow** is an idealised flow process that can be reversed exactly, so that the fluid and its surroundings return to their original states with no net change. This requires **no friction** (ideal, non-viscous fluid), no heat transfer across a finite temperature difference, no shocks or sudden expansion and no mixing. Reversible adiabatic flow is isentropic. Real flows are irreversible because viscous friction converts mechanical energy into heat (the head loss h_L).

II.2 Reynolds number and Froude number

QB II.8 AKTU 2023-24 | BME302 | Q1(e) | 2 Marks AKTU 2022-23 | KME302 | Q1(e) | 2 Marks AKTU 2021-22 | KME302 | Q1(d) | 2 Marks Write the expression for the Reynolds number. State its significance and applications.

SOLUTION

$$\mathrm{Re} = \frac{\text{Inertia force}}{\text{Viscous force}} = \frac{\rho V D}{\mu} = \frac{V D}{\nu}$$

where D is a characteristic length (pipe diameter).

Significance: it decides the type of flow. In a pipe, $\mathrm{Re} < 2000$: laminar; 2000–4000: transition; $\mathrm{Re} > 4000$: turbulent.

Applications: predicting laminar/turbulent flow and the friction factor in pipes ($f = 64/\mathrm{Re}$ for laminar); boundary layer transition; model testing of fully submerged bodies (submarines, aircraft, flow in pipes) under Reynolds' model law.

QB II.9 AKTU 2025-26 | BME302 | Q1(d) | 2 Marks AKTU 2021-22 | KME302 | Q1(c) | 2 Marks Define the Froude number and discuss its significance.

SOLUTION

The **Froude number** is the square root of the ratio of inertia force to gravity force:

$$\mathrm{Fr} = \sqrt{\frac{F_i}{F_g}} = \frac{V}{\sqrt{gL}}$$

In an open channel L is the depth y, so $\mathrm{Fr} = V/\sqrt{gy}$.

Significance: it governs flows with a free surface where gravity dominates. $\mathrm{Fr} < 1$: sub-critical (tranquil); $\mathrm{Fr} = 1$: critical; $\mathrm{Fr} > 1$: supercritical (rapid). Froude's model law is used for model tests of ships, spillways, weirs, open channels and hydraulic jumps.

II.3 Flow lines

QB II.10 AKTU 2022-23 | KCE303 | Q1(c) | 2 Marks Differentiate between a path line and a streak line.

SOLUTION

Path line	Streak line
Actual path traced by **one fluid particle** over a period of time	Line joining, at an instant, **all particles that have passed through a fixed point**
Lagrangian idea: follow one particle	Observed by injecting dye or smoke continuously at one point
Example: track of a single floating cork (long-exposure photo)	Example: smoke from a chimney, dye filament in water

In steady flow, path lines, streak lines and streamlines coincide.

II.4 Continuity equation

QB II.11 AKTU 2023-24 | BME302 | Q4(b) | 7 Marks AKTU 2025-26 | BME302 | Q2(b) | 7 Marks AKTU 2021-22 | KME302 | Q4(b) | 10 Marks Derive the continuity equation in Cartesian coordinates for three-dimensional flow, and reduce it for steady incompressible flow.

SOLUTION

The continuity equation expresses **conservation of mass** for a small fixed fluid element.

Assumptions.

1. Mass is conserved (no source or sink inside the element).
2. The element is a fixed rectangular box of sides dx, dy, dz; its volume $dx\,dy\,dz$ is constant.
3. The fluid is a continuum with density $\rho\,(\,x, y, z, t\,)$ and velocity components u, v, w.
4. The element is small, so the change of a quantity across it is given by the first term of a Taylor series.

Sketch: a rectangular parallelepiped $ABCDEFGH$ with sides dx, dy, dz; mass flux ρu entering face $ABCD$ (at x) and $\rho u + \frac{\partial\,(\,\rho u\,)}{\partial x} dx$ leaving face $EFGH$ (at $x + dx$).

Step 1. Mass entering the left face per second (x-direction, face area $dy\,dz$):

$$\text{Mass in}_x = \rho u\,dy\,dz$$

Step 2. Mass leaving the right face per second:

$$\text{Mass out}_x = \left[\rho u + \frac{\partial\,(\,\rho u\,)}{\partial x}\,dx\right] dy\,dz$$

Step 3. Net outflow in the x-direction:

$$\text{Net outflow}_x = \frac{\partial\,(\,\rho u\,)}{\partial x}\,dx\,dy\,dz$$

Step 4. Similarly in the y- and z-directions (face areas $dx\,dz$ and $dx\,dy$; fluxes ρv and ρw):

$$\text{Net outflow}_y = \frac{\partial\,(\,\rho v\,)}{\partial y}\,dx\,dy\,dz, \qquad \text{Net outflow}_z = \frac{\partial\,(\,\rho w\,)}{\partial z}\,dx\,dy\,dz$$

Step 5. Total net outflow per second:

$$\left[\frac{\partial\,(\,\rho u\,)}{\partial x} + \frac{\partial\,(\,\rho v\,)}{\partial y} + \frac{\partial\,(\,\rho w\,)}{\partial z}\right] dx\,dy\,dz$$

Step 6. Rate of increase of mass inside the element. Mass inside $= \rho\,dx\,dy\,dz$; only ρ can change:

$$\frac{\partial \rho}{\partial t}\,dx\,dy\,dz$$

Step 7. Mass balance. Net outflow per second = rate of decrease of mass inside, so net outflow + rate of increase = 0:

$$\frac{\partial \rho}{\partial t}\,dx\,dy\,dz + \left[\frac{\partial\,(\,\rho u\,)}{\partial x} + \frac{\partial\,(\,\rho v\,)}{\partial y} + \frac{\partial\,(\,\rho w\,)}{\partial z}\right] dx\,dy\,dz = 0$$

Step 8. Divide by $dx\,dy\,dz$. General continuity equation (steady or unsteady, compressible or incompressible):

$$\frac{\partial \rho}{\partial t} + \frac{\partial\,(\,\rho u\,)}{\partial x} + \frac{\partial\,(\,\rho v\,)}{\partial y} + \frac{\partial\,(\,\rho w\,)}{\partial z} = 0$$

Step 9. Steady flow. $\partial\rho/\partial t = 0$:

$$\frac{\partial\,(\,\rho u\,)}{\partial x} + \frac{\partial\,(\,\rho v\,)}{\partial y} + \frac{\partial\,(\,\rho w\,)}{\partial z} = 0$$

Step 10. Steady incompressible flow. ρ is constant and comes out of each derivative:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

For two-dimensional incompressible flow: $\dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} = 0$. In vector form, $\nabla \cdot \mathbf{V} = 0$.

Significance. Any velocity field describing a liquid flow must satisfy Step 10; it is used to check whether a given flow is “possible” and to find a missing velocity component.

QB II.12 AKTU 2023-24 | BME302 | Q4(a) | 7 Marks A pipe AB of 1.2 m diameter carries water at 3 m/s. It is followed by pipe BC of 1.5 m diameter. At C the pipe branches into CD and CE. Branch CD is 0.8 m in diameter and carries one-third of the flow in AB. The velocity in CE is 2.5 m/s. Find (i) the discharge in AB, (ii) the velocity in BC, (iii) the velocity in CD and (iv) the diameter of CE.

SOLUTION

Given: $D_1 = 1.2\,\text{m}$, $V_1 = 3\,\text{m/s}$ (AB); $D_2 = 1.5\,\text{m}$ (BC); $D_3 = 0.8\,\text{m}$, $Q_3 = Q_1/3$ (CD); $V_4 = 2.5\,\text{m/s}$ (CE).

To find: Q_1, V_2, V_3, D_4.

Sketch: pipe AB (1.2 m) joined at B to a wider pipe BC (1.5 m), which splits at C into branch CD (0.8 m) and branch CE.

Principle: continuity for incompressible flow, $Q = AV$; at the junction, $Q_1 = Q_3 + Q_4$.

(i) Discharge in AB

1. $A_1 = \dfrac{\pi}{4} \times 1.2^2 = 1.131\,\text{m}^2$
2. $Q_1 = A_1 V_1 = 1.131 \times 3 = \ 3.393\,\text{m}^3/\text{s}$

(ii) Velocity in BC

3. $A_2 = \dfrac{\pi}{4} \times 1.5^2 = 1.767\,\text{m}^2$
4. $V_2 = \dfrac{Q_1}{A_2} = \dfrac{3.393}{1.767} = \ 1.92\,\text{m/s}$

(iii) Velocity in CD

5. $Q_3 = \dfrac{Q_1}{3} = \dfrac{3.393}{3} = 1.131\,\text{m}^3/\text{s}$; $A_3 = \dfrac{\pi}{4} \times 0.8^2 = 0.5027\,\text{m}^2$
6. $V_3 = \dfrac{1.131}{0.5027} = \ 2.25\,\text{m/s}$

(iv) Diameter of CE

7. $Q_4 = Q_1 - Q_3 = 3.393 - 1.131 = 2.262\,\text{m}^3/\text{s}$
8. $A_4 = \dfrac{Q_4}{V_4} = \dfrac{2.262}{2.5} = 0.9048\,\text{m}^2$
9. $D_4 = \sqrt{\dfrac{4A_4}{\pi}} = \sqrt{\dfrac{4 \times 0.9048}{\pi}} = \ 1.073\,\text{m}$

Answer: $Q_{AB} = 3.393\,\text{m}^3/\text{s}$; $V_{BC} = 1.92\,\text{m/s}$; $V_{CD} = 2.25\,\text{m/s}$; $D_{CE} \approx 1.07\,\text{m}$.

QB II.13 AKTU 2022-23 | KCE303 | Q2(b) | 10 Marks For a steady, incompressible, two-dimensional flow, the y-component of velocity is $v = 2y^2 + 2x - 2y$. Find a possible x-component u.

SOLUTION

Given: $v = 2y^2 + 2x - 2y$; flow steady, incompressible, two-dimensional.

To find: $u\,(x, y)$.

Principle. Every possible incompressible 2D flow must satisfy the continuity equation (from mass conservation):

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

1. Differentiate v with respect to y (treating x as constant): $\frac{\partial v}{\partial y}=4y-2$
2. From continuity: $\frac{\partial u}{\partial x}=-\frac{\partial v}{\partial y}=-\left(4y-2\right)=2-4y$
3. Integrate with respect to x (treating y as constant); the "constant" of integration may be any function of y:

$$u=\int\left(2-4y\right)\,dx=2x-4xy+f\left(y\right)$$

4. The simplest possible component is obtained with $f\left(y\right)=0$:

$$u=2x-4xy+f\left(y\right),\quad \text{e.g. } u=2x-4xy$$

5. Check: $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=\left(2-4y\right)+\left(4y-2\right)=0$. Continuity is satisfied.

Answer: $u=2x-4xy+f\left(y\right)$; taking $f\left(y\right)=0$, a possible x-component is $u=2x-4xy$.

Significance. Continuity alone fixes u only up to an arbitrary function of y; any choice of $f\left(y\right)$ (e.g. a constant) gives a possible flow. Fixing $f\left(y\right)$ needs another condition, such as irrotationality or a boundary value.

II.5 Stream function and velocity potential (asked in AKTU papers; not named in the current Unit II syllabus)

QB II.14 AKTU 2022-23 | KME302 | Q4(a) | 10 Marks Explain velocity potential and stream function. Show the form of the continuity equation for three-dimensional flow in Cartesian coordinates.

SOLUTION

The sign convention of R.K. Bansal is used: $u=-\partial\phi/\partial x$ and $u=-\partial\psi/\partial y$.

1. Velocity potential function ϕ. A scalar function of space (and time) whose negative derivative in any direction gives the velocity component in that direction:

$$u=-\frac{\partial\phi}{\partial x},\qquad v=-\frac{\partial\phi}{\partial y},\qquad w=-\frac{\partial\phi}{\partial z}$$

(The minus sign shows that flow is in the direction of decreasing ϕ.)

- **Continuity:** substituting into $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0$ gives the **Laplace equation**

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}=0$$

Any ϕ that satisfies it is a possible steady incompressible flow.

- **Rotation:** $\omega_z=\frac{1}{2}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)=\frac{1}{2}\left(-\frac{\partial^2\phi}{\partial x\,\partial y}+\frac{\partial^2\phi}{\partial y\,\partial x}\right)=0$. So ϕ exists **only for irrotational flow**; such flow is called potential flow.
- Lines of constant ϕ are **equipotential lines**.

2. Stream function ψ. A scalar function of space (and time), defined for two-dimensional flow, whose partial derivative in any direction gives the velocity component at right angles (counter-clockwise) to that direction:

$$\frac{\partial\psi}{\partial x}=v,\qquad \frac{\partial\psi}{\partial y}=-u$$

- **Continuity:** $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=-\frac{\partial^2\psi}{\partial x\,\partial y}+\frac{\partial^2\psi}{\partial y\,\partial x}=0$. So ψ **always satisfies continuity**; its existence means the flow is possible.
- **Rotation:** $\omega_z=\frac{1}{2}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)=\frac{1}{2}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}\right)$. If the flow is irrotational, ψ satisfies the Laplace equation $\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}=0$.
- Lines of constant ψ are **streamlines**; the difference $\psi_2-\psi_1$ equals the discharge per unit width between two streamlines.

3. Continuity equation in three-dimensional Cartesian form. From mass conservation for an element $dx\,dy\,dz$ (net mass outflow per second + rate of increase of mass inside = 0):

$$\frac{\partial\rho}{\partial t}+\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0$$

Derivation outline: net outflow in x is $\frac{\partial(\rho u)}{\partial x}dx\,dy\,dz$, similarly in y and z; rate of increase of mass is $\frac{\partial\rho}{\partial t}dx\,dy\,dz$; adding and dividing by $dx\,dy\,dz$ gives the result (full steps as in QB II.11).

Special forms:

Case	Continuity equation
Steady, compressible	$\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0$
Incompressible	$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0$
In terms of ϕ	$\nabla^2\phi=0$

Significance. ϕ and ψ reduce the velocity field to a single scalar function; together they form the flow net used to analyse ideal flow round bodies, under weirs and through dams.

QB II.15 AKTU 2022-23 | KME302 | Q4(b) | 10 Marks Explain the properties of the velocity potential and the stream function. The stream function of a flow is $\psi = x^2 - y^2$. Find the velocity potential ϕ.

SOLUTION

Sign convention (Bansal): $u = -\partial\phi/\partial x$, $v = -\partial\phi/\partial y$; $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$.

1. Properties of the velocity potential ϕ

1. Its negative gradient gives the velocity: $u = -\partial\phi/\partial x$, $v = -\partial\phi/\partial y$, $w = -\partial\phi/\partial z$.
2. It exists only for **irrotational** flow (rotation computed from ϕ is automatically zero).
3. If it satisfies the **Laplace equation** $\nabla^2\phi = 0$, the flow satisfies continuity (possible steady incompressible flow).
4. Lines of $\phi = \text{constant}$ are equipotential lines; they are perpendicular to streamlines.
5. It can be defined for 3D flow.

2. Properties of the stream function ψ

1. Its derivatives give the velocity: $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$ (2D flow).
2. It **always satisfies continuity**, so its existence means a possible flow.
3. If it satisfies $\nabla^2\psi = 0$, the flow is **irrotational**.
4. Lines of $\psi = \text{constant}$ are streamlines; there is no flow across them.
5. $\psi_2 - \psi_1$ = discharge per unit width between two streamlines.
6. It can be defined for rotational as well as irrotational flow.

3. Numerical

Given: $\psi = x^2 - y^2$. **To find:** ϕ.

1. Velocity components: $u = -\dfrac{\partial\psi}{\partial y} = -(-2y) = 2y$; $v = \dfrac{\partial\psi}{\partial x} = 2x$
2. Check irrotationality (ϕ must exist): $\dfrac{\partial v}{\partial x} = 2$, $\dfrac{\partial u}{\partial y} = 2$, so $\omega_z = \frac{1}{2}(2 - 2) = 0$. The flow is irrotational; ϕ exists. (Also $\nabla^2\psi = 2 - 2 = 0$.)
3. From $u = -\dfrac{\partial\phi}{\partial x}$: $\dfrac{\partial\phi}{\partial x} = -2y \Rightarrow \phi = -2xy + f(y)$
4. Differentiate with respect to y: $\dfrac{\partial\phi}{\partial y} = -2x + f'(y)$
5. From $v = -\dfrac{\partial\phi}{\partial y}$: $-(-2x + f'(y)) = 2x \Rightarrow f'(y) = 0 \Rightarrow f(y) = C$
6. Taking the constant as zero:

$$\phi = -2xy$$

7. Check: $-\frac{\partial \phi}{\partial x} = 2y = u$ and $-\frac{\partial \phi}{\partial y} = 2x = v$. Correct.

Answer: $\phi = -2xy$ (plus an arbitrary constant). The streamlines $x^2 - y^2 = \text{const}$ and equipotentials $xy = \text{const}$ are both rectangular hyperbolas and cut each other at right angles.

QB II.16 AKTU 2021-22 | KME302 | Q2(b) | 10 Marks The velocity potential of a two-dimensional flow is $\phi = x^2 - y^2$. Find (i) the velocity components u and v and (ii) the stream function ψ.

SOLUTION

Sign convention (Bansal): $u = -\partial\phi/\partial x$, $v = -\partial\phi/\partial y$; $\partial\psi/\partial x = v$, $\partial\psi/\partial y = -u$.

Given: $\phi = x^2 - y^2$. **To find:** u, v and ψ.

(i) Velocity components

1. $u = -\frac{\partial \phi}{\partial x} = -2x$
2. $v = -\frac{\partial \phi}{\partial y} = -(-2y) = 2y$
3. Check continuity: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = -2 + 2 = 0$ (equivalently $\nabla^2\phi = 2 - 2 = 0$). The flow is a possible incompressible flow, so ψ exists.

(ii) Stream function

4. From $\frac{\partial \psi}{\partial x} = v = 2y$: integrate with respect to x, $\psi = 2xy + g(y)$
5. Differentiate with respect to y: $\frac{\partial \psi}{\partial y} = 2x + g'(y)$
6. This must equal $-u = 2x$: $2x + g'(y) = 2x \Rightarrow g'(y) = 0 \Rightarrow g(y) = C$
7. Taking the constant as zero:

$$\psi = 2xy$$

8. Check: $-\frac{\partial \psi}{\partial y} = -2x = u$ and $\frac{\partial \psi}{\partial x} = 2y = v$. Correct.

Answer: $u = -2x$, $v = 2y$; $\psi = 2xy$. This is flow towards a stagnation point at the origin (flow in a right-angled corner); streamlines $xy = \text{const}$ are hyperbolas.

QB II.17 AKTU 2025-26 | BME302 | Q4(a) | 7 Marks The stream function of a flow is $\psi = 2xy$. Find (i) the velocity at the point $P(2, 3)$ and (ii) the velocity potential ϕ.

SOLUTION

Sign convention (Bansal): $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$; $u = -\partial\phi/\partial x$, $v = -\partial\phi/\partial y$.

Given: $\psi = 2xy$; point $P(2, 3)$ (coordinates in m).

To find: (i) velocity at P; (ii) ϕ.

(i) Velocity at P

1. $u = -\dfrac{\partial \psi}{\partial y} = -2x$; $v = \dfrac{\partial \psi}{\partial x} = 2y$
2. At $P(2, 3)$: $u = -2 \times 2 = -4\ \text{m/s}$; $v = 2 \times 3 = 6\ \text{m/s}$
3. Resultant: $V = \sqrt{u^2 + v^2} = \sqrt{(-4)^2 + 6^2} = \sqrt{52} = 7.211\ \text{m/s}$
4. Direction: $\tan \alpha = \left|\dfrac{v}{u}\right| = \dfrac{6}{4} \Rightarrow \alpha = 56.31^\circ$ above the negative x-axis, i.e. 123.7° from the positive x-axis.

(ii) Velocity potential

5. Check irrotationality: $\dfrac{\partial v}{\partial x} = 0$, $\dfrac{\partial u}{\partial y} = 0$, so $\omega_z = 0$. The flow is irrotational; ϕ exists.
6. From $u = -\dfrac{\partial \phi}{\partial x} = -2x$: $\dfrac{\partial \phi}{\partial x} = 2x \Rightarrow \phi = x^2 + f(y)$
7. From $v = -\dfrac{\partial \phi}{\partial y} = 2y$: $-f'(y) = 2y \Rightarrow f(y) = -y^2 + C$
8. Taking the constant as zero:

$$\phi = x^2 - y^2$$

Answer: at $P(2, 3)$, $u = -4\ \text{m/s}$, $v = 6\ \text{m/s}$, resultant $V \approx 7.21\ \text{m/s}$ at 123.7° to the x-axis; $\phi = x^2 - y^2$.

QB II.18 AKTU 2022-23 | KCE303 | Q4(a) | 10 Marks Prove that streamlines and equipotential lines are orthogonal to each other.

SOLUTION

Two families of curves are orthogonal if, at every point where they meet, the product of their slopes is -1. We find the slope dy/dx of each line.

Sign convention (Bansal): $u = -\dfrac{\partial \phi}{\partial x}$, $v = -\dfrac{\partial \phi}{\partial y}$; $u = -\dfrac{\partial \psi}{\partial y}$, $v = \dfrac{\partial \psi}{\partial x}$.

Assumptions: steady, two-dimensional, incompressible and irrotational flow (so both ϕ and ψ exist).

Sketch: a flow net: a family of streamlines ψ_1, ψ_2, ψ_3 crossed at right angles by equipotential lines ϕ_1, ϕ_2, ϕ_3, forming small curvilinear squares; at an intersection, the velocity V is tangent to the streamline and normal to the equipotential line.

Step 1. Equipotential line ($\phi = \text{constant}$). $\phi = \phi(x, y)$, so its total differential is

$$d\phi = \frac{\partial \phi}{\partial x}\, dx + \frac{\partial \phi}{\partial y}\, dy$$

Step 2. Substitute $\dfrac{\partial \phi}{\partial x} = -u$, $\dfrac{\partial \phi}{\partial y} = -v$:

$$d\phi = -u\, dx - v\, dy$$

Step 3. Along an equipotential line ϕ is constant, so $d\phi = 0$:

$$-u\,dx - v\,dy = 0 \quad \Rightarrow \quad \left(\frac{dy}{dx}\right)_{\phi} = -\frac{u}{v}$$

Step 4. Streamline ($\psi = \text{constant}$). Total differential:

$$d\psi = \frac{\partial \psi}{\partial x}\,dx + \frac{\partial \psi}{\partial y}\,dy = v\,dx - u\,dy$$

Step 5. Along a streamline $d\psi = 0$:

$$v\,dx - u\,dy = 0 \quad \Rightarrow \quad \left(\frac{dy}{dx}\right)_{\psi} = \frac{v}{u}$$

(This is the familiar streamline equation $dx/u = dy/v$: the streamline is tangent to the velocity.)

Step 6. Product of slopes:

$$\left(\frac{dy}{dx}\right)_{\phi} \times \left(\frac{dy}{dx}\right)_{\psi} = \left(-\frac{u}{v}\right) \times \frac{v}{u} = -1$$

Result.

$$\left(\frac{dy}{dx}\right)_{\phi} \times \left(\frac{dy}{dx}\right)_{\psi} = -1$$

Hence streamlines and equipotential lines intersect at right angles everywhere: they are **orthogonal**. (At a stagnation point, $u = v = 0$, the slopes are undefined and the result does not apply.)

Significance. This property is the basis of the **flow net**, a grid of streamlines and equipotential lines drawn to solve ideal-flow problems such as seepage under a dam or flow round a body. Velocity is tangent to streamlines and normal to equipotential lines.

QB II.19 AKTU 2022-23 | KCE303 | Q4(b) | 10 Marks The velocity components of a flow are $u = 4xy$ and $v = c - 2y^2$, where c is a constant. Explain the method of finding the stream function from the velocity components, and find ψ.

SOLUTION

Sign convention (Bansal): $\frac{\partial \psi}{\partial x} = v$, $\frac{\partial \psi}{\partial y} = -u$.

1. Method of finding ψ from u and v

1. **Check continuity:** $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$. A stream function exists only if the flow satisfies continuity.
2. **Integrate one relation:** from $\frac{\partial \psi}{\partial y} = -u$, integrate with respect to y, holding x constant. The "constant" of integration is an unknown function $f(x)$.
3. **Differentiate the result** with respect to x and equate it to v (since $\frac{\partial \psi}{\partial x} = v$). This gives $f'(x)$.

4. **Integrate** $f'(x)$ to find $f(x)$; the remaining additive constant can be taken as zero (only differences of ψ matter).
5. **Check** by differentiating the final ψ to recover u and v.

(Equally, one may start with $\frac{\partial \psi}{\partial x} = v$ and find a function $g(y)$.)

2. Numerical

Given: $u = 4xy$, $v = c - 2y^2$.

1. Continuity: $\frac{\partial u}{\partial x} = 4y$, $\frac{\partial v}{\partial y} = -4y$; sum $= 0$. Continuity is satisfied, so ψ exists.
2. $\frac{\partial \psi}{\partial y} = -u = -4xy$. Integrating with respect to y: $\psi = -2xy^2 + f(x)$
3. $\frac{\partial \psi}{\partial x} = -2y^2 + f'(x)$, which must equal $v = c - 2y^2$
4. $-2y^2 + f'(x) = c - 2y^2 \Rightarrow f'(x) = c \Rightarrow f(x) = cx + C_1$
5. Taking $C_1 = 0$:

$$\psi = cx - 2xy^2$$

6. Check: $-\frac{\partial \psi}{\partial y} = 4xy = u$; $\frac{\partial \psi}{\partial x} = c - 2y^2 = v$. Correct.

Answer: $\psi = cx - 2xy^2$ (plus an arbitrary constant).

Remark. $\frac{\partial v}{\partial x} = 0$ and $\frac{\partial u}{\partial y} = 4x$, so $\omega_z = \frac{1}{2}(0 - 4x) = -2x \neq 0$. The flow is rotational: a stream function exists, but a velocity potential does not.

QB II.20 AKTU 2022-23 | KCE303 | Q1(e) | 2 Marks Explain sink flow.

SOLUTION

Sink flow is a two-dimensional ideal flow in which fluid moves **radially inward** from all directions towards a point (a line normal to the plane), where it is removed at a constant rate q per unit depth (strength of the sink). It is the reverse of source flow.

Radial velocity at radius r: $u_r = -\frac{q}{2\pi r}$, $u_\theta = 0$. Streamlines are radial lines and equipotential lines are concentric circles. Example: water draining from a large flat tank through a central hole (without swirl).

II.6 Dimensional analysis and similarity (CO2 of BME302; not listed in the unit topics)

QB II.21 AKTU 2022-23 | KCE303 | Q1(i) | 2 Marks State the requirement for dynamic similarity.

SOLUTION

Dynamic similarity between a model and its prototype exists when the **ratios of corresponding forces** (inertia, viscous, gravity, pressure, surface tension, elastic) at all corresponding points are the same:

$$\frac{(F_i)_m}{(F_i)_p} = \frac{(F_v)_m}{(F_v)_p} = \frac{(F_g)_m}{(F_g)_p} = \ldots = F_r$$

It requires **geometric and kinematic similarity** first, and then equality of the governing dimensionless number(s): e.g. $\mathrm{Re}_m = \mathrm{Re}_p$ for pipe flow and submerged bodies, $\mathrm{Fr}_m = \mathrm{Fr}_p$ for free-surface flows.

QB II.22 AKTU 2022-23 | KCE303 | Q1(j) | 2 Marks Name four important dimensionless numbers used in fluid mechanics.

SOLUTION

1. **Reynolds number** $\mathrm{Re} = \dfrac{\rho V L}{\mu}$ (inertia / viscous force): pipe flow, submerged bodies.
2. **Froude number** $\mathrm{Fr} = \dfrac{V}{\sqrt{gL}}$ (inertia / gravity, square root): open channels, ships, spillways.
3. **Euler number** $\mathrm{Eu} = \dfrac{V}{\sqrt{p/\rho}}$ (inertia / pressure, square root): pressure-dominated flows, cavitation.
4. **Mach number** $M = \dfrac{V}{C}$ (inertia / elastic, square root): compressible, high-speed flow.

(Also the **Weber number** $\mathrm{We} = \dfrac{V}{\sqrt{\sigma/(\rho L)}}$ for surface tension effects.)

QB II.23 AKTU 2021-22 | KME302 | Q4(a) | 10 Marks The resistance R of a partially submerged body depends on its velocity V, length l, the viscosity μ and density ρ of the fluid, and g. Find an expression for R using Buckingham's π theorem.

SOLUTION

Buckingham's π theorem: if a physical relation has n variables containing m fundamental dimensions, it can be written as a relation among $(n-m)$ independent dimensionless groups (π terms).

Given: $R = f(V, l, \mu, \rho, g)$, i.e. $f_1(R, V, l, \mu, \rho, g) = 0$.

Step 1. Dimensions of the variables.

Variable	Symbol	Dimensions
Resistance (force)	R	MLT^{-2}
Velocity	V	LT^{-1}
Length	l	L
Dynamic viscosity	μ	$ML^{-1}T^{-1}$
Density	ρ	ML^{-3}
Acceleration due to gravity	g	LT^{-2}

Step 2. Number of π terms. $n = 6$, $m = 3$ (M, L, T), so $n - m = 3$:

$$f_1\left(\pi_1, \pi_2, \pi_3\right) = 0$$

Step 3. Repeating variables. Choose $m = 3$ variables that together contain M, L, T and do not form a dimensionless group: l (geometry), V (flow), ρ (fluid property).

Step 4. First π term: $\pi_1 = l^{a_1} V^{b_1} \rho^{c_1} R$

$$M^0 L^0 T^0 = L^{a_1} \left(LT^{-1}\right)^{b_1} \left(ML^{-3}\right)^{c_1} MLT^{-2}$$

- M: $0 = c_1 + 1 \Rightarrow c_1 = -1$
- T: $0 = -b_1 - 2 \Rightarrow b_1 = -2$
- L: $0 = a_1 + b_1 - 3c_1 + 1 = a_1 - 2 + 3 + 1 \Rightarrow a_1 = -2$

$$\pi_1 = \frac{R}{\rho V^2 l^2}$$

Step 5. Second π term: $\pi_2 = l^{a_2} V^{b_2} \rho^{c_2} \mu$

$$M^0 L^0 T^0 = L^{a_2} \left(LT^{-1}\right)^{b_2} \left(ML^{-3}\right)^{c_2} ML^{-1}T^{-1}$$

- M: $0 = c_2 + 1 \Rightarrow c_2 = -1$
- T: $0 = -b_2 - 1 \Rightarrow b_2 = -1$
- L: $0 = a_2 + b_2 - 3c_2 - 1 = a_2 - 1 + 3 - 1 \Rightarrow a_2 = -1$

$$\pi_2 = \frac{\mu}{\rho V l}$$

Step 6. Third π term: $\pi_3 = l^{a_3} V^{b_3} \rho^{c_3} g$

$$M^0 L^0 T^0 = L^{a_3} \left(LT^{-1}\right)^{b_3} \left(ML^{-3}\right)^{c_3} LT^{-2}$$

- M: $0 = c_3 \Rightarrow c_3 = 0$
- T: $0 = -b_3 - 2 \Rightarrow b_3 = -2$
- L: $0 = a_3 + b_3 - 3c_3 + 1 = a_3 - 2 + 1 \Rightarrow a_3 = 1$

$$\pi_3 = \frac{lg}{V^2}$$

Step 7. Final relation. $f_1\left(\frac{R}{\rho V^2 l^2}, \frac{\mu}{\rho V l}, \frac{lg}{V^2}\right) = 0$, so

$$R = \rho\, l^2 V^2\, \phi\left[\frac{\mu}{\rho V l},\ \frac{lg}{V^2}\right]$$

Significance. $\frac{\mu}{\rho V l} = \frac{1}{\mathrm{Re}}$ and $\frac{lg}{V^2} = \frac{1}{\mathrm{Fr}^2}$, so $R = \rho l^2 V^2\, \phi\,(\,\mathrm{Re}, \mathrm{Fr}\,)$. The resistance of a ship depends on both viscous (Reynolds) and wave-making (Froude) effects; since both cannot be matched in one model, ship models are run on Froude's law and the skin friction is corrected separately.

QB II.24 AKTU 2022-23 | KCE303 | Q2(e) | 10 Marks The discharge Q over a rectangular weir depends on the head H, g, the length L of the weir, the height Z of the weir crest and the kinematic viscosity ν. Find an expression for Q by dimensional analysis.

SOLUTION

Method: Buckingham's π theorem.

Given: $Q = f\,(\,H, g, L, Z, \nu\,)$, i.e. $f_1\,(\,Q, H, g, L, Z, \nu\,) = 0$.

Step 1. Dimensions.

Variable	Symbol	Dimensions
Discharge	Q	L^3T^{-1}
Head over the weir	H	L
Acceleration due to gravity	g	LT^{-2}
Length of weir	L	L
Height of weir crest	Z	L
Kinematic viscosity	ν	L^2T^{-1}

Step 2. Number of π terms. $n = 6$. Mass does not appear, so only L and T are involved: $m = 2$. Number of π terms $= n - m = 4$:

$$f_1\,(\,\pi_1, \pi_2, \pi_3, \pi_4\,) = 0$$

Step 3. Repeating variables. $m = 2$: choose H (geometry) and g (flow/gravity); together they contain L and T.

Step 4. $\pi_1 = H^{a_1} g^{b_1} Q$

$$L^0T^0 = L^{a_1}\left(LT^{-2}\right)^{b_1} L^3T^{-1}$$

- T: $0 = -2b_1 - 1 \Rightarrow b_1 = -\frac{1}{2}$
- L: $0 = a_1 + b_1 + 3 \Rightarrow a_1 = -\frac{5}{2}$

$$\pi_1 = \frac{Q}{H^{5/2}\sqrt{g}}$$

Step 5. $\pi_2 = H^{a_2} g^{b_2} L$. T: $b_2 = 0$; L: $a_2 + 1 = 0 \Rightarrow a_2 = -1$.

$$\pi_2 = \frac{L}{H}$$

Step 6. $\pi_3 = H^{a_3} g^{b_3} Z$. Similarly $b_3 = 0$, $a_3 = -1$.

$$\pi_3 = \frac{Z}{H}$$

Step 7. $\pi_4 = H^{a_4} g^{b_4} \nu$

$$L^0 T^0 = L^{a_4} \left(L T^{-2} \right)^{b_4} L^2 T^{-1}$$

- T: $0 = -2b_4 - 1 \Rightarrow b_4 = -\frac{1}{2}$
- L: $0 = a_4 + b_4 + 2 \Rightarrow a_4 = -\frac{3}{2}$

$$\pi_4 = \frac{\nu}{H^{3/2}\sqrt{g}}$$

Step 8. Final relation. $f_1 \left(\frac{Q}{H^{5/2}\sqrt{g}}, \frac{L}{H}, \frac{Z}{H}, \frac{\nu}{H^{3/2}\sqrt{g}} \right) = 0$, so

$$Q = \sqrt{g}\ H^{5/2}\, \phi \left[\frac{L}{H}, \frac{Z}{H}, \frac{\nu}{H^{3/2}\sqrt{g}} \right]$$

Significance. The result agrees with the weir formula $Q = \frac{2}{3} C_d L \sqrt{2g}\, H^{3/2}$: since ϕ may contain the factor L/H, $Q = \sqrt{g}\ H^{3/2} L \times \phi'$, where ϕ' (i.e. C_d) depends on Z/H and the viscous group, which is a Reynolds-type number. For large weirs the viscous effect is small.

QB II.25 AKTU 2022-23 | KCE303 | Q7(b) | 10 Marks A 1 : 64 model of an open channel is built on Froude similarity. Manning's n for the prototype is 0.014. Find Manning's n for the model.

SOLUTION

Given: scale ratio $L_r = \frac{L_p}{L_m} = 64$ (undistorted model); Froude similarity; $n_p = 0.014$.

To find: n_m.

Step 1. Froude model law. $\mathrm{Fr}_m = \mathrm{Fr}_p$:

$$\frac{V_m}{\sqrt{gL_m}} = \frac{V_p}{\sqrt{gL_p}} \quad \Rightarrow \quad V_r = \frac{V_p}{V_m} = \sqrt{L_r}$$

Step 2. Manning's formula for each:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

where R is the hydraulic radius (a length) and S the bed slope.

Step 3. Ratio of velocities. Geometric similarity gives $R_r = L_r$ and $S_r = 1$ (slope is a ratio of lengths):

$$V_r = \frac{1}{n_r} R_r^{2/3} S_r^{1/2} = \frac{L_r^{2/3}}{n_r}$$

Step 4. Equate Steps 1 and 3:

$$\sqrt{L_r} = \frac{L_r^{2/3}}{n_r} \quad \Rightarrow \quad n_r = \frac{n_p}{n_m} = L_r^{2/3\,-\,1/2} = L_r^{1/6}$$

Step 5. Numerical.

1. $L_r^{1/6} = 64^{1/6} = 2$
2. $n_m = \dfrac{n_p}{L_r^{1/6}} = \dfrac{0.014}{2} = 0.007$

Answer: Manning's n for the model $= 0.007$, half the prototype value.

Note. A Manning's n of 0.007 is smoother than practical model materials (smooth glass or perspex give about 0.009–0.010). Such a model would be too rough; in practice the model is distorted (larger vertical scale) or its slope is adjusted to obtain correct flow behaviour.

UNIT III. Pipe flow, boundary layer, drag and lift

III.1 Laminar flow through pipes and between plates

QB III.1 AKTU 2023-24 | BME302 | Q5(b) | 7 Marks Derive the expression for the velocity distribution for laminar flow of water through a circular pipe.

SOLUTION

In laminar flow the fluid moves in concentric cylindrical layers. A force balance on a cylinder of fluid, combined with Newton's law of viscosity, gives the velocity at any radius.

Assumptions. Steady, laminar, fully developed flow of an incompressible Newtonian fluid in a straight horizontal pipe of radius R; the pressure is uniform over a cross-section; no slip at the wall.

Sketch: a horizontal pipe with a fluid cylinder of radius r and length Δx on the axis; pressure p on the left face, $p + \left(\partial p/\partial x\right)\Delta x$ on the right face, shear τ on the curved surface acting backwards; beside it a parabolic velocity profile, zero at the wall and $u_{\max}$ on the axis.

Step 1. Forces on the element.

$$p\,\pi r^2 - \left(p + \frac{\partial p}{\partial x}\Delta x\right)\pi r^2 - \tau\,\left(2\pi r\,\Delta x\right) = 0$$

Step 2. Simplify. Cancel $p\,\pi r^2$ and divide by $\pi r^2 \Delta x$:

$$-\frac{\partial p}{\partial x} = \frac{2\tau}{r} \quad \Rightarrow \quad \tau = -\frac{\partial p}{\partial x}\frac{r}{2}$$

Step 3. Newton's law of viscosity. With $y = R - r$ measured from the wall, $dy = -dr$:

$$\tau = \mu \frac{du}{dy} = -\mu \frac{du}{dr}$$

Step 4. Equate the two expressions for τ.

$$-\mu \frac{du}{dr} = -\frac{\partial p}{\partial x}\frac{r}{2} \quad \Rightarrow \quad \frac{du}{dr} = \frac{1}{2\mu}\frac{\partial p}{\partial x} r$$

Step 5. Integrate. $\partial p/\partial x$ does not depend on r:

$$u = \frac{1}{4\mu}\frac{\partial p}{\partial x} r^2 + C$$

Step 6. Boundary condition. At $r = R$, $u = 0$ (no slip):

$$C = -\frac{1}{4\mu}\frac{\partial p}{\partial x} R^2$$

Step 7. Substitute C.

$$u = -\frac{1}{4\mu}\frac{\partial p}{\partial x}\left(R^2 - r^2 \right)$$

Step 8. Maximum velocity. u is largest on the axis ($r = 0$):

$$u_{\max} = -\frac{1}{4\mu}\frac{\partial p}{\partial x} R^2$$

Result.

$$u = -\frac{1}{4\mu}\frac{\partial p}{\partial x}\left(R^2 - r^2 \right) = u_{\max}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

Significance. The profile is a paraboloid of revolution: zero at the wall, maximum at the axis. Integrating it gives the mean velocity $V = u_{\max}/2$ and the Hagen–Poiseuille equation $p_1 - p_2 = 32\mu VL/D^2$.

QB III.2 AKTU 2025-26 | BME302 | Q5(b) | 7 Marks An oil of viscosity 0.7 N·s/m² and specific gravity 1.3 flows through a 100 mm diameter pipe. The shear stress at the pipe wall is 196.2 N/m². Find (i) the pressure gradient and (ii) the average velocity. Check that the flow is laminar.

SOLUTION

Given: $\mu = 0.7\ \text{N·s/m}^2$; $S = 1.3$, so $\rho = 1300\ \text{kg/m}^3$; $D = 0.1\ \text{m}$, $R = 0.05\ \text{m}$; $\tau_0 = 196.2\ \text{N/m}^2$.

To find: $\partial p/\partial x$, $\bar{u}$, and Re.

1. Wall shear stress in laminar pipe flow: $\tau_0 = -\frac{\partial p}{\partial x}\frac{R}{2}$
2. $-\frac{\partial p}{\partial x} = \frac{2\tau_0}{R} = \frac{2 \times 196.2}{0.05} = 7848\ \text{N/m}^3$, so $\frac{\partial p}{\partial x} = -7848\ \text{N/m}^2\ \text{per m}$ (pressure falls along the flow)

3. Maximum velocity: $u_{\max} = -\dfrac{1}{4\mu}\dfrac{\partial p}{\partial x}R^2 = \dfrac{7848 \times 0.05^2}{4 \times 0.7} = 7.007\ \text{m/s}$
4. Average velocity: $\bar{u} = \dfrac{u_{\max}}{2} = \dfrac{7.007}{2} = 3.504\ \text{m/s}$
5. Reynolds number: $\text{Re} = \dfrac{\rho \bar{u} D}{\mu} = \dfrac{1300 \times 3.504 \times 0.1}{0.7} = 650.7$

Answer: pressure gradient $-7848\ \text{N/m}^2$ per metre; average velocity $3.504\ \text{m/s}$; $\text{Re} = 650.7 < 2000$, so the flow is laminar and the laminar formulas used are valid.

QB III.3 AKTU 2022-23 | KCE303 | Q6(b) | 10 Marks In laminar flow through a 300 mm diameter pipe, the maximum velocity is 2 m/s. Find (i) the average velocity and the radial position where it occurs, and (ii) the velocity 50 mm from the wall.

SOLUTION

Given: $D = 0.3\ \text{m}$, $R = 0.15\ \text{m}$; $u_{\max} = 2\ \text{m/s}$; laminar flow.

To find: (i) $\bar{u}$ and the radius r where $u = \bar{u}$; (ii) u at 50 mm from the wall.

Basis. For laminar pipe flow $u = u_{\max}\left[1 - (r/R)^2\right]$ and $\bar{u} = u_{\max}/2$.

(i) Average velocity and its location

1. $\bar{u} = \dfrac{u_{\max}}{2} = \dfrac{2}{2} = 1\ \text{m/s}$
2. Put $u = \bar{u}$: $u_{\max}\left[1 - \left(\dfrac{r}{R}\right)^2\right] = \dfrac{u_{\max}}{2} \Rightarrow \left(\dfrac{r}{R}\right)^2 = \dfrac{1}{2}$
3. $r = \dfrac{R}{\sqrt{2}} = 0.7071 \times 0.15 = 0.1061\ \text{m} = 106.1\ \text{mm}$ from the axis
4. Distance from the wall: $R - r = 150 - 106.1 = 43.9\ \text{mm}$

(ii) Velocity 50 mm from the wall

1. $r = R - y = 0.15 - 0.05 = 0.10\ \text{m}$
2. $u = 2\left[1 - \left(\dfrac{0.10}{0.15}\right)^2\right] = 2\,(1 - 0.4444) = 1.111\ \text{m/s}$

Answer: $\bar{u} = 1\ \text{m/s}$, occurring at $r = 0.707R = 106.1\ \text{mm}$ from the axis (43.9 mm from the wall); velocity 50 mm from the wall $= 1.111\ \text{m/s}$.

QB III.4 AKTU 2022-23 | KME302 | Q5(b) | 10 Marks Derive the momentum correction factor and the kinetic energy correction factor for laminar flow through a circular pipe.

SOLUTION

The velocity in a pipe is not uniform, but in the momentum and energy equations we use the mean velocity V. Correction factors account for the difference.

Definitions.

- Momentum correction factor: $\beta = \dfrac{\text{true momentum flux}}{\text{momentum flux at } V} = \dfrac{1}{AV^2}\int_A u^2\,dA$

- Kinetic energy correction factor: $\alpha = \dfrac{\text{true KE flux}}{\text{KE flux at } V} = \dfrac{1}{AV^3}\int_A u^3\,dA$

Sketch: pipe cross-section of radius R with a thin ring of radius r and width dr; its area is $dA = 2\pi r\,dr$; beside it the parabolic laminar profile.

Data for laminar flow. $u = u_{\max}\left(1 - r^2/R^2\right)$, $V = u_{\max}/2$, $A = \pi R^2$.

Useful substitution. Put $s = r^2/R^2$; then $r\,dr = \dfrac{R^2}{2}\,ds$ and s goes from 0 to 1.

(a) Momentum correction factor β

Step 1. Momentum flux through the ring. Mass flow $\rho u\,dA$ carries velocity u, so the momentum flux is $\rho u^2\,dA$.

Step 2. Integrate over the section.

$$\int_A u^2\,dA = \int_0^R u_{\max}^2\left(1 - \frac{r^2}{R^2}\right)^2 2\pi r\,dr$$

$$= 2\pi u_{\max}^2\,\frac{R^2}{2}\int_0^1 (1-s)^2\,ds = \pi u_{\max}^2 R^2 \times \frac{1}{3}$$

Step 3. Momentum flux at the mean velocity.

$$AV^2 = \pi R^2\left(\frac{u_{\max}}{2}\right)^2 = \frac{\pi R^2 u_{\max}^2}{4}$$

Step 4. Ratio.

$$\beta = \frac{\pi u_{\max}^2 R^2/3}{\pi R^2 u_{\max}^2/4} = \frac{4}{3}$$

$$\beta = \frac{4}{3} \approx 1.33$$

(b) Kinetic energy correction factor α

Step 1. KE flux through the ring. Mass flow $\rho u\,dA$ carries kinetic energy $u^2/2$ per unit mass, so the KE flux is $\frac{1}{2}\rho u^3\,dA$.

Step 2. Integrate.

$$\int_A u^3\,dA = 2\pi u_{\max}^3\,\frac{R^2}{2}\int_0^1 (1-s)^3\,ds = \pi u_{\max}^3 R^2 \times \frac{1}{4}$$

Step 3. KE flux at the mean velocity.

$$AV^3 = \pi R^2\left(\frac{u_{\max}}{2}\right)^3 = \frac{\pi R^2 u_{\max}^3}{8}$$

Step 4. Ratio.

$$\alpha = \frac{\pi u_{\max}^3 R^2/4}{\pi R^2 u_{\max}^3/8} = 2$$

$$\alpha = 2$$

Significance. In laminar flow the true kinetic energy is twice $V^2/(2g)$ and the true momentum is $4/3$ of ρQV, so the velocity head in Bernoulli's equation must be written $\alpha V^2/(2g)$. In turbulent flow the profile is flat, and $\alpha \approx 1.05$, $\beta \approx 1.02$, so they are usually taken as 1.

QB III.5 AKTU 2022-23 | KCE303 | Q2(d) | 10 Marks A fluid is in laminar motion between two parallel plates, driven by the motion of one plate and a pressure gradient, such that the net discharge across any section is zero. Find (i) the location and magnitude of the minimum velocity and (ii) sketch the velocity distribution.

SOLUTION

Set-up. Two parallel plates a distance t apart. The lower plate ($y = 0$) is fixed; the upper plate ($y = t$) moves with velocity U in the x direction. A pressure gradient $\partial p/\partial x$ also acts. Flow is steady, laminar and fully developed; per unit width.

Step 1. Force balance on an element $\Delta x \times \Delta y$ (unit width). Pressure forces and shear forces on the top and bottom faces balance:

$$\left[p - \left(p + \frac{\partial p}{\partial x}\Delta x\right)\right]\Delta y + \left[\left(\tau + \frac{\partial \tau}{\partial y}\Delta y\right) - \tau\right]\Delta x = 0 \quad \Rightarrow \quad \frac{\partial \tau}{\partial y} = \frac{\partial p}{\partial x}$$

Step 2. Newton's law. $\tau = \mu\, du/dy$, so

$$\mu\frac{d^2u}{dy^2} = \frac{\partial p}{\partial x}$$

Step 3. Integrate twice.

$$u = \frac{1}{2\mu}\frac{\partial p}{\partial x}y^2 + C_1 y + C_2$$

Step 4. Boundary conditions. At $y = 0$, $u = 0$: $C_2 = 0$. At $y = t$, $u = U$: $C_1 = \dfrac{U}{t} - \dfrac{1}{2\mu}\dfrac{\partial p}{\partial x}t$.

Hence

$$u = \frac{Uy}{t} - \frac{1}{2\mu}\frac{\partial p}{\partial x}\left(ty - y^2\right)$$

Step 5. Discharge per unit width.

$$q = \int_0^t u\, dy = \frac{Ut}{2} - \frac{1}{2\mu}\frac{\partial p}{\partial x}\left(\frac{t^3}{2} - \frac{t^3}{3}\right) = \frac{Ut}{2} - \frac{t^3}{12\mu}\frac{\partial p}{\partial x}$$

Step 6. Zero net discharge. Put $q = 0$:

$$\frac{\partial p}{\partial x} = \frac{6\mu U}{t^2}$$

This is positive: the pressure rises in the direction of plate motion (an adverse gradient) and pushes fluid back.

Step 7. Velocity profile. Substitute into Step 4:

$$u = \frac{Uy}{t} - \frac{3U}{t^2}\left(ty - y^2\right) \quad \Rightarrow \quad u = U\left[3\left(\frac{y}{t}\right)^2 - 2\frac{y}{t}\right]$$

(i) Minimum velocity

Step 8. Set $du/dy = 0$:

$$\frac{du}{dy} = U\left(\frac{6y}{t^2} - \frac{2}{t}\right) = 0 \quad \Rightarrow \quad y = \frac{t}{3}$$

($d^2u/dy^2 = 6U/t^2 > 0$, so this is a minimum.)

Step 9. Its value:

$$u_{\min} = U\left[3 \times \frac{1}{9} - 2 \times \frac{1}{3}\right] = -\frac{U}{3}$$

The minus sign means the fluid near the fixed plate flows backwards, opposite to the moving plate.

(ii) Velocity distribution

Step 10. Zeros of u: $3\left(y/t\right)^2 - 2\left(y/t\right) = 0 \Rightarrow y = 0$ and $y = 2t/3$.

y/t	**0**	**1/3**	**2/3**	**1**
u/U	0	$-1/3$	0	1

Sketch: a parabola between the plates. It starts at $u = 0$ on the fixed plate, bulges backwards to $-U/3$ at $y = t/3$, crosses zero at $y = 2t/3$, and reaches $+U$ at the moving plate. The area of the backward lobe ($0 < y < 2t/3$) equals the area of the forward lobe ($2t/3 < y < t$), giving zero net flow.

Significance. This is the situation in a closed-end journal bearing gap or in a dragged film against a closed end: the moving surface drags fluid forward near itself, and the pressure built up returns it near the fixed surface.

III.2 Resistance to flow, turbulence and losses

QB III.6 AKTU 2025-26 | BME302 | Q1(e) | 2 Marks Write the Darcy–Weisbach equation in terms of the discharge Q.

SOLUTION

The Darcy–Weisbach equation is $h_f = \dfrac{fLV^2}{2gD}$. Substituting $V = \dfrac{4Q}{\pi D^2}$:

$$h_f = \frac{8fLQ^2}{\pi^2 g D^5} \approx \frac{fLQ^2}{12.1\,D^5}$$

Here f is the Darcy friction factor ($f = 64/\mathrm{Re}$ in laminar flow), L the length and D the diameter. At a fixed discharge the loss varies as $1/D^5$.

QB III.7 AKTU 2025-26 | BME302 | Q5(a) | 7 Marks Explain (i) the frictional loss in pipe flow and (ii) Prandtl's mixing length theory for turbulent shear stress.

SOLUTION

(i) Frictional loss in pipe flow

When a fluid flows through a pipe, the wall shear stress τ_0 resists the motion. Part of the energy of the fluid is converted into heat, and the total head falls along the pipe. This fall is the **frictional (major) loss** h_f.

1. Force balance on a length L of pipe: $\left(p_1 - p_2\right)\dfrac{\pi D^2}{4} = \tau_0\,\pi D L$, so $h_f = \dfrac{p_1 - p_2}{\rho g} = \dfrac{4\tau_0 L}{\rho g D}$.
2. Experiments show $\tau_0 \propto \rho V^2$; writing $\tau_0 = \dfrac{f}{8}\rho V^2$ gives the **Darcy–Weisbach equation**:

$$h_f = \frac{f L V^2}{2 g D}$$

3. Laws of fluid friction: in **laminar** flow the loss is proportional to V, independent of wall roughness, and $f = 64/\mathrm{Re}$; in **turbulent** flow the loss varies roughly as $V^{1.75}$ to V^2, and f depends on Re and on the relative roughness k/D (Moody chart).

(ii) Prandtl's mixing length theory

In turbulent flow, the velocity fluctuates: $u = \bar{u} + u'$, $v = v'$. Lumps of fluid moving across layers carry momentum and create an extra (Reynolds) shear stress $\tau_t = -\rho\,\overline{u'v'}$.

Sketch: a mean velocity profile $\bar{u}\,(\,y\,)$ near a wall; a fluid lump at level y moves up a distance l to level $y + l$, where the mean velocity is larger by $l\,d\bar{u}/dy$.

1. **Assumption.** A lump travels a distance l (the **mixing length**) across the flow before mixing with its new surroundings, keeping its original velocity on the way.
2. On arrival, its velocity differs from the local mean by $\Delta u = l\dfrac{d\bar{u}}{dy}$, so $u' \approx l\dfrac{d\bar{u}}{dy}$.
3. By continuity, the transverse fluctuation is of the same order: $v' \approx l\dfrac{d\bar{u}}{dy}$.
4. Hence the turbulent shear stress:

$$\tau_t = \rho\, l^2 \left(\frac{d\bar{u}}{dy}\right)^2$$

5. Near a wall Prandtl took $l = \kappa y$, with $\kappa = 0.4$ (von Kármán constant). With $\tau \approx \tau_0 = \rho u_*^2$, $d\bar{u}/dy = u_*/\,(\,\kappa y\,)$; integrating gives the **logarithmic velocity law** $\bar{u}/u_* = 5.75\,\log_{10} y + C$.

Significance. Unlike viscosity, the mixing length is a property of the flow, not of the fluid. The theory explains why turbulent stresses are much larger than viscous ones and gives the velocity laws used for turbulent pipe flow.

QB III.8 AKTU 2021-22 | KME302 | Q1(e) | 2 Marks Explain eddy viscosity.

SOLUTION

Boussinesq expressed the turbulent shear stress in the same form as Newton's law of viscosity:

$$\tau_t = \eta \frac{d\bar{u}}{dy}$$

where η is the **eddy viscosity** (N·s/m²). It represents momentum transfer by the mixing of eddies, not molecular friction. Unlike μ, it is **not a fluid property**: it depends on the flow, the turbulence level and the distance from the wall, and is usually hundreds of times larger than μ. From mixing length theory, $\eta = \rho\, l^2\, |d\bar{u}/dy|$.

QB III.9 AKTU 2021-22 | KME302 | Q1(f) | 2 Marks Define the laminar sub-layer.

SOLUTION

In turbulent flow over a wall, a very thin layer next to the wall remains laminar, because turbulent fluctuations are damped by the wall. This layer is the **laminar (viscous) sub-layer**. Its thickness is

$$\delta' = \frac{11.6\,\nu}{u_*}, \qquad u_* = \sqrt{\tau_0/\rho}$$

If the roughness height $k < 0.25\,\delta'$ the surface is hydraulically smooth; if $k > 6\,\delta'$ it is hydraulically rough.

QB III.10 AKTU 2022-23 | KME302 | Q5(a) | 10 Marks Explain (i) major and minor losses, (ii) a siphon, (iii) the boundary layer on a flat plate, (iv) the velocity distribution in turbulent flow and (v) an equivalent pipe.

SOLUTION

(i) Major and minor losses

- **Major loss** is the loss of head due to friction along the pipe length, given by the Darcy–Weisbach equation $h_f = \dfrac{fLV^2}{2gD}$. It is large in long pipes.
- **Minor losses** are losses due to changes in the velocity or direction of flow at fittings: they are caused by eddies and separation. They are written as $h = k\dfrac{V^2}{2g}$:

Minor loss	Expression
Sudden enlargement	$h_e = \frac{(V_1 - V_2)^2}{2g}$
Sudden contraction	$h_c = \left(\frac{1}{C_c} - 1\right)^2 \frac{V_2^2}{2g} \approx 0.5\frac{V_2^2}{2g}$
Entrance to pipe	$0.5\frac{V^2}{2g}$
Exit from pipe	$\frac{V^2}{2g}$
Bends, valves, fittings	$k\frac{V^2}{2g}$

In long pipes (length more than about 1000 diameters) minor losses are small and are often neglected.

(ii) Siphon

A siphon is a long bent pipe that carries liquid from a higher reservoir to a lower one over a ridge (summit) higher than the upper water level. *Sketch:* upper reservoir A, pipe rising over a summit S, falling to lower reservoir B.

- Flow is driven by the level difference H: $H = h_L + V^2/(2g)$, where h_L is the sum of the entry and friction losses and $V^2/(2g)$ is lost at exit.
- The pressure at the summit is below atmospheric. Applying Bernoulli between the upper surface and the summit (height h_s above the upper level): $\frac{p_s}{\rho g} = \frac{p_{\text{atm}}}{\rho g} - h_s - \frac{V^2}{2g} - h_L$.
- The summit pressure must stay above the vapour pressure (in practice above about 2.5 m of water absolute), so the summit height above the upper level is limited to about 7–8 m for water. Otherwise dissolved air comes out and the flow stops.
- Uses: draining tanks and channels, carrying water over hills, emptying canals.

(iii) Boundary layer on a flat plate

When a fluid of velocity U flows over a flat plate, the fluid in contact with the plate has zero velocity (no slip). In a thin layer near the plate the velocity rises from 0 to U; this layer is the **boundary layer**. Its thickness δ is where $u = 0.99U$.

Sketch: plate with flow from left; a thin laminar boundary layer starting at the leading edge, a transition zone, then a thicker turbulent boundary layer with a laminar sub-layer beneath it.

- **Laminar boundary layer** near the leading edge, for $\mathrm{Re}_x = Ux/\nu < 5 \times 10^5$; $\delta/x = 5/\sqrt{\mathrm{Re}_x}$ (Blasius).
- **Transition** region, then a **turbulent boundary layer** for $\mathrm{Re}_x > 5 \times 10^5$; $\delta/x = 0.37/\mathrm{Re}_x^{1/5}$.
- Below the turbulent layer is a thin **laminar sub-layer**.
- Outside the boundary layer the flow can be treated as ideal (inviscid).

(iv) Velocity distribution in turbulent flow

Using Prandtl's mixing length ($l = 0.4y$) the velocity near the wall is logarithmic:

$$\frac{u_{\max} - \bar{u}}{u_*} = 5.75 \log_{10} \frac{R}{y} \quad \left(\text{velocity defect law, smooth and rough pipes} \right)$$

$$\text{Smooth: } \frac{\bar{u}}{u_*} = 5.75 \log_{10} \frac{u_* y}{\nu} + 5.5, \qquad \text{Rough: } \frac{\bar{u}}{u_*} = 5.75 \log_{10} \frac{y}{k} + 8.5$$

A simple empirical form is the one-seventh power law $u/u_{\max} = \left(y/R \right)^{1/7}$. The turbulent profile is much flatter than the laminar parabola: the mean velocity is about $0.8\, u_{\max}$ (compared with $0.5\, u_{\max}$ in laminar flow), because mixing evens out the velocity across the core.

(v) Equivalent pipe

An **equivalent pipe** is a single pipe of uniform diameter D that carries the same discharge with the same total head loss as a compound pipe (pipes of lengths L_1, L_2, L_3 and diameters D_1, D_2, D_3 in series). With the same f and Q, and neglecting minor losses:

$$\frac{fLQ^2}{12.1\, D^5} = \frac{fL_1Q^2}{12.1\, D_1^5} + \frac{fL_2Q^2}{12.1\, D_2^5} + \frac{fL_3Q^2}{12.1\, D_3^5}$$

$$\frac{L}{D^5} = \frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5} \qquad \left(\text{Dupuit's equation, } L = L_1 + L_2 + L_3 \right)$$

QB III.11 AKTU 2021-22 | KME302 | Q5(a) | 10 Marks Explain (i) a siphon, (ii) pipes in series and (iii) the total energy line.

SOLUTION

(i) Siphon

A siphon is a long bent pipe used to carry liquid from a reservoir at a higher level to one at a lower level when the two are separated by a ridge higher than the upper liquid surface.

Sketch: reservoir A (upper), the pipe rising from A to the summit S at height h_s above A's surface, then falling to reservoir B at a level H below A.

1. **Discharge.** Bernoulli between the surfaces of A and B:

$$H = h_L + \frac{V^2}{2g} \quad \left(\text{friction + entry + exit losses} \right)$$

2. **Summit pressure.** Bernoulli between surface A and summit S:

$$\frac{p_{\text{atm}}}{\rho g} = \frac{p_s}{\rho g} + h_s + \frac{V^2}{2g} + h_{L,\, AS} \quad \Rightarrow \quad \frac{p_s}{\rho g} = \frac{p_{\text{atm}}}{\rho g} - h_s - \frac{V^2}{2g} - h_{L,\, AS}$$

So the summit is below atmospheric pressure.

3. **Limitation.** If p_s falls to the vapour pressure, air and vapour are released, an air lock forms and the flow stops. For water the summit is kept less than about 7–8 m above the upper level (absolute pressure not less than about 2.5 m of water).
4. **Uses.** Carrying water over a hill, draining a tank or channel, emptying a reservoir without a bottom outlet.

(ii) Pipes in series (compound pipe)

Pipes of different lengths and diameters joined end to end form a compound pipe.

Sketch: reservoir A, pipe 1 (L_1, D_1), pipe 2 (L_2, D_2), pipe 3 (L_3, D_3), reservoir B; head difference H.

1. **Continuity:** the same discharge flows through each pipe:

$$Q = \frac{\pi}{4} D_1^2 V_1 = \frac{\pi}{4} D_2^2 V_2 = \frac{\pi}{4} D_3^2 V_3$$

2. **Energy:** the total head difference equals the sum of all losses:

$$H = \frac{f_1 L_1 V_1^2}{2gD_1} + \frac{f_2 L_2 V_2^2}{2gD_2} + \frac{f_3 L_3 V_3^2}{2gD_3} + \text{minor losses}$$

3. If minor losses are neglected and f is the same: $H = \dfrac{fQ^2}{12.1}\left(\dfrac{L_1}{D_1^5} + \dfrac{L_2}{D_2^5} + \dfrac{L_3}{D_3^5}\right)$
4. Such a system can be replaced by an **equivalent pipe** with $L/D^5 = \sum L_i/D_i^5$.

(iii) Total energy line (TEL)

The **total energy line** is a line joining the points that represent the total head $\dfrac{p}{\rho g} + \dfrac{V^2}{2g} + z$ at successive sections of a pipe, plotted above the pipe axis. The **hydraulic gradient line (HGL)** joins the piezometric heads $\dfrac{p}{\rho g} + z$, and lies below the TEL by $V^2/(2g)$.

Sketch: a reservoir feeding a pipe with a sudden enlargement, ending in a lower reservoir; the TEL starts at the upper water surface, drops by $0.5V^2/(2g)$ at entry, slopes down steadily along each pipe (friction), drops suddenly at the enlargement, and ends at the lower water surface; the HGL runs parallel to it, $V^2/(2g)$ lower.

Properties:

1. The TEL always slopes **downwards** in the direction of flow (unless a pump adds energy); its fall between two sections equals the head loss.
2. Its slope along a uniform pipe is h_f/L.
3. It drops suddenly at minor losses (entry, enlargement, valves).
4. Where the HGL falls below the pipe, the pressure there is below atmospheric (as at a siphon summit).

Significance. The TEL and HGL show at a glance where energy is lost and where the pressure may become dangerously low.

QB III.12 AKTU 2022-23 | KCE303 | Q5(b) | 10 Marks Two reservoirs with a difference in water level of 60 m are connected by a pipe 200 mm in diameter and 4000 m long. Water is drawn off from the pipe at a point 1500 m from the upper reservoir. Taking $f = 0.024$ and neglecting minor losses, find the discharge in each section of the pipe. *(The offtake is printed as 500 L/s. A 200 mm pipe of this length cannot carry that much, so the solution explains this and works the problem with 50 L/s.)*

SOLUTION

Given: level difference $H = 60\,\text{m}$; $D = 0.2\,\text{m}$; total length 4000 m; offtake at J, with $L_1 = 1500\,\text{m}$ (upper reservoir A to J) and $L_2 = 2500\,\text{m}$ (J to lower reservoir B); $f = 0.024$ (Darcy); minor losses neglected.

To find: discharges Q_1 (A to J) and Q_2 (in J–B).

Sketch: reservoir A, 60 m above reservoir B, joined by one pipe; at J, 1500 m from A, a branch draws off q. Take the datum at the water level of B, and let h_J be the piezometric head at J.

Pipe resistance. $h_f = \dfrac{8fLQ^2}{\pi^2 g D^5} = kQ^2$ with

$$\frac{8f}{\pi^2 g D^5} = \frac{8 \times 0.024}{\pi^2 \times 9.81 \times 0.2^5} = 6.197\,\text{s}^2/\text{m}^6$$

1. $k_1 = 6.197 \times 1500 = 9296\,\text{s}^2/\text{m}^5$; $k_2 = 6.197 \times 2500 = 15\,493\,\text{s}^2/\text{m}^5$

Step A. Check the printed offtake of 500 L/s.

2. With no offtake, the pipe carries $Q = \sqrt{\dfrac{60}{k_1 + k_2}} = \sqrt{\dfrac{60}{24\,789}} = 0.0492\,\text{m}^3/\text{s} = 49.2\,\text{L/s}$.
3. The most that A alone can send to J is when h_J falls to the level of B ($Q_2 = 0$): $Q_1 = \sqrt{\dfrac{60}{k_1}} = \sqrt{\dfrac{60}{9296}} = 0.0803\,\text{m}^3/\text{s} = 80.3\,\text{L/s}$.
4. For any offtake above 80.3 L/s, h_J must fall below the level of B, so **both** reservoirs feed the junction: $Q_1 + Q_2 = q$, with $60 + x = k_1 Q_1^2$ and $x = k_2 Q_2^2$, where x is the depth of h_J below the level of B.
5. For $q = 0.5\,\text{m}^3/\text{s}$, solving $\sqrt{\dfrac{60 + x}{9296}} + \sqrt{\dfrac{x}{15\,493}} = 0.5$ by trial gives $x \approx 704\,\text{m}$ ($Q_1 = 0.287$, $Q_2 = 0.213\,\text{m}^3/\text{s}$).
6. A piezometric head about 704 m below the lower water surface means a pressure in the pipe far below a perfect vacuum (gauge pressure cannot go below about $-10.3\,\text{m}$ of water). This is impossible: the water would vaporise and the flow would break. Equivalently, to push 500 L/s through the first 1500 m alone would need $k_1 \times 0.5^2 = 2324\,\text{m}$ of head, against only 60 m available, at a velocity of $0.5/0.03142 = 15.9\,\text{m/s}$.

So the 500 L/s figure is a misprint; the problem is solved with $q = 50\,\text{L/s} = 0.05\,\text{m}^3/\text{s}$.

Step B. Direction of flow in J–B for $q = 50$ L/s.

7. $q = 50\ \mathrm{L/s} < 80.3\ \mathrm{L/s}$, so A alone can supply the offtake and still send water on: h_J stays above the level of B, and water flows from J **into** B. Continuity at J: $Q_1 = Q_2 + q$.

Step C. Solve.

8. Energy, A to J: $60 - h_J = k_1 Q_1^2$; J to B: $h_J - 0 = k_2 Q_2^2$. Adding: $k_1 Q_1^2 + k_2 Q_2^2 = 60$
9. Put $Q_1 = Q_2 + 0.05$: $9296\ \left(Q_2 + 0.05\right)^2 + 15\,493\, Q_2^2 = 60$
10. Expand: $24\,789\, Q_2^2 + 929.6\, Q_2 + 23.24 - 60 = 0 \Rightarrow 24\,789\, Q_2^2 + 929.6\, Q_2 - 36.76 = 0$
11. $Q_2 = \dfrac{-929.6 + \sqrt{929.6^2 + 4 \times 24\,789 \times 36.76}}{2 \times 24\,789} = 0.02408\ \mathrm{m^3/s} = 24.08\ \mathrm{L/s}$ (J to B)
12. $Q_1 = Q_2 + q = 0.02408 + 0.05 = 0.07408\ \mathrm{m^3/s} = 74.08\ \mathrm{L/s}$ (A to J)

Check.

13. $h_{f1} = 9296 \times 0.07408^2 = 51.02\ \mathrm{m}$; $h_{f2} = 15\,493 \times 0.02408^2 = 8.98\ \mathrm{m}$; sum $= 60.0\ \mathrm{m}$. Head at J is 8.98 m above the level of B, confirming flow into B.
14. Velocities: $V_1 = 0.07408/0.03142 = 2.36\ \mathrm{m/s}$, $V_2 = 0.02408/0.03142 = 0.77\ \mathrm{m/s}$, both reasonable.

Answer: with an offtake of 50 L/s: $Q_1 = 74.1\ \mathrm{L/s}$ from the upper reservoir to the junction, 50 L/s drawn off, and $Q_2 = 24.1\ \mathrm{L/s}$ flowing on into the lower reservoir.

Note. The printed 500 L/s cannot be delivered by this pipe: even with the junction head lowered to the lower water level the upper reservoir supplies only 80.3 L/s, and the extra needed from both reservoirs would require an impossible vacuum of about 700 m of water.

QB III.13 AKTU 2022-23 | KME302 | Q1(f) | 2 Marks Explain water hammer briefly.

SOLUTION

When a valve at the end of a long pipe is closed suddenly, the moving water is brought to rest abruptly. Its momentum is converted into a large pressure rise, which travels up and down the pipe as a pressure wave with a hammering noise. This is **water hammer**. For sudden closure ($t < 2L/C$) in a rigid pipe the pressure rise is

$$p = \rho V C, \qquad C = \sqrt{K/\rho}$$

It can burst pipes and damage fittings; it is controlled by slow valve closure, relief valves and surge tanks.

III.3 Boundary layer

QB III.14 AKTU 2022-23 | KCE303 | Q1(h) | 2 Marks Define the laminar boundary layer.

SOLUTION

When a fluid flows over a flat plate, a thin layer forms near the surface in which the velocity rises from zero (no slip) to the free-stream value U. Near the leading edge this layer is laminar: the fluid moves in smooth layers with no mixing. This is the **laminar boundary layer**. It exists while $\mathrm{Re}_x = Ux/\nu < 5 \times 10^5$, and its thickness grows as

$$\frac{\delta}{x} = \frac{5}{\sqrt{\mathrm{Re}_x}} \quad \text{(Blasius)}$$

QB III.15 AKTU 2023-24 | BME302 | Q2(c) | 7 Marks The velocity distribution in a boundary layer is $u/U = 2\,(\,y/\delta\,) - (\,y/\delta\,)^2$. Find the displacement thickness and the momentum thickness.

SOLUTION

Given: $\dfrac{u}{U} = 2\dfrac{y}{\delta} - \left(\dfrac{y}{\delta}\right)^2$ for $0 \le y \le \delta$.

To find: displacement thickness δ^* and momentum thickness θ.

Substitution. Let $\eta = y/\delta$, so $dy = \delta\, d\eta$ and η runs from 0 to 1. Then $u/U = 2\eta - \eta^2$ and $1 - u/U = 1 - 2\eta + \eta^2 = (\,1-\eta\,)^2$.

(a) Displacement thickness

1. Definition: $\delta^* = \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy$
2. $\delta^* = \delta \int_0^1 \left(1 - 2\eta + \eta^2\right) d\eta = \delta \left[\eta - \eta^2 + \frac{\eta^3}{3}\right]_0^1$
3. $\delta^* = \delta\left(1 - 1 + \dfrac{1}{3}\right) = \dfrac{\delta}{3}$

(b) Momentum thickness

1. Definition: $\theta = \int_0^{\delta} \frac{u}{U}\left(1 - \frac{u}{U}\right) dy$
2. Product: $\left(2\eta - \eta^2\right)\left(1 - 2\eta + \eta^2\right) = 2\eta - 5\eta^2 + 4\eta^3 - \eta^4$
3. $\theta = \delta \int_0^1 \left(2\eta - 5\eta^2 + 4\eta^3 - \eta^4\right) d\eta = \delta \left[\eta^2 - \frac{5\eta^3}{3} + \eta^4 - \frac{\eta^5}{5}\right]_0^1$
4. $\theta = \delta\left(1 - \dfrac{5}{3} + 1 - \dfrac{1}{5}\right) = \delta \times \dfrac{15 - 25 + 15 - 3}{15} = \dfrac{2\delta}{15}$

Answer: $\delta^* = \delta/3 = 0.333\,\delta$ and $\theta = 2\delta/15 = 0.133\,\delta$. The shape factor $H = \delta^*/\theta = 2.5$, typical of a laminar boundary layer.

QB III.16 AKTU 2021-22 | KME302 | Q2(c) | 10 Marks Derive the expression for the energy thickness of a boundary layer.

SOLUTION

Definition. The **energy thickness** δ^{**} is the distance by which the wall would have to be displaced so that the free stream, flowing at velocity U through that thickness, carries kinetic energy equal to the loss of kinetic energy caused by the boundary layer.

Sketch: a flat plate with a boundary layer of thickness δ; free stream U outside; a thin strip of thickness dy at height y moving with velocity u; a dashed line at height δ^{**} above the plate representing the equivalent layer.

Assumptions. Steady, two-dimensional, incompressible flow; consider unit width ($b = 1$) of the plate; free-stream velocity U outside the layer.

Step 1. Mass flow through the strip. Area of the strip $= b\,dy$; mass flow rate:

$$dm = \rho\, u\, b\, dy$$

Step 2. Kinetic energy actually carried by this mass.

$$\frac{1}{2}\, dm\, u^2 = \frac{1}{2}\rho\, u\, b\, dy\, u^2$$

Step 3. Kinetic energy this same mass would carry without the boundary layer (moving at U):

$$\frac{1}{2}\, dm\, U^2 = \frac{1}{2}\rho\, u\, b\, dy\, U^2$$

Step 4. Loss of kinetic energy through the strip.

$$d\,(\text{ KE loss }) = \frac{1}{2}\rho\, u\, b\left(U^2 - u^2\right) dy$$

Step 5. Total loss over the boundary layer.

$$\text{KE loss} = \frac{1}{2}\rho\, b \int_0^{\delta} u\left(U^2 - u^2\right) dy$$

Step 6. KE carried by free-stream fluid through thickness δ^{**}. Mass flow $\rho U b \delta^{**}$, with kinetic energy:

$$\frac{1}{2}\left(\rho U b\, \delta^{**}\right) U^2 = \frac{1}{2}\rho\, b\, U^3\, \delta^{**}$$

Step 7. Equate Steps 5 and 6.

$$\frac{1}{2}\rho\, b\, U^3\, \delta^{**} = \frac{1}{2}\rho\, b \int_0^{\delta} u\left(U^2 - u^2\right) dy$$

Step 8. Divide by $\frac{1}{2}\rho\, b\, U^3$.

Result.

$$\delta^{**} = \int_0^{\delta} \frac{u}{U}\left(1 - \frac{u^2}{U^2}\right) dy$$

Example of use. For the linear profile $u/U = y/\delta$:
$\delta^{**} = \delta \int_0^1 \eta \left(1 - \eta^2\right) d\eta = \delta \left(\frac{1}{2} - \frac{1}{4}\right) = \frac{\delta}{4}$.

Significance. The energy thickness measures the rate at which kinetic energy is dissipated in the boundary layer. Along with δ^* (mass deficit) and θ (momentum deficit), it is used in integral methods; always $\delta > \delta^* > \delta^{**} > \theta$.

QB III.17 AKTU 2023-24 | BME302 | Q5(a) | 7 Marks AKTU 2025-26 | BME302 | Q2(c) | 7 Marks AKTU 2022-23 | KME302 | Q2(c) | 10 Marks

Explain boundary layer separation with a sketch. Explain the effect of the pressure gradient on separation, and the methods of preventing (controlling) it.

SOLUTION

(i) Boundary layer separation

When a fluid flows over a curved surface (or a surface where the pressure rises downstream), the fluid in the boundary layer loses energy to friction. If the pressure increases in the direction of flow, this slow fluid near the wall cannot move against the rising pressure: it stops and then reverses. The boundary layer then leaves the surface. This is **boundary layer separation**, and the point where it starts is the **point of separation**.

Sketch: flow over a cylinder or a curved surface; velocity profiles drawn at several stations: (a) full profile upstream, (b) profile becoming less full, (c) at the separation point S the profile is vertical at the wall, $\left(\partial u/\partial y\right)_{y=0} = 0$, (d) downstream of S the profile shows reverse flow near the wall; a separated region of eddies (wake) behind S.

Mechanism.

1. Upstream of the maximum thickness of a body the pressure falls; the fluid accelerates.
2. Downstream the pressure rises; the fluid in the boundary layer, already slowed by friction, decelerates further.
3. At some point its momentum near the wall is used up: the velocity gradient at the wall becomes zero. Beyond this point the flow reverses and eddies form.

Criterion for separation. With wall shear stress $\tau_0 = \mu \left(\frac{\partial u}{\partial y}\right)_{y=0}$:

Condition at the wall	State of the flow
$\left(\frac{\partial u}{\partial y}\right)_{y=0} > 0$	Attached flow, no separation
$\left(\frac{\partial u}{\partial y}\right)_{y=0} = 0$	Point of separation (on the verge)
$\left(\frac{\partial u}{\partial y}\right)_{y=0} < 0$	Separated flow, reverse flow near the wall

(ii) Effect of the pressure gradient

Near the wall $u \approx 0$, so the boundary layer equation gives $\mu \left(\frac{\partial^2 u}{\partial y^2} \right)_{y=0} = \frac{dp}{dx}$. The sign of dp/dx fixes the curvature of the profile at the wall:

1. **Favourable gradient,** $dp/dx < 0$ (pressure falls downstream, accelerating flow): the pressure helps push the slow fluid forward; the profile is full, the boundary layer grows slowly, and **separation does not occur**.
2. **Zero gradient,** $dp/dx = 0$ (flat plate): the boundary layer grows steadily but does not separate.
3. **Adverse gradient,** $dp/dx > 0$ (pressure rises downstream, decelerating flow): the rising pressure opposes the slow fluid; the profile develops a point of inflection, the wall gradient decreases, and **separation can occur** where $\left(\partial u/\partial y \right)_{y=0} = 0$.

So separation occurs only in an adverse pressure gradient. A turbulent boundary layer has more momentum near the wall and resists separation longer than a laminar one.

Effects of separation. Large pressure (form) drag, loss of energy in eddies, loss of lift and stall on aerofoils, vibration and noise, poor diffuser efficiency.

(iii) Methods of preventing (controlling) separation

1. **Streamlining the body:** gradual tapering reduces the adverse pressure gradient (aerofoil and teardrop shapes).
2. **Suction of the boundary layer:** slow fluid is sucked into the body through slots or a porous surface, leaving fresh high-energy fluid at the wall.
3. **Blowing (injection):** high-velocity fluid is blown tangentially through a slot to re-energise the boundary layer.
4. **Slotted wings and leading-edge slats:** high-pressure air from below the wing flows through a slot to the upper surface and energises the boundary layer (used at take-off and landing).
5. **Rotating cylinder at the leading edge / rotating the body:** the moving wall adds momentum to the boundary layer (Magnus-effect devices).
6. **Vortex generators and tripping the layer turbulent:** small vanes or roughness make the layer turbulent, giving it more momentum near the wall (dimples on a golf ball).
7. **Guide vanes in bends and gentle diffuser angles:** keep the adverse gradient small in internal flows.

Significance. Controlling separation reduces drag and energy loss, delays stall of wings, and improves the efficiency of diffusers, bends and turbomachines.

III.4 Drag and lift

QB III.18 AKTU 2021-22 | KME302 | Q5(b) | 10 Marks A kite 0.8 m × 0.8 m weighs 3.924 N. It makes an angle of 12° with the horizontal, and the string makes 45° with the horizontal. The pull on the string is 24.525 N when the wind blows at 30 km/h. Taking the density of air as 1.25 kg/m³, find the coefficients of drag and lift. *(The angles are printed unclearly; 12° and 45° are taken.)*

SOLUTION

The printed angles are unclear; as stated in the question, 12° (kite) and 45° (string) are used.

Given: $A = 0.8 \times 0.8 = 0.64\ \text{m}^2$; $W = 3.924\ \text{N}$; kite at 12°; string at $\theta = 45^\circ$ to the horizontal; $T = 24.525\ \text{N}$; $U = 30\ \text{km/h} = \dfrac{30 \times 1000}{3600} = 8.333\ \text{m/s}$; $\rho = 1.25\ \text{kg/m}^3$.

To find: C_D and C_L.

Sketch: the kite plate inclined at 12° to the wind, wind blowing horizontally; forces on the kite: drag F_D horizontally downwind, lift F_L vertically upward, weight W downward, string tension T pulling down and upwind at 45° below the horizontal.

Assumption. The kite is in equilibrium; drag acts along the wind (horizontal) and lift normal to it (vertical); the coefficients are based on the plate area $A = 0.64\ \text{m}^2$, so the 12° angle fixes the flow pattern (the values of C_D, C_L) but does not enter the calculation.

1. Horizontal equilibrium: $F_D = T\cos 45^\circ = 24.525 \times 0.7071 = 17.34\ \text{N}$
2. Vertical equilibrium: $F_L = W + T\sin 45^\circ = 3.924 + 24.525 \times 0.7071 = 21.27\ \text{N}$
3. Dynamic force: $\dfrac{1}{2}\rho A U^2 = \dfrac{1}{2} \times 1.25 \times 0.64 \times 8.333^2 = 27.78\ \text{N}$
4. $C_D = \dfrac{F_D}{\frac{1}{2}\rho A U^2} = \dfrac{17.34}{27.78} = 0.624$
5. $C_L = \dfrac{F_L}{\frac{1}{2}\rho A U^2} = \dfrac{21.27}{27.78} = 0.766$

Answer: coefficient of drag $C_D = 0.624$; coefficient of lift $C_L = 0.766$.

QB III.19 AKTU 2022-23 | KCE303 | Q7(a) | 10 Marks Stokes' law gives the drag on a sphere as $F_D = 3\pi\mu DU$. State the range of Reynolds number for which it is valid, and derive the drag coefficient $C_D = 24/\text{Re}$.

SOLUTION

Stokes' law. For very slow (creeping) flow past a sphere, inertia forces are negligible compared with viscous forces. Stokes solved the flow exactly and found the total drag

$$F_D = 3\pi\mu DU$$

made up of skin-friction drag $2\pi\mu DU$ (two-thirds) and pressure drag $\pi\mu DU$ (one-third). There is no separation: the streamlines close smoothly behind the sphere.

Sketch: a sphere of diameter D in a uniform stream U; streamlines symmetric front and back, with no wake; drag F_D acting downstream.

Range of validity. Stokes' law holds for

$$\mathrm{Re} = \frac{\rho U D}{\mu} < 1 \qquad (\text{ strictly up to about } 0.2\text{; used in practice up to } 1\ )$$

Typical cases: settling of fine dust, silt, fog droplets, and the falling-sphere viscometer.

Derivation of C_D

Step 1. Definition of the drag coefficient.

$$F_D = C_D\, A\, \frac{\rho U^2}{2}$$

where A is the projected (frontal) area of the sphere.

Step 2. Projected area.

$$A = \frac{\pi D^2}{4}$$

Step 3. Make C_D the subject.

$$C_D = \frac{F_D}{\frac{\pi D^2}{4} \cdot \frac{\rho U^2}{2}} = \frac{8 F_D}{\pi \rho U^2 D^2}$$

Step 4. Substitute Stokes' drag $F_D = 3\pi\mu D U$.

$$C_D = \frac{8 \times 3\pi\mu D U}{\pi \rho U^2 D^2}$$

Step 5. Cancel π, one U and one D.

$$C_D = \frac{24\,\mu}{\rho U D}$$

Step 6. Recognise the Reynolds number $\mathrm{Re} = \rho U D/\mu$.

Result.

$$C_D = \frac{24}{\mathrm{Re}} \qquad (\ \mathrm{Re} < 1\ )$$

Significance. In Stokes flow the drag is directly proportional to velocity (not U^2), so C_D falls steeply as Re rises. Equating F_D plus buoyancy to the weight gives the terminal velocity $U = \dfrac{D^2\left(\rho_s - \rho\right) g}{18\mu}$, the basis of the falling-sphere viscometer. Above $\mathrm{Re} \approx 1$ inertia matters and the measured C_D exceeds $24/\mathrm{Re}$ (Oseen: $C_D = \dfrac{24}{\mathrm{Re}}\left(1 + \dfrac{3}{16}\mathrm{Re}\right)$).

UNIT IV. Impact of jets and hydraulic turbines

IV.1 Impact of jets

QB IV.1 **AKTU 2025-26 | BME302 | Q6(b) | 7 Marks** A jet of water 50 mm in diameter with a velocity of 40 m/s strikes a fixed, symmetrical curved plate at its centre. The jet is deflected through 120°. Find (i) the rate of flow and (ii) the force exerted by the jet on the plate in the direction of the jet.

SOLUTION

Given: $d = 50\ \text{mm} = 0.05\ \text{m}$, $V = 40\ \text{m/s}$, angle of deflection $= 120^\circ$, $\rho = 1000\ \text{kg/m}^3$.

To find: (i) discharge Q; (ii) force F_x in the direction of the jet.

Sketch: a horizontal jet strikes the centre of a fixed symmetrical curved plate; the jet splits into two halves that leave the two tips, each turned through 120°, i.e. each leaving at $\theta = 180^\circ - 120^\circ = 60^\circ$ to the x-axis (backwards).

For a jet striking a fixed symmetrical curved plate at its centre, momentum along x gives (Bansal):

$$F_x = \rho a V[V - (-V\cos\theta)] = \rho a V^2 (1 + \cos\theta)$$

1. Jet area: $a = \frac{\pi}{4}d^2 = \frac{\pi}{4}(0.05)^2 = 1.9635 \times 10^{-3}\ \text{m}^2$
2. Rate of flow: $Q = aV = 1.9635 \times 10^{-3} \times 40 = 0.07854\ \text{m}^3/\text{s}\ \ (= 78.54\ \text{L/s})$
3. Outlet angle: $\theta = 180^\circ - 120^\circ = 60^\circ$
4. Force: $F_x = \rho a V^2 (1 + \cos\theta) = 1000 \times 1.9635 \times 10^{-3} \times 40^2 \times (1 + \cos 60^\circ)$
   $= 1000 \times 1.9635 \times 10^{-3} \times 1600 \times 1.5 = 4712\ \text{N}$

Answer: $Q = 0.0785\ \text{m}^3/\text{s}$; $F_x = 4712\ \text{N} \approx 4.71\ \text{kN}$.

The force is 1.5 times that on a fixed flat plate ($\rho a V^2 = 3142\ \text{N}$), because the curved plate turns the water partly backwards.

QB IV.2 **AKTU 2021-22 | KME302 | Q6(b) | 10 Marks** A jet of water 60 mm in diameter with a velocity of 18 m/s strikes a smooth curved plate at its centre. The plate moves at 6 m/s in the direction of the jet, and the jet is deflected through 165°. Find (i) the thrust on the plate in the direction of the jet, (ii) the power of the jet and (iii) the efficiency of the jet. *(The angle is printed unclearly; 165° is taken.)*

SOLUTION

The deflection angle is unclear in the printed paper; 165° is used, as stated.

Given: $d = 60\ \text{mm} = 0.06\ \text{m}$, $V = 18\ \text{m/s}$, $u = 6\ \text{m/s}$, deflection $= 165^\circ$, plate smooth, single plate.

To find: (i) F_x; (ii) power (work done per second on the plate); (iii) efficiency.

Sketch: a jet strikes the centre of a symmetrical curved plate moving away at u; relative to the plate the water enters at $(V-u)$ and leaves both tips at $(V-u)$, at angle $\theta = 180^\circ - 165^\circ = 15^\circ$ to the backward x-direction.

Theory. For a single moving plate the mass striking per second is $\rho a\,(V-u)$ (the plate runs away from the jet). The relative velocity changes from $(V-u)$ forwards to $(V-u)\cos\theta$ backwards along x:

$$F_x = \rho a\,(V-u)\,[(V-u) + (V-u)\cos\theta] = \rho a\,(V-u)^2\,(1+\cos\theta)$$

1. $a = \dfrac{\pi}{4}(0.06)^2 = 2.8274\times10^{-3}\ \text{m}^2$; $\theta = 180^\circ - 165^\circ = 15^\circ$
2. Thrust:
   $F_x = \rho a\,(V-u)^2\,(1+\cos\theta) = 1000\times2.8274\times10^{-3}\times(18-6)^2\times(1+\cos15^\circ)$
   $F_x = 2.8274\times144\times1.9659 = 800.4\ \text{N}$
3. Power (work done per second on the plate):
   $P = F_x\,u = 800.4\times6 = 4803\ \text{W} = 4.80\ \text{kW}$
4. Kinetic energy of the jet per second:
   $\text{KE/s} = \dfrac{1}{2}(\rho a V)\,V^2 = \dfrac{1}{2}\times1000\times2.8274\times10^{-3}\times18^3 = 8245\ \text{W}$
5. Efficiency: $\eta = \dfrac{P}{\text{KE/s}} = \dfrac{4803}{8245} = 0.5825 = 58.25\%$

Answer: $F_x = 800.4\ \text{N}$; power $= 4.80\ \text{kW}$; $\eta = 58.3\%$.

A single plate cannot use the whole jet, which is why practical machines use a series of vanes on a wheel.

QB IV.3 AKTU 2016-17 | ME401 | Q2(b) | 10 Marks A jet of water 20 mm in diameter with a velocity of 20 m/s strikes the centre of a square plate hinged at its upper edge, and the plate swings through 15°. Find (i) the weight of the plate and (ii) the force needed at the lower edge to keep the plate vertical.

SOLUTION

Given: $d = 20\ \text{mm} = 0.02\ \text{m}$, $V = 20\ \text{m/s}$, angle of swing $\theta = 15^\circ$; plate square, uniform, hinged at its upper edge O; the horizontal jet strikes the centre of the plate when it hangs vertical.

To find: (i) weight W of the plate; (ii) force P at the lower edge to keep the plate vertical.

Sketch: plate OA of side L hangs from hinge O. The horizontal jet is at depth $x = L/2$ below O. Under the jet the plate swings to OA', at θ to the vertical; the jet now strikes it at point B, and the weight W acts at the plate centre G.

Derivation (plate in the swung position).

1. The jet is horizontal and the plate is at θ to the vertical, so the jet makes angle $(90^\circ-\theta)$ with the plate. The force normal to the plate is
   $$F_n = \rho a V^2 \sin(90^\circ-\theta) = \rho a V^2\cos\theta$$
2. The jet stays at depth x below O, so the distance of B from O along the plate is $\text{OB} = x/\cos\theta$.

3. Moment of F_n about O: $F_n \times \mathrm{OB} = \rho a V^2 \cos\theta \times \dfrac{x}{\cos\theta} = \rho a V^2 x$
4. The weight acts at G, with $\mathrm{OG} = L/2 = x$; its moment arm about O is $x \sin\theta$, so its moment is $Wx \sin\theta$.
5. Equilibrium: $Wx \sin\theta = \rho a V^2 x$, so

$$\sin\theta = \frac{\rho a V^2}{W}$$

Calculation.

1. $a = \dfrac{\pi}{4}(0.02)^2 = 3.1416 \times 10^{-4}\ \mathrm{m}^2$
2. $\rho a V^2 = 1000 \times 3.1416 \times 10^{-4} \times 20^2 = 125.66\ \mathrm{N}$
3. i. $W = \dfrac{\rho a V^2}{\sin\theta} = \dfrac{125.66}{\sin 15^\circ} = \dfrac{125.66}{0.2588} = 485.5\ \mathrm{N}$
4. ii. With the plate held vertical the jet is normal to it, so $F = \rho a V^2 = 125.66\ \mathrm{N}$ acting at the centre ($L/2$ from O). A horizontal force P at the lower edge (L from O) must balance its moment (the weight passes through O and gives no moment):

$$P \times L = F \times \frac{L}{2} \quad \Rightarrow \quad P = \frac{F}{2} = \frac{125.66}{2} = 62.83\ \mathrm{N}$$

Answer: $W = 485.5\ \mathrm{N}$; $P = 62.8\ \mathrm{N}$ at the lower edge, opposing the jet.

The size of the plate cancels out, so the result holds for any square plate struck at its centre.

QB IV.4 AKTU 2018-19 | RME601 | Q2(a) | 7 Marks A jet of water with a velocity of 11.5 m/s strikes a smooth vane moving at 4.5 m/s, and is deflected through 115°. Find (i) the angle of the jet at inlet for no shock, (ii) the absolute velocity of the jet at exit, in magnitude and direction, and (iii) the work done per second per unit weight of water striking per second. *(The deflection angle is printed unclearly; 115° is taken.)*

SOLUTION

The deflection angle is unclear in the printed paper; 115° is used, as stated. As in the standard Bansal problem, the vane is taken as **symmetrical**, so its inlet and outlet tip angles are equal: $\theta = \phi = \dfrac{180^\circ - 115^\circ}{2} = 32.5^\circ$.

Given: $V_1 = 11.5\ \mathrm{m/s}$, $u_1 = u_2 = u = 4.5\ \mathrm{m/s}$, $\theta = \phi = 32.5^\circ$, smooth vane ($V_{r2} = V_{r1}$).

To find: (i) jet angle α at inlet; (ii) V_2 and its direction; (iii) work done per unit weight.

Sketch: inlet triangle: V_1 at α to the direction of motion, u along the motion, V_{r1} closing the triangle along the vane tip at θ. Outlet triangle: V_{r2} backwards along the outlet tip at ϕ, u forwards, V_2 the resultant.

1. No shock at inlet: $\tan\theta = \dfrac{V_{f1}}{V_{w1} - u} = \dfrac{V_1 \sin\alpha}{V_1 \cos\alpha - u}$, i.e.

$$\frac{11.5 \sin\alpha}{11.5 \cos\alpha - 4.5} = \tan 32.5^\circ = 0.6371$$

2. Solving by trial: $\alpha = 20.36^\circ$ (check: $V_{w1} = 11.5 \cos 20.36^\circ = 10.78\ \text{m/s}$, $V_{f1} = 11.5 \sin 20.36^\circ = 4.002\ \text{m/s}$, $\dfrac{4.002}{10.78 - 4.5} = 0.6371$)
3. $V_{r1} = \dfrac{V_{f1}}{\sin\theta} = \dfrac{4.002}{\sin 32.5^\circ} = 7.448\ \text{m/s} = V_{r2}$
4. Outlet whirl: $V_{w2} = V_{r2}\cos\phi - u = 7.448 \cos 32.5^\circ - 4.5 = 6.281 - 4.5 = 1.781\ \text{m/s}$ (positive, so opposite to the vane motion)
5. Outlet flow: $V_{f2} = V_{r2}\sin\phi = 7.448 \sin 32.5^\circ = 4.002\ \text{m/s}$
6. ii. $V_2 = \sqrt{V_{w2}^2 + V_{f2}^2} = \sqrt{1.781^2 + 4.002^2} = 4.380\ \text{m/s}$; direction $\beta = \tan^{-1}\dfrac{V_{f2}}{V_{w2}} = \tan^{-1}\dfrac{4.002}{1.781} = 66.0^\circ$ with the direction opposite to the vane motion (i.e. 114.0° with the direction of motion)
7. iii. V_{w2} is opposite to u, so the $+$ sign applies: $\dfrac{\text{Work}}{\text{unit weight}} = \dfrac{(V_{w1} + V_{w2})\,u}{g} = \dfrac{(10.78 + 1.781) \times 4.5}{9.81} = 5.763\ \text{N·m/N}$

Answer: $\alpha = 20.4^\circ$; $V_2 = 4.38\ \text{m/s}$ at 66.0° to the backward direction of motion; work $= 5.76\ \text{N·m/N}$.

QB IV.5 AKTU 2018-19 | RME601 | Q3(a) | 7 Marks Explain the term "jet of water". Derive the expression for the efficiency of a series of moving curved vanes when a jet strikes them at one tip.

SOLUTION

Jet of water. A jet is a stream of water issuing from a nozzle into the atmosphere with high velocity. All its pressure energy has been converted into kinetic energy, so the pressure in the jet is atmospheric. When it strikes a plate or vane, its momentum changes and a force (dynamic thrust, impact of jet) acts on the plate. This is the working principle of impulse turbines such as the Pelton wheel.

Derivation: series of moving curved vanes, jet at one tip.

Sketch: an unsymmetrical curved vane moving at u along x. Inlet triangle: V_1 at α to u; V_{r1} along the inlet tip at θ. Outlet triangle: V_{r2} along the outlet tip at ϕ (backwards), u forwards, resultant V_2 at β, with whirl V_{w2} and flow V_{f2}.

Assumptions. Series of vanes on a wheel, so the whole jet $\rho a V_1$ is used; smooth vane ($V_{r2} = V_{r1}$); shock-free entry and exit; $u_1 = u_2 = u$.

1. **Inlet triangle.** $V_{w1} = V_1\cos\alpha$, $V_{f1} = V_1\sin\alpha$, and for no shock
$$\tan\theta = \frac{V_{f1}}{V_{w1} - u}, \qquad V_{r1} = \sqrt{(V_{w1} - u)^2 + V_{f1}^2}$$
2. **Outlet triangle.** $V_{r2} = V_{r1}$. The water leaves along the tip at ϕ, backwards:
$$V_{w2} = V_{r2}\cos\phi - u, \qquad V_{f2} = V_{r2}\sin\phi$$
3. **Mass per second.** For a series of vanes, $\dot{m} = \rho a V_1$.

4. **Force along the motion.** Initial velocity along x is V_{w1}; final is $-V_{w2}$ (when V_{w2} is opposite to u):

$$F_x = \rho a V_1 \left[V_{w1} - \left(-V_{w2} \right) \right] = \rho a V_1 \left(V_{w1} + V_{w2} \right)$$

   If the outlet whirl is in the direction of u, its sign is minus. So in general $F_x = \rho a V_1 \left(V_{w1} \pm V_{w2} \right)$.
5. **Work done per second.** The vanes move u metres each second:

$$\text{Work/s} = F_x u = \rho a V_1 \left(V_{w1} \pm V_{w2} \right) u$$

6. **Input.** Kinetic energy of the jet per second $= \frac{1}{2} \left(\rho a V_1 \right) V_1^2$.
7. **Efficiency.**

$$\eta = \frac{\rho a V_1 \left(V_{w1} \pm V_{w2} \right) u}{\frac{1}{2} \rho a V_1^3} = \frac{2 \left(V_{w1} \pm V_{w2} \right) u}{V_1^2}$$

Result.

$$\eta = \frac{2 \left(V_{w1} \pm V_{w2} \right) u}{V_1^2}$$

$+$ when V_{w2} is opposite to u, $-$ when it is along u, and $V_{w2} = 0$ when the water leaves at 90° to the motion. Work per unit weight is $\left(V_{w1} \pm V_{w2} \right) u/g$.

Significance. The efficiency is highest when the vane reverses the whirl strongly and the water leaves with little absolute velocity; this is the basis of turbine blade design.

QB IV.6 AKTU 2016-17 | ME401 | Q2(f) | 10 Marks A lawn sprinkler has two nozzles of 4 mm diameter each, at distances of 30 cm and 20 cm from the axis of rotation on opposite sides. The total flow through the sprinkler is 120 cm³/s, and both nozzles discharge downward (perpendicular to the arm). Find the angular speed at which the sprinkler rotates freely.

SOLUTION

Given: nozzle diameter $d = 4\ \text{mm} = 0.004\ \text{m}$; $r_A = 30\ \text{cm} = 0.30\ \text{m}$, $r_B = 20\ \text{cm} = 0.20\ \text{m}$ (opposite sides of the axis); total $Q = 120\ \text{cm}^3/\text{s} = 120 \times 10^{-6}\ \text{m}^3/\text{s}$.

To find: angular speed ω for free rotation (no friction, no load).

Sketch: a straight arm AOB pivoted at O, A at 0.30 m on one side and B at 0.20 m on the other. Both jets point downward, perpendicular to the arm, so they are tangential to their circles of rotation. Their reactions push A up and B up, so the two jets turn the arm in **opposite** senses. Jet A (longer arm) wins; the sprinkler turns in the sense driven by A.

Assumptions. Equal nozzles, so each carries $q = Q/2$; free rotation means net torque on the arm is zero (Bansal standard case).

1. Flow per nozzle: $q = \dfrac{120}{2} = 60\ \text{cm}^3/\text{s} = 60 \times 10^{-6}\ \text{m}^3/\text{s}$
2. Nozzle area: $a = \dfrac{\pi}{4} \left(0.004 \right)^2 = 1.2566 \times 10^{-5}\ \text{m}^2$
3. Jet velocity relative to the nozzle: $V = \dfrac{q}{a} = \dfrac{60 \times 10^{-6}}{1.2566 \times 10^{-5}} = 4.775\ \text{m/s}$

4. Absolute jet velocities. Nozzle A moves opposite to its jet at ωr_A; nozzle B moves in the direction of its jet at ωr_B:

$$V_A = V - \omega r_A, \qquad V_B = V + \omega r_B$$

5. Torque from the rate of change of angular momentum (jet A drives, jet B opposes):

$$T = \rho q\ (\ V - \omega r_A\ )\ r_A - \rho q\ (\ V + \omega r_B\ )\ r_B$$

6. Free rotation, $T = 0$:

$$(\ V - \omega r_A\ )\ r_A = (\ V + \omega r_B\ )\ r_B \quad \Rightarrow \quad \omega = \frac{V(\ r_A - r_B\ )}{r_A^2 + r_B^2}$$

7. $\omega = \dfrac{4.775 \times (\ 0.30 - 0.20\ )}{0.30^2 + 0.20^2} = \dfrac{0.4775}{0.13} = 3.673\ \text{rad/s}$
8. In rpm: $N = \dfrac{60\omega}{2\pi} = \dfrac{60 \times 3.673}{2\pi} = 35.07\ \text{rpm}$
9. Check: $(\ 4.775 - 0.30 \times 3.673\ ) \times 0.30 = 1.102$ and $(\ 4.775 + 0.20 \times 3.673\ ) \times 0.20 = 1.102$, so the torques balance.

Answer: $\omega = 3.67\ \text{rad/s}$ ($\approx 35\ \text{rpm}$), in the sense driven by the 30 cm nozzle.

IV.2 Classification and general ideas

QB IV.7 AKTU 2025-26 | BME302 | Q1(f) | 2 Marks AKTU 2022-23 | KME302 | Q1(g) | 2 Marks Distinguish between impulse and reaction turbines.

SOLUTION

Impulse turbine	Reaction turbine
Whole head converted to kinetic energy in a nozzle before the runner	Only part converted before the runner; the rest (pressure) in the runner
Runner at atmospheric pressure; casing has no hydraulic function	Runner full of water at varying pressure; casing must be closed
Partial admission (water strikes a few buckets)	Full admission all round the runner
No draft tube; high head, small discharge (Pelton)	Draft tube used; medium/low head, large discharge (Francis, Kaplan)

QB IV.8 AKTU 2018-19 | RME601 | Q1(d) | 2 Marks Explain why impulse turbines need a high head and low flow, while reaction turbines suit the opposite conditions.

SOLUTION

An impulse turbine converts the whole head into a jet, $V_1 = C_v\sqrt{2gH}$, and its buckets must run at $u \approx 0.46V_1$; a high head gives a fast jet and a useful speed. Its flow is limited by one or a few jets (partial admission), so it suits small Q. A reaction turbine is

fed all round its periphery (full admission), so it passes large Q, and its runner speed ratio $u_1/\sqrt{2gH}$ is higher, so it gives a good speed even at low H. In terms of specific speed $N_s = N\sqrt{P}/H^{5/4}$: high H, low Q gives low N_s (Pelton); low H, high Q gives high N_s (Francis, Kaplan).

QB IV.9 AKTU 2022-23 | KME302 | Q6(a) | 10 Marks Explain (i) the difference between a turbine and a pump, (ii) the governing of hydraulic turbines and (iii) the function of the draft tube.

SOLUTION

(i) Turbine and pump.

Turbine	Pump
Converts hydraulic energy of water into mechanical energy at the shaft	Converts mechanical energy of the shaft into hydraulic (pressure) energy of the liquid
Prime mover: drives a generator	Driven by a motor or engine
Flow from high head to low head; passages converge (accelerating flow)	Flow from low to high pressure; passages diverge (decelerating flow)
Work done by the fluid on the runner, $\left(V_{w1}u_1 \pm V_{w2}u_2\right)/g$	Work done by the impeller on the fluid, $\left(V_{w2}u_2 - V_{w1}u_1\right)/g$
Higher efficiency (accelerating flow, little separation)	Lower efficiency (adverse pressure gradient, separation)
Water enters at the outer diameter (Francis)	Liquid enters at the eye (inner diameter)

(ii) Governing of hydraulic turbines.

Governing means keeping the speed of the turbine constant when the load changes, because the generator must run at synchronous speed $N = 120f/p$ to keep the frequency constant. It is done automatically by changing the discharge with an **oil-pressure governor**.

Sketch: centrifugal governor (flyballs) driven from the turbine shaft; its sleeve moves a lever connected to a relay (control) valve; the valve admits oil from an oil sump and gear pump to one side of a servomotor (relay cylinder) piston; the piston rod moves the spear (Pelton) or the guide-vane regulating ring (Francis/Kaplan).

1. Load decreases, so the turbine speed increases.
2. The flyballs move outwards and lift the sleeve.
3. The lever opens the relay valve; pressurised oil enters one side of the servomotor.
4. The servomotor piston moves the spear forward (Pelton) or closes the guide vanes (Francis), reducing the discharge.
5. The speed falls back to normal and the sleeve and relay valve return to the neutral position.
6. When the load increases, every step is reversed and the discharge is increased.

For a Pelton wheel a **deflector** first turns the jet away on sudden load rejection, and the spear closes slowly to avoid water hammer in the penstock. In a Francis turbine a **relief valve** (pressure regulator) opens to bypass water; in a Kaplan turbine the runner

blades are also turned (double regulation).

(iii) Draft tube.

A draft tube is a gradually diverging pipe connecting the outlet of a reaction-turbine runner to the tailrace, its exit being below the tailrace level. Its functions are:

1. It allows the turbine to be placed above the tailrace (for access and maintenance) **without losing** the head H_s between runner and tailrace, because it creates a suction (pressure below atmospheric) at the runner exit.
2. It recovers most of the exit kinetic energy $V_2^2/(2g)$ as pressure energy, because the velocity falls from V_2 to V_3 in the diverging tube. This raises the net head on the runner and the output.

Applying Bernoulli between runner exit (2) and tube outlet (3):

$$\frac{p_2}{\rho g} = H_a - H_s - \left(\frac{V_2^2 - V_3^2}{2g} - h_f\right), \qquad \eta_d = \frac{\left(V_2^2 - V_3^2\right)/(2g) - h_f}{V_2^2/(2g)}$$

Types: conical (best, η_d up to about 90%), simple elbow, elbow with circular inlet and rectangular outlet, Moody spreading tube. The cone half-angle is kept below about 8° to avoid separation, and p_2 must stay above vapour pressure to avoid cavitation.

QB IV.10 AKTU 2018-19 | RME601 | Q1(a) | 2 Marks Distinguish between fluid mechanics and fluid machinery.

SOLUTION

Fluid mechanics is the science of fluids at rest and in motion: pressure, buoyancy, kinematics, Bernoulli and momentum equations, viscous flow, boundary layers. **Fluid machinery** is the application of these laws to machines that exchange energy between a fluid and a rotor: turbines (fluid to shaft) and pumps/compressors (shaft to fluid). Fluid mechanics gives the principles (e.g. the momentum equation), and fluid machinery uses them to design and test machines (e.g. Euler's equation, velocity triangles, efficiencies).

QB IV.11 AKTU 2018-19 | RME601 | Q1(b) | 2 Marks Give the reason why turbines are generally more efficient than pumps.

SOLUTION

In a turbine the flow through the passages is **accelerating** (converging passages, pressure falling along the flow, favourable pressure gradient), so the boundary layer stays attached and losses are small. In a pump the flow is **decelerating** (diverging passages, pressure rising along the flow, adverse pressure gradient), so the boundary layer separates easily and eddy losses are large. Pumps also have extra losses in the volute/diffuser, which converts velocity head to pressure. Hence turbine efficiencies of 90–95% versus 75–85% for pumps.

QB IV.12 AKTU 2018-19 | RME601 | Q1(c) | 2 Marks Define the whirl velocity in a turbine and state its significance.

SOLUTION

Whirl velocity V_w is the tangential component of the absolute velocity of water at the runner inlet or outlet, i.e. the component in the direction of blade motion: $V_{w1} = V_1 \cos\alpha$. **Significance:** only the whirl component has a moment about the shaft, so only it does work. By Euler's equation, work per unit weight $= \left(V_{w1}u_1 \pm V_{w2}u_2 \right)/g$. The flow component V_f only carries the discharge. For maximum work in Francis/Kaplan turbines the outlet whirl is made zero ($V_{w2} = 0$).

QB IV.13 AKTU 2016-17 | ME401 | Q1(g) | 2 Marks Explain the principle of a turbomachine.

SOLUTION

A turbomachine transfers energy between a continuously flowing fluid and a rotating element (runner/impeller) carrying blades. The blades change the **angular momentum** (whirl) of the fluid; by Newton's second law the rate of change of angular momentum equals the torque on the rotor:

$$T = \rho Q \left(V_{w1}r_1 \pm V_{w2}r_2 \right), \qquad \frac{\text{Work}}{\text{unit weight}} = \frac{V_{w1}u_1 \pm V_{w2}u_2}{g}$$

If the fluid gives energy to the rotor it is a turbine; if the rotor gives energy to the fluid it is a pump or compressor.

QB IV.14 AKTU 2021-22 | KME302 | Q1(h) | 2 Marks Explain the function of a penstock in a hydroelectric power plant.

SOLUTION

A **penstock** is the large-diameter pipe (steel or reinforced concrete) that carries water under pressure from the reservoir (or surge tank/forebay) to the turbine. Its function is to deliver the water with the least head loss, $h_f = fLV^2/\left(2gD\right)$, so that the net head $H = H_g - h_f$ is as large as possible. It is fitted with a gate/valve at entry, and a surge tank protects it from water hammer when the turbine is suddenly shut down.

IV.3 Pelton wheel

QB IV.15 AKTU 2023-24 | BME302 | Q2(d) | 7 Marks AKTU 2025-26 | BME302 | Q2(d) | 7 Marks AKTU 2022-23 | KME302 | Q2(d) | 10 Marks AKTU 2016-17 | ME401 | Q2(d) | 10 Marks Draw the velocity triangles for a Pelton wheel. Derive the expressions for the work done per second, the hydraulic efficiency and the maximum hydraulic efficiency.

SOLUTION

The Pelton wheel is a tangential-flow impulse turbine. A jet from the nozzle strikes the splitter of each double-cup bucket at the pitch circle and is turned back through about 160°–165°.

Velocity triangles.

Sketch (inlet): jet velocity V_1 and bucket velocity u lie on the same straight line (tangent to the pitch circle), so the "triangle" is a straight line: $V_{w1} = V_1$, $V_{r1} = V_1 - u$, $\alpha = 0$, $\theta = 0$, $V_{f1} = 0$.

Sketch (outlet): V_{r2} leaves along the bucket tip at angle ϕ to the backward direction of motion; u is drawn forwards from its tip; the closing side is V_2 at angle β; its components are V_{w2} (backwards) and V_{f2}.

Assumptions. Whole jet used by successive buckets; $u_1 = u_2 = u = \pi DN/60$; $V_{r2} = kV_{r1}$ ($k = 1$ for smooth buckets); V_{w2} opposite to u.

1. **Jet and bucket speeds.**
$$V_1 = C_v\sqrt{2gH}, \qquad u = \frac{\pi DN}{60}$$
2. **Inlet.** $V_{w1} = V_1$, $V_{r1} = V_1 - u$.
3. **Outlet.** $V_{r2} = kV_{r1} = k\,(\,V_1 - u\,)$ and
$$V_{w2} = V_{r2}\cos\phi - u$$
4. **Total change of whirl.**
$$V_{w1} + V_{w2} = V_1 + k\,(\,V_1 - u\,)\,\cos\phi - u = \,(\,V_1 - u\,)\,\,(\,1 + k\cos\phi\,)$$
5. **Force on the buckets.** Mass per second $\rho a V_1$; impulse-momentum along the motion:
$$F_x = \rho a V_1\,(\,V_{w1} + V_{w2}\,)$$
6. **Work done per second.**
$$\text{W.D./s} = \rho a V_1\,(\,V_{w1} + V_{w2}\,)\,u = \rho a V_1\,(\,V_1 - u\,)\,\,(\,1 + k\cos\phi\,)\,u$$
Per unit weight: $(\,V_{w1} + V_{w2}\,)\,u/g$.
7. **Hydraulic efficiency.** Input is the jet kinetic energy per second $\frac{1}{2}\rho a V_1 \cdot V_1^2$:
$$\eta_h = \frac{\rho a V_1\,(\,V_{w1} + V_{w2}\,)\,u}{\frac{1}{2}\rho a V_1^3} = \frac{2\,(\,V_{w1} + V_{w2}\,)\,u}{V_1^2}$$
$$\eta_h = \frac{2\,(\,V_1 - u\,)\,\,(\,1 + k\cos\phi\,)\,u}{V_1^2}$$

8. **Condition for maximum.** For a given wheel V_1, k and ϕ are constant. Differentiate with respect to u:

$$\frac{d\eta_h}{du} = \frac{2\,(\,1 + k\cos\phi\,)}{V_1^2}\,(\,V_1 - 2u\,) = 0 \quad \Rightarrow \quad u = \frac{V_1}{2}$$

$d^2\eta_h/du^2 = -4\,(\,1 + k\cos\phi\,)/V_1^2 < 0$, so this is a maximum.

9. **Maximum efficiency.** Put $u = V_1/2$:

$$\eta_{h,\,\max} = \frac{2\,(\,V_1/2\,)\,(\,1 + k\cos\phi\,)\,(\,V_1/2\,)}{V_1^2}$$

$$\eta_{h,\,\max} = \frac{1 + k\cos\phi}{2} \quad \text{at} \quad u = \frac{V_1}{2}$$

Significance. With $k = 1$ and $\phi = 15^\circ$, $\eta_{h,\,\max} = (\,1 + 0.9659\,)/2 = 98.3\%$. The bucket should run at half the jet speed; in practice $u \approx 0.46V_1$ because of friction and windage. The jet is not turned through a full 180°, so that the leaving water does not strike the next bucket.

QB IV.16 AKTU 2025-26 | BME302 | Q6(a) | 7 Marks AKTU 2021-22 | KME302 | Q2(d) | 10 Marks AKTU 2016-17 | ME401 | Q2(e) | 10 Marks Explain the governing of a Pelton turbine with a neat sketch.

SOLUTION

Governing is the automatic regulation of the discharge so that the speed of the turbine stays constant (the synchronous speed of the generator, $N = 120f/p$) whatever the load. In a Pelton wheel the discharge is changed by moving the **spear (needle)** in the nozzle. Large wheels use an **oil-pressure governor**.

Main parts.

1. Oil sump with a gear pump driven from the turbine shaft, supplying oil under pressure.
2. Servomotor (relay cylinder) with a double-acting piston; its rod is connected to the spear rod.
3. Relay (control) valve: a spool valve that sends oil to one side of the servomotor piston.
4. Centrifugal governor (flyballs) driven by the turbine shaft, with a sleeve.
5. Lever connecting the governor sleeve to the relay-valve spindle, pivoted at a fulcrum.
6. Deflector at the nozzle mouth, and pipes connecting all these parts.

Sketch: draw the nozzle with the spear and the deflector at the left; the spear rod runs back to the servomotor piston. Above the servomotor draw the relay valve with two pipes to the two sides of the servomotor piston and a return pipe to the oil sump. The gear pump feeds oil from the sump to the relay valve. On the right draw the flyball governor on a vertical spindle; its sleeve drives a lever whose other end moves the relay-valve spool.

Working on a load decrease.

1. Load on the generator falls, so the speed of the wheel rises.
2. The flyballs move outward and the sleeve rises.
3. The lever turns about its fulcrum and moves the relay-valve spool, opening the port that sends pressure oil to the left side of the servomotor piston; oil on the right side returns to the sump.
4. The piston moves right and pushes the spear forward, reducing the nozzle area and hence the discharge.
5. The power falls to match the load and the speed returns to normal; the sleeve and the spool return to the neutral position, closing both ports.

Working on a load increase. Every step is reversed: the speed falls, the flyballs move in, oil goes to the other side of the piston, the spear moves back and the discharge increases.

Sudden load rejection (deflector). If the spear closed the nozzle suddenly, the large mass of water in the long penstock would be stopped quickly and cause **water hammer** (a dangerous pressure rise). So the governor first swings the **deflector** into the jet, diverting it away from the buckets to the tailrace, and the spear then closes slowly. Using spear and deflector together is called **double regulation**.

Significance. Governing keeps the frequency of the electrical supply constant, and the deflector protects the penstock from water hammer.

QB IV.17 AKTU 2016-17 | ME401 | Q2(c) | 10 Marks Design a Pelton wheel for a head of 80 m and a speed of 300 rpm. The wheel develops 103 kW of shaft power. Take $C_v = 0.98$, speed ratio 0.45 and overall efficiency 0.80.

SOLUTION

Given: $H = 80\ \text{m}$, $N = 300\ \text{rpm}$, $P = 103\ \text{kW} = 103 \times 10^3\ \text{W}$, $C_v = 0.98$, $K_u = 0.45$, $\eta_o = 0.80$.

To find: jet velocity, bucket speed, pitch diameter D, jet diameter d, discharge Q, jet ratio, number of buckets, bucket size. (Single jet assumed.)

1. $\sqrt{2gH} = \sqrt{2 \times 9.81 \times 80} = 39.62\ \text{m/s}$
2. Jet velocity: $V_1 = C_v\sqrt{2gH} = 0.98 \times 39.62 = 38.83\ \text{m/s}$
3. Bucket speed: $u = K_u\sqrt{2gH} = 0.45 \times 39.62 = 17.83\ \text{m/s}$
4. Pitch diameter: $D = \dfrac{60u}{\pi N} = \dfrac{60 \times 17.83}{\pi \times 300} = 1.135\ \text{m}$
5. Discharge: $Q = \dfrac{P}{\rho g H \eta_o} = \dfrac{103 \times 10^3}{1000 \times 9.81 \times 80 \times 0.80} = 0.1641\ \text{m}^3/\text{s}$
6. Jet diameter: $\dfrac{\pi}{4}d^2 V_1 = Q \Rightarrow d = \sqrt{\dfrac{4Q}{\pi V_1}} = \sqrt{\dfrac{4 \times 0.1641}{\pi \times 38.83}} = 0.0733\ \text{m} = 73.3\ \text{mm}$
7. Jet ratio: $m = \dfrac{D}{d} = \dfrac{1.135}{0.0733} = 15.5$ (more than 10, so a single jet is acceptable)
8. Number of buckets (Tygun): $z = 15 + \dfrac{D}{2d} = 15 + \dfrac{15.47}{2} = 22.7$, say 23 buckets
9. Bucket size (usual proportions): width $\approx 5d = 5 \times 73.3 = 367\ \text{mm}$; depth $\approx 1.2d = 88\ \text{mm}$

10. Check: $N_s = \dfrac{N\sqrt{P}}{H^{5/4}} = \dfrac{300\sqrt{103}}{80^{1.25}} = 12.7$, within the single-jet Pelton range (8.5–30).

Answer: $D = 1.135\ \text{m}$, $d = 73.3\ \text{mm}$, $Q = 0.164\ \text{m}^3/\text{s}$, $m = 15.5$, $z = 23$, bucket $\approx 367\ \text{mm} \times 88\ \text{mm}$.

QB IV.18 AKTU 2018-19 | RME601 | Q3(b) | 7 Marks Explain why a notch is cut in the lip of a Pelton bucket. A Pelton wheel works under a head of 550 m and develops 10 000 kW at 450 rpm. Wheel efficiency 85%, $C_v = 0.98$, speed ratio 0.45. Find the discharge, the wheel diameter and the nozzle diameter.

SOLUTION

Why a notch. Each bucket has a notch cut in its outer lip, on the splitter line. As a bucket comes into the jet, its lip would otherwise cut across the jet and deflect part of the water before the bucket reaches the proper position, and the jet would strike the back of the lip. The notch lets the jet pass freely to the preceding bucket until the new bucket is in the correct position (nearly perpendicular to the jet), so the jet enters smoothly and the full jet acts on the bucket; this reduces shock, splashing and loss.

Numerical.

Given: $H = 550\ \text{m}$, $P = 10\,000\ \text{kW} = 10^7\ \text{W}$, $N = 450\ \text{rpm}$, $\eta_o = 0.85$ (the "wheel efficiency" is taken as the overall efficiency, shaft power / water power), $C_v = 0.98$, $K_u = 0.45$; single jet assumed.

To find: Q, D, d.

1. $\sqrt{2gH} = \sqrt{2 \times 9.81 \times 550} = 103.88\ \text{m/s}$
2. $V_1 = C_v\sqrt{2gH} = 0.98 \times 103.88 = 101.80\ \text{m/s}$
3. Discharge: $Q = \dfrac{P}{\rho g H \eta_o} = \dfrac{10^7}{1000 \times 9.81 \times 550 \times 0.85} = 2.180\ \text{m}^3/\text{s}$
4. $u = K_u\sqrt{2gH} = 0.45 \times 103.88 = 46.75\ \text{m/s}$
5. Wheel (pitch) diameter: $D = \dfrac{60u}{\pi N} = \dfrac{60 \times 46.75}{\pi \times 450} = 1.984\ \text{m}$
6. Nozzle (jet) diameter: $d = \sqrt{\dfrac{4Q}{\pi V_1}} = \sqrt{\dfrac{4 \times 2.180}{\pi \times 101.80}} = 0.1651\ \text{m} = 165.1\ \text{mm}$
7. Check: $D/d = 1.984/0.1651 = 12.0$ (normal), and $N_s = \dfrac{450\sqrt{10\,000}}{550^{1.25}} = 16.9$ (single-jet Pelton).

Answer: $Q = 2.18\ \text{m}^3/\text{s}$; $D = 1.98\ \text{m}$; $d = 165\ \text{mm}$.

IV.4 Reaction turbines (Francis)

QB IV.19 AKTU 2022-23 | KME302 | Q6(b) | 10 Marks A Francis turbine with an overall efficiency of 75% develops 148.25 kW under a head of 7.62 m. The peripheral velocity is $0.26\sqrt{2gH}$ and the radial velocity of flow at inlet is $0.96\sqrt{2gH}$. The wheel runs at 150 rpm and the hydraulic

losses are 22% of the available energy. Assuming radial discharge, find (i) the guide blade angle, (ii) the wheel vane angle at inlet, (iii) the diameter of the wheel at inlet and (iv) the width of the wheel at inlet.

SOLUTION

Given: $\eta_o = 0.75$, $P = 148.25\,\text{kW}$, $H = 7.62\,\text{m}$, $u_1 = 0.26\sqrt{2gH}$, $V_{f1} = 0.96\sqrt{2gH}$, $N = 150\,\text{rpm}$, hydraulic losses $= 22\%$, so $\eta_h = 1 - 0.22 = 0.78$; radial discharge, $V_{w2} = 0$.

To find: (i) α; (ii) θ; (iii) D_1; (iv) B_1.

Sketch: inlet triangle with u_1 along the tangent, V_1 at α, V_{r1} at θ; here $V_{w1} > u_1$, so $\theta < 90^\circ$.

1. $\sqrt{2gH} = \sqrt{2 \times 9.81 \times 7.62} = 12.227\,\text{m/s}$
2. $u_1 = 0.26 \times 12.227 = 3.179\,\text{m/s}$; $V_{f1} = 0.96 \times 12.227 = 11.738\,\text{m/s}$
3. Radial discharge: $\eta_h = \dfrac{V_{w1}u_1}{gH}$, so $V_{w1} = \dfrac{\eta_h gH}{u_1} = \dfrac{0.78 \times 9.81 \times 7.62}{3.179} = 18.34\,\text{m/s}$
4. i. $\tan\alpha = \dfrac{V_{f1}}{V_{w1}} = \dfrac{11.738}{18.34} = 0.640$, so $\alpha = 32.62^\circ$
5. ii. $\tan\theta = \dfrac{V_{f1}}{V_{w1} - u_1} = \dfrac{11.738}{18.34 - 3.179} = 0.7742$, so $\theta = 37.75^\circ$
6. iii. $D_1 = \dfrac{60u_1}{\pi N} = \dfrac{60 \times 3.179}{\pi \times 150} = 0.4048\,\text{m}$
7. Discharge: $Q = \dfrac{P}{\rho g H \eta_o} = \dfrac{148.25 \times 10^3}{1000 \times 9.81 \times 7.62 \times 0.75} = 2.644\,\text{m}^3/\text{s}$
8. iv. $B_1 = \dfrac{Q}{\pi D_1 V_{f1}} = \dfrac{2.644}{\pi \times 0.4048 \times 11.738} = 0.1772\,\text{m}$

Answer: $\alpha = 32.6^\circ$, $\theta = 37.8^\circ$, $D_1 = 0.405\,\text{m}$, $B_1 = 0.177\,\text{m}$.

Note. These data (a real AKTU question, also in Bansal) are physically unrealistic. The inlet absolute velocity is $V_1 = \sqrt{18.34^2 + 11.738^2} = 21.78\,\text{m/s}$, larger than $\sqrt{2gH} = 12.23\,\text{m/s}$; its kinetic head $V_1^2/(2g) = 24.2\,\text{m}$ exceeds the whole available head of $7.62\,\text{m}$, which is impossible. The ratios are also far from normal Francis values ($K_u \approx 0.6$–0.9, $K_f \approx 0.15$–0.30); the two coefficients appear to have been interchanged. In the exam, solve as printed, as above.

QB IV.20 AKTU 2023-24 | BME302 | Q6(a) | 7 Marks A reaction turbine runs at 450 rpm under a head of 120 m. Its inlet diameter is 1.2 m and the flow area is 0.4 m². The absolute and relative velocities at inlet make 20° and 60° with the tangential velocity. Find (i) the volume flow rate and (ii) the power developed.

SOLUTION

Given: $N = 450\,\text{rpm}$, $H = 120\,\text{m}$, $D_1 = 1.2\,\text{m}$, flow area $A = 0.4\,\text{m}^2$, $\alpha = 20^\circ$, $\theta = 60^\circ$. Assume radial discharge ($V_{w2} = 0$), as usual for a reaction turbine.

To find: (i) Q; (ii) power developed by the runner.

Sketch: inlet triangle: u_1 along the tangent; V_1 at 20° and V_{r1} at 60° to it, meeting at a common height V_{f1}; $V_{w1} > u_1$ since $\theta < 90^\circ$.

1. $u_1 = \dfrac{\pi D_1 N}{60} = \dfrac{\pi \times 1.2 \times 450}{60} = 28.27\ \text{m/s}$
2. From the triangle: $V_{f1} = V_{w1}\tan\alpha$ and $V_{f1} = (V_{w1} - u_1)\tan\theta$. Equate: $V_{w1}\tan 20^\circ = (V_{w1} - 28.27)\tan 60^\circ$, so $V_{w1} = \dfrac{u_1 \tan 60^\circ}{\tan 60^\circ - \tan 20^\circ} = \dfrac{28.27 \times 1.7321}{1.7321 - 0.3640} = 35.80\ \text{m/s}$
3. $V_{f1} = V_{w1}\tan 20^\circ = 35.80 \times 0.3640 = 13.03\ \text{m/s}$
4. i. $Q = A\,V_{f1} = 0.4 \times 13.03 = 5.212\ \text{m}^3/\text{s}$
5. ii. $P = \rho Q V_{w1} u_1 = 1000 \times 5.212 \times 35.80 \times 28.27 = 5.275 \times 10^6\ \text{W} = 5.27\ \text{MW}$
6. Check: $\eta_h = \dfrac{V_{w1}u_1}{gH} = \dfrac{35.80 \times 28.27}{9.81 \times 120} = 0.860$ (86%, reasonable).

Answer: $Q = 5.21\ \text{m}^3/\text{s}$; $P = 5.27\ \text{MW}$ (runner power, radial discharge).

QB IV.21 AKTU 2016-17 | ME401 | Q3 | 15 Marks An inward-flow reaction turbine has an external diameter of 1 m and an inlet breadth of 200 mm. The velocity of flow at inlet is 1.5 m/s and the blade thickness takes up 15% of the flow area. The runner turns at 200 rpm and the guide blades make 15° with the wheel tangent. Find the mass of water flowing per second, draw the inlet velocity triangle, and find (i) the runner vane angle at inlet, (ii) the wheel velocity at inlet, (iii) the absolute velocity of water leaving the guide vanes and (iv) the relative velocity of water entering the runner.

SOLUTION

Given: $D_1 = 1\ \text{m}$, $B_1 = 200\ \text{mm} = 0.2\ \text{m}$, $V_{f1} = 1.5\ \text{m/s}$, blade thickness blocks 15% of the area (effective area $= 0.85\pi D_1 B_1$), $N = 200\ \text{rpm}$, $\alpha = 15^\circ$.

To find: $\dot{m}$; inlet triangle; (i) θ; (ii) u_1; (iii) V_1; (iv) V_{r1}.

Mass flow rate.

1. Gross area: $\pi D_1 B_1 = \pi \times 1 \times 0.2 = 0.6283\ \text{m}^2$
2. Effective area: $A = 0.85 \times 0.6283 = 0.5341\ \text{m}^2$
3. $Q = AV_{f1} = 0.5341 \times 1.5 = 0.8011\ \text{m}^3/\text{s}$
4. $\dot{m} = \rho Q = 1000 \times 0.8011 = 801.1\ \text{kg/s}$

Inlet velocity triangle.

5. ii. $u_1 = \dfrac{\pi D_1 N}{60} = \dfrac{\pi \times 1 \times 200}{60} = 10.47\ \text{m/s}$
6. $V_{w1} = \dfrac{V_{f1}}{\tan\alpha} = \dfrac{1.5}{\tan 15^\circ} = \dfrac{1.5}{0.2679} = 5.598\ \text{m/s}$
7. iii. $V_1 = \dfrac{V_{f1}}{\sin\alpha} = \dfrac{1.5}{\sin 15^\circ} = \dfrac{1.5}{0.2588} = 5.796\ \text{m/s}$

Sketch: draw $u_1 = 10.47\ \text{m/s}$ horizontally (along the wheel tangent). From its starting point draw $V_1 = 5.80\ \text{m/s}$ at 15° above it; its horizontal projection $V_{w1} = 5.60\ \text{m/s}$ is **shorter** than u_1, and its vertical height is $V_{f1} = 1.5\ \text{m/s}$. Join the tip of u_1 to the tip of V_1: this is V_{r1}, which leans backwards, so the vane angle θ (measured from the direction of u_1) is obtuse.

8. Since $V_{w1} < u_1$: $\tan(180^\circ - \theta) = \dfrac{V_{f1}}{u_1 - V_{w1}} = \dfrac{1.5}{10.47 - 5.598} = \dfrac{1.5}{4.874} = 0.3078$

9. $180^\circ - \theta = 17.11^\circ$, so (i) $\theta = 162.9^\circ$ with the direction of motion (the vane is inclined 17.1° backwards)
10. iv. $V_{r1} = \sqrt{V_{f1}^2 + (u_1 - V_{w1})^2} = \sqrt{1.5^2 + 4.874^2} = 5.099\ \text{m/s}$
11. For information, with radial discharge the work per unit weight is $\dfrac{V_{w1}u_1}{g} = \dfrac{5.598 \times 10.47}{9.81} = 5.98\ \text{N·m/N}$.

Answer: $\dot{m} = 801\ \text{kg/s}$; $\theta = 162.9^\circ$ (i.e. 17.1° backward); $u_1 = 10.47\ \text{m/s}$; $V_1 = 5.80\ \text{m/s}$; $V_{r1} = 5.10\ \text{m/s}$.

This is a fast runner ($V_{w1} < u_1$), with backward-curved vanes at inlet.

QB IV.22 AKTU 2018-19 | RME601 | Q2(b) | 7 Marks For a Francis turbine with constant flow velocity, no blade friction and radial discharge, show that the hydraulic efficiency is $\eta_h = \dfrac{2}{2 + \dfrac{\tan^2\alpha}{1 - \tan\alpha\cot\theta}}$, where α is the guide blade angle and θ the runner vane angle at inlet. Show also that for radial vanes at inlet $\eta_h = \dfrac{2}{2 + \tan^2\alpha}$. *(The expressions are unclear in the printed paper; the standard results are used.)*

SOLUTION

The printed expressions are unclear; the standard results stated in the question are proved.

Assumptions. Constant velocity of flow $V_{f1} = V_{f2} = V_f$; no friction losses in the runner, so the only loss is the kinetic energy carried away at exit; radial discharge, $V_{w2} = 0$, so $V_2 = V_{f2} = V_f$.

Sketch: inlet triangle: u_1 along the tangent, V_1 at α, V_{r1} at θ; V_{w1} is the horizontal projection of V_1, and $V_{w1} - u_1$ that of V_{r1}. Outlet triangle: right-angled, $V_2 = V_f$ radial.

1. **Energy balance.** Head supplied = work done per unit weight + exit loss:
$$H = \frac{V_{w1}u_1}{g} + \frac{V_2^2}{2g} = \frac{V_{w1}u_1}{g} + \frac{V_f^2}{2g}$$
2. **Hydraulic efficiency.**
$$\eta_h = \frac{V_{w1}u_1/g}{H} = \frac{V_{w1}u_1}{V_{w1}u_1 + V_f^2/2}$$
3. **Inlet triangle, guide angle.** $\tan\alpha = \dfrac{V_f}{V_{w1}}$, so
$$V_f = V_{w1}\tan\alpha$$
4. **Inlet triangle, vane angle.** $\tan\theta = \dfrac{V_f}{V_{w1} - u_1}$, so $V_{w1} - u_1 = V_f\cot\theta$ and
$$u_1 = V_{w1} - V_f\cot\theta = V_{w1} - V_{w1}\tan\alpha\cot\theta = V_{w1}(1 - \tan\alpha\cot\theta)$$

5. **Substitute.** $V_{w1}u_1 = V_{w1}^2\,(\,1 - \tan\alpha\cot\theta\,)$ and $V_f^2/2 = V_{w1}^2\tan^2\alpha/2$:

$$\eta_h = \frac{V_{w1}^2\,(\,1 - \tan\alpha\cot\theta\,)}{V_{w1}^2\,(\,1 - \tan\alpha\cot\theta\,) + \tfrac{1}{2}V_{w1}^2\tan^2\alpha}$$

6. **Simplify.** Cancel V_{w1}^2, divide numerator and denominator by $(\,1 - \tan\alpha\cot\theta\,)$, then multiply both by 2:

$$\eta_h = \frac{1}{1 + \dfrac{\tan^2\alpha}{2\,(\,1 - \tan\alpha\cot\theta\,)}}$$

$$\eta_h = \frac{2}{2 + \dfrac{\tan^2\alpha}{1 - \tan\alpha\cot\theta}}$$

7. **Radial vanes at inlet.** $\theta = 90^\circ$, so $\cot\theta = 0$ (and $V_{w1} = u_1$):

$$\eta_h = \frac{2}{2 + \tan^2\alpha}$$

Significance. A small guide-blade angle α gives a small exit velocity and a high hydraulic efficiency; e.g. $\alpha = 15^\circ$ with radial vanes gives $\eta_h = 2/\,(\,2 + 0.0718\,) = 96.5\%$.

QB IV.23 AKTU 2018-19 | RME601 | Q4(b) | 7 Marks An inward-flow reaction turbine works under an average head of 160 m with a discharge of 80 m³/s. The inlet and outlet diameters are 4 m and 2 m. The runner blade angle at inlet is 120°, the velocity of flow at outlet (radial discharge) is 15 m/s, the breadth is constant and the hydraulic efficiency is 90%. Find the power produced (MW) and the speed of the runner (rpm).

SOLUTION

Given: $H = 160\ \text{m}$, $Q = 80\ \text{m}^3/\text{s}$, $D_1 = 4\ \text{m}$, $D_2 = 2\ \text{m}$, $\theta = 120^\circ$, $V_{f2} = 15\ \text{m/s}$, $V_{w2} = 0$, $B_1 = B_2$, $\eta_h = 0.90$.

To find: power P (MW) and speed N (rpm).

Sketch: $\theta = 120^\circ > 90^\circ$, so $V_{w1} < u_1$ and V_{r1} leans backwards, making 60° with the reverse of u_1.

1. Constant breadth, continuity:
   $\pi D_1 B V_{f1} = \pi D_2 B V_{f2} \Rightarrow V_{f1} = V_{f2}\dfrac{D_2}{D_1} = 15 \times \dfrac{2}{4} = 7.5\ \text{m/s}$
2. Power produced (runner power):
   $P = \eta_h\,\rho g Q H = 0.90 \times 1000 \times 9.81 \times 80 \times 160 = 113.0 \times 10^6\ \text{W} = 113.0\ \text{MW}$
3. Work relation: $\eta_h = \dfrac{V_{w1}u_1}{gH} \Rightarrow V_{w1}u_1 = 0.90 \times 9.81 \times 160 = 1412.6\ \text{m}^2/\text{s}^2$
4. Inlet triangle: $\tan\,(\,180^\circ - \theta\,) = \dfrac{V_{f1}}{u_1 - V_{w1}} \Rightarrow u_1 - V_{w1} = \dfrac{7.5}{\tan 60^\circ} = 4.330\ \text{m/s}$
5. Put $V_{w1} = u_1 - 4.330$: $u_1\,(\,u_1 - 4.330\,) = 1412.6$, i.e. $u_1^2 - 4.330u_1 - 1412.6 = 0$
   $u_1 = \dfrac{4.330 + \sqrt{4.330^2 + 4 \times 1412.6}}{2} = \dfrac{4.330 + 75.30}{2} = 39.81\ \text{m/s}$; $V_{w1} = 35.48\ \text{m/s}$

6. Speed: $N = \dfrac{60 u_1}{\pi D_1} = \dfrac{60 \times 39.81}{\pi \times 4} = 190.1 \text{ rpm}$
7. (Breadth, for information: $B = \dfrac{Q}{\pi D_1 V_{f1}} = \dfrac{80}{\pi \times 4 \times 7.5} = 0.849 \text{ m}$.)

Answer: $P = 113 \text{ MW}$; $N \approx 190 \text{ rpm}$.

IV.5 Specific speed, unit quantities and performance

QB IV.24 AKTU 2022-23 | KME302 | Q1(h) | 2 Marks Define the specific speed of a turbine.

SOLUTION

Specific speed N_s of a turbine is the speed of a geometrically similar turbine that would develop **1 kW** of power under a head of **1 m**:

$$N_s = \frac{N\sqrt{P}}{H^{5/4}} \qquad (N \text{ in rpm},\ P \text{ in kW},\ H \text{ in m})$$

It depends only on the shape of the runner and is used to select the type of turbine: Pelton 8.5–51, Francis 51–255, Kaplan 255–860.

QB IV.25 AKTU 2021-22 | KME302 | Q1(g) | 2 Marks Define the unit power of a turbine.

SOLUTION

Unit power P_u is the power developed by a turbine working under a head of **1 m**. Since $P \propto H^{3/2}$ for a given turbine,

$$P_u = \frac{P}{H^{3/2}}$$

It is used to predict the power of the same turbine at another head: $P_2 = P_1 \left(H_2/H_1 \right)^{3/2}$.

QB IV.26 AKTU 2021-22 | KME302 | Q6(a) | 10 Marks Derive the expressions for (i) unit discharge and (ii) unit speed of a turbine.

SOLUTION

Unit quantities give the performance of a turbine under a head of 1 m. They allow the performance of the **same** turbine (same D) at a different head to be predicted, since the head at a site varies. Assumptions: the turbine runs with geometrically similar velocity triangles (same speed ratio and flow ratio) and the same efficiency at both heads.

(i) Unit speed N_u. It is the speed of the turbine under a head of 1 m.

1. Jet or flow velocity is proportional to $\sqrt{2gH}$, and with the same speed ratio K_u the blade speed is $u = K_u\sqrt{2gH}$, so $u \propto \sqrt{H}$.

2. Blade speed from rpm: $u = \dfrac{\pi D N}{60}$. For a given turbine D is constant, so $u \propto N$.
3. Hence $N \propto \sqrt{H}$, i.e. $N = K_1\sqrt{H}$, where K_1 is a constant.
4. Definition: when $H = 1\,\text{m}$, $N = N_u$. So $N_u = K_1 \times 1 = K_1$.
5. Therefore

$$N = N_u\sqrt{H} \quad \Rightarrow \quad N_u = \frac{N}{\sqrt{H}}$$

(ii) Unit discharge Q_u. It is the discharge through the turbine under a head of 1 m.

1. $Q = A \times V_f$, where A is the flow area (fixed for a given turbine) and V_f is the velocity of flow.
2. With the same flow ratio, $V_f = K_f\sqrt{2gH}$, so $V_f \propto \sqrt{H}$.
3. Hence $Q \propto \sqrt{H}$, i.e. $Q = K_2\sqrt{H}$.
4. Definition: when $H = 1\,\text{m}$, $Q = Q_u$, so $K_2 = Q_u$.
5. Therefore

$$Q = Q_u\sqrt{H} \quad \Rightarrow \quad Q_u = \frac{Q}{\sqrt{H}}$$

Unit power (for completeness). $P = \eta\rho g Q H \propto \sqrt{H} \times H = H^{3/2}$, so $P_u = P/H^{3/2}$.

Use. If a turbine works under H_1 and H_2, its unit quantities are the same:

$$\frac{N_1}{\sqrt{H_1}} = \frac{N_2}{\sqrt{H_2}}, \qquad \frac{Q_1}{\sqrt{H_1}} = \frac{Q_2}{\sqrt{H_2}}, \qquad \frac{P_1}{H_1^{3/2}} = \frac{P_2}{H_2^{3/2}}$$

For example, a turbine running at 450 rpm under 100 m will run at $450\sqrt{81/100} = 405\,\text{rpm}$ under 81 m. Unit quantities are also used to plot the main characteristic curves of turbines.

QB IV.27 AKTU 2016-17 | ME401 | Q5 | 15 Marks A turbine works under a head of 30 m at 300 rpm with a discharge of 10 m³/s and an efficiency of 90%. Find (i) the specific speed, (ii) the power generated and (iii) the suitable type of turbine.

SOLUTION

Given: $H = 30\,\text{m}$, $N = 300\,\text{rpm}$, $Q = 10\,\text{m}^3/\text{s}$, $\eta_o = 0.90$.

To find: (i) N_s; (ii) P; (iii) type of turbine.

Theory. Water power $= \rho g Q H/1000$ kW; shaft power $P = \eta_o \times$ water power. Specific speed $N_s = N\sqrt{P}/H^{5/4}$ (SI: rpm, kW, m) is the speed of a geometrically similar turbine developing 1 kW under 1 m head; its value fixes the runner shape.

1. Water power: $\text{W.P.} = \dfrac{\rho g Q H}{1000} = \dfrac{1000 \times 9.81 \times 10 \times 30}{1000} = 2943\,\text{kW}$
2. ii. Power generated: $P = \eta_o \times \text{W.P.} = 0.90 \times 2943 = 2649\,\text{kW} \approx 2.65\,\text{MW}$
3. $\sqrt{P} = \sqrt{2648.7} = 51.47$; $H^{5/4} = 30^{1.25} = 70.21$
4. i. $N_s = \dfrac{N\sqrt{P}}{H^{5/4}} = \dfrac{300 \times 51.47}{70.21} = 219.9$ (SI units)

5. iii. Selection table: Pelton 8.5–51, Francis 51–255, Kaplan 255–860. $N_s = 220$ lies in 51–255, so a **Francis turbine** (fast-runner Francis, near the upper end of its range) is suitable.
6. Check by head: $30\,\mathrm{m}$ is a low-to-medium head with a fairly large discharge; at the upper end of the Francis range a fast Francis is chosen, and a Kaplan would also be considered if the head varied widely.

Answer: $N_s \approx 220$; $P \approx 2649\,\mathrm{kW}$; a (fast) Francis turbine.

QB IV.28 AKTU 2016-17 | ME401 | Q2(g) | 10 Marks Draw the characteristic curves of reaction turbines.

SOLUTION

Characteristic curves show how a turbine performs when one parameter is changed while others are kept fixed. They are plotted from tests, using unit quantities so that the results do not depend on the test head. The important parameters are speed N, head H, discharge Q, power P, overall efficiency η_o and gate opening. Three types are drawn.

1. Main (constant head) characteristic curves. Head is kept constant; for each gate opening the speed is varied by changing the load, and Q_u, P_u, η_o are plotted against $N_u = N/\sqrt{H}$.

- *Sketch (a) Q_u vs N_u:* one curve for each gate opening (e.g. 40%, 60%, 80%, 100%), higher curves for larger openings. For a Francis turbine Q_u **falls slightly** as N_u rises (the rotating runner chokes the flow); for a Kaplan turbine it **rises slightly**; for a Pelton wheel the lines are horizontal.
- *Sketch (b) P_u vs N_u:* a family of inverted parabolas, one per gate opening, each starting at zero at $N_u = 0$, rising to a peak and falling to zero at the runaway speed.
- *Sketch (c) η_o vs N_u:* similar inverted parabolas; the peak of the full-opening curve is the design point.

2. Operating (constant speed) characteristic curves. The speed is kept constant (synchronous speed of the generator), and η_o and Q are plotted against power P (or % full load).

- *Sketch:* η_o rises from zero (at no-load the turbine needs some discharge just to overcome losses, so the curve starts at a small P) and flattens near full load. The Kaplan curve is **flat** (high efficiency from about 30% load, because the blades are adjustable); the Francis curve is **more peaked** and falls quickly at part load; the propeller turbine curve is the most peaked.
- Q vs P is almost a straight line starting from a finite no-load discharge.

3. Constant efficiency (Muschel or iso-efficiency) curves. Derived from the main characteristics: points of equal efficiency on the Q_u–N_u and η_o–N_u curves are transferred to one Q_u vs N_u (or P_u vs N_u) graph and joined.

- *Sketch:* a set of closed, nested oval curves like a contour map (a "shell"), the innermost oval being the zone of maximum efficiency; lines of constant gate opening cross them. Away from the centre the curves open out.

Significance. The main curves give the best speed and runaway speed; the operating curves show part-load behaviour and help choose between Francis and Kaplan; the Muschel curves show the region of best efficiency for planning operation of the plant.

IV.6 Cavitation in turbines

QB IV.29 **AKTU 2023-24 | BME302 | Q6(b) | 7 Marks** **AKTU 2018-19 | RME601 | Q4(a) | 7 Marks** Explain cavitation in turbines, its effects and the methods of avoiding it. Define Thoma's cavitation factor.

SOLUTION

Cavitation is the formation, growth and sudden collapse of vapour bubbles in a flowing liquid at places where the local pressure falls to the **vapour pressure** of the liquid at that temperature.

Where and how in turbines. In reaction turbines the pressure is lowest at the **runner blade outlet** and at the **inlet of the draft tube**, because the draft tube creates suction and the velocity there is high. If $p \le p_v$, the water boils locally and vapour bubbles (with released dissolved air) form. The bubbles are carried to regions of higher pressure, where they collapse violently. The surrounding water rushes into the cavities, producing very high local pressures (up to several thousand atmospheres) and high-frequency impacts on the metal surface.

Effects.

1. Pitting and erosion of runner blades and draft-tube surfaces (the metal is hammered and becomes rough and spongy).
2. Noise and vibration of the machine.
3. Fall in discharge, power output and efficiency, because the vapour pockets disturb the flow.
4. Higher maintenance cost and shorter life of the runner.

Methods of avoiding cavitation.

1. Keep the pressure everywhere above the vapour pressure: set the runner at a suitable (small) height H_s above the tailrace, or below it for high-N_s turbines.
2. Design the runner and draft tube so that $\sigma > \sigma_c$ (see below), avoiding sharp curves and sudden changes of velocity.
3. Use cavitation-resistant materials (stainless steel, aluminium bronze) or coatings on blades; polish the surfaces.

4. Avoid running the turbine far from its design point (very low or very high loads), and admit air into the draft tube if needed.

Thoma's cavitation factor. Applying Bernoulli between the runner exit and the tailrace gives the margin of pressure above vapour pressure. Thoma defined

$$\sigma = \frac{H_a - H_v - H_s}{H} = \frac{\dfrac{p_a}{\rho g} - \dfrac{p_v}{\rho g} - H_s}{H}$$

where H_a = atmospheric pressure head (m of water), H_v = vapour pressure head, H_s = suction head (height of runner exit above tailrace) and H = net head. For cavitation-free operation, $\sigma \geq \sigma_c$, the critical value found by test, which increases with specific speed (Bansal: Francis $\sigma_c = 0.625\ (\ N_s/380.78\ )^{\ 2}$, Kaplan $\sigma_c = 0.3 + 0.0024\ (\ N_s/380.78\ )^{\ 2.73}$).

Significance. For a given σ_c, the maximum permissible suction head is $H_s = H_a - H_v - \sigma_c H$; high-head and high-$N_s$ turbines must therefore be set low, often below the tailrace.

UNIT V. Centrifugal and reciprocating pumps

V.1 Centrifugal pump: basics and efficiencies

QB V.1 AKTU 2021-22 | KME302 | Q1(i) | 2 Marks Define a pump.

SOLUTION

A **pump** is a hydraulic machine that converts mechanical energy (from a motor or engine) into hydraulic energy of a liquid, mainly pressure energy, so that the liquid can be lifted to a higher level or moved through pipes against resistance. The energy given per unit weight is the head H_m, and the power given to the liquid is $\rho g Q H_m$. Pumps are classed as **rotodynamic** (centrifugal, mixed and axial flow) and **positive displacement** (reciprocating, gear, screw).

QB V.2 AKTU 2023-24 | BME302 | Q1(f) | 2 Marks Define the mechanical efficiency of a pump.

SOLUTION

The **mechanical efficiency** η_m is the ratio of the power delivered by the impeller to the liquid to the power supplied at the pump shaft:

$$\eta_m = \frac{\text{Power at the impeller}}{\text{Shaft power}} = \frac{\rho Q\, V_{w2} u_2}{P}$$

It accounts for losses in bearings, glands and disc friction. Typical values are 0.95–0.98, and $\eta_o = \eta_{\text{mano}} \times \eta_m$.

QB V.3 AKTU 2025-26 | BME302 | Q1(g) | 2 Marks AKTU 2016-17 | ME401 | Q1(f) | 2 Marks Explain the priming of a centrifugal pump.

SOLUTION

Priming is filling the suction pipe, the casing and the delivery pipe up to the delivery valve with the liquid to be pumped, so that no air remains. It is needed because the head developed by an impeller is in metres of the fluid inside it. In air, the pressure developed is only about $1/800$ of that in water, which is far too small to lift water up the suction pipe, so an unprimed pump runs but delivers nothing. Priming is done by pouring water through a funnel, by a vacuum pump, or by using a foot valve; flooded-suction and self-priming pumps need no separate priming.

QB V.4 AKTU 2018-19 | RME601 | Q5(a) | 7 Marks Explain why priming is needed before a centrifugal pump can deliver. Explain the significance of similarity parameters for centrifugal pumps.

SOLUTION

(i) Why priming is needed

1. **Meaning.** Priming is filling the suction pipe, casing and delivery pipe (up to the delivery valve) completely with the liquid, so that all air is driven out.
2. **Pressure rise depends on density.** When liquid turns with the impeller, the pressure rise between eye and rim is (forced vortex)

$$p_2 - p_1 = \rho \frac{u_2^2 - u_1^2}{2}$$

   so for a given speed it is proportional to the density ρ of the fluid in the impeller.
3. **Effect of air.** The density of air is about $1/800$ of that of water. An impeller running in air produces a pressure rise about 800 times smaller, equal to only a few centimetres of water. This cannot lift water up the suction pipe to the eye.
4. **Result.** An unprimed pump runs, heats up and may damage the glands and wearing rings, but it delivers no water. Even a small air pocket in the eye can break the suction.
5. **How it is done.** Water is poured through a funnel with the air cock open, or a vacuum (ejector) pump draws the air out; a foot valve at the bottom of the suction pipe keeps the pipe full between runs.

Sketch: a pump above a sump, suction pipe with foot valve and strainer, funnel and air cock on top of the casing, delivery valve closed.

(ii) Significance of similarity parameters

For geometrically similar pumps working at similar velocity triangles, the following dimensionless groups (from dimensional analysis) are the same for model and prototype:

Parameter	Expression	What it fixes
Flow (capacity) coefficient	$\dfrac{Q}{ND^3}$	Discharge
Head coefficient	$\dfrac{gH_m}{N^2D^2}$	Head
Power coefficient	$\dfrac{P}{\rho N^3 D^5}$	Power
Specific speed	$N_s = \dfrac{N\sqrt{Q}}{H_m^{3/4}}$	Type of impeller

Significance:

1. **Model testing.** A small model is tested in the laboratory and the results are scaled up to the full-size pump with these groups, saving cost and time.
2. **Change of speed (affinity laws).** For the same pump (D fixed): $Q \propto N$, $H_m \propto N^2$, $P \propto N^3$, so performance at a new speed can be predicted.
3. **Change of size.** For the same speed, $Q \propto D^3$, $H_m \propto D^2$, $P \propto D^5$.
4. **Selection.** The specific speed, which follows from these groups, tells whether a radial, mixed or axial impeller suits the required Q and H_m.
5. **Multistage design.** The number of stages and the impeller diameter for a new duty are found by keeping N_s and $gH_m/\left(N^2D^2\right)$ equal to those of a tested pump.

Significance in one line: priming makes the pump able to develop its head at all, and the similarity parameters let one tested pump predict the behaviour of a whole family of similar pumps.

QB V.5 AKTU 2022-23 | KME302 | Q2(e) | 10 Marks (i) Derive the expression for the specific speed of a centrifugal pump. (ii) Explain the construction and working of a centrifugal pump.

SOLUTION

(i) Specific speed of a centrifugal pump

Definition. The specific speed N_s of a pump is the speed of a geometrically similar pump that delivers $1\ \mathrm{m^3/s}$ against a head of $1\ \mathrm{m}$.

Assumptions. The pumps are geometrically similar, so their velocity triangles are similar and every velocity is proportional to the blade velocity u; efficiencies are equal.

1. **Discharge.** $Q = \pi DBV_f$. For similar pumps $B \propto D$, so the flow area $\propto D^2$. Also $V_f \propto u$ and $u = \pi DN/60 \propto DN$:

$$Q \propto D^2 \cdot DN = D^3 N \qquad (1)$$

2. **Head.** The head is proportional to u^2/g, so

$$u \propto \sqrt{H_m} \quad \Rightarrow \quad DN \propto \sqrt{H_m} \quad \Rightarrow \quad D \propto \frac{\sqrt{H_m}}{N} \qquad (2)$$

3. **Eliminate D.** Put $D^3 \propto H_m^{3/2}/N^3$ from (2) into (1):

$$Q \propto \frac{H_m^{3/2}}{N^3} N = \frac{H_m^{3/2}}{N^2} \quad \Rightarrow \quad \frac{N^2 Q}{H_m^{3/2}} = \text{constant}$$

4. **Fix the constant.** When $Q = 1\ \text{m}^3/\text{s}$ and $H_m = 1\ \text{m}$, $N = N_s$, so the constant is N_s^2:

$$N_s^2 = \frac{N^2 Q}{H_m^{3/2}}$$

5. **Take the square root.**

$$N_s = \frac{N\sqrt{Q}}{H_m^{3/4}} \qquad \left(N \text{ in rpm},\ Q \text{ in } \text{m}^3/\text{s},\ H_m \text{ in m} \right)$$

Significance. Low N_s (10–80) means a radial impeller for high head and small discharge; N_s 80–160 is mixed flow; N_s 160–500 is axial flow. For a multistage pump the head of one stage is used.

(ii) Construction and working of a centrifugal pump

Sketch: a sump with a suction pipe rising to the eye of the impeller; foot valve and strainer at the bottom of the suction pipe; a backward-curved impeller on a shaft inside a spiral (volute) casing; the delivery pipe leaving the casing tangentially with a delivery valve; arrows showing axial entry at the eye and radial outflow at the rim.

Construction (main parts):

1. **Impeller.** A wheel with a set of backward-curved vanes, keyed to a shaft driven by an electric motor. It is the only part that gives energy to the liquid.
2. **Casing.** An air-tight spiral passage around the impeller whose area increases towards the outlet. It collects the liquid and converts kinetic energy into pressure energy. Types: volute, vortex and casing with guide blades (diffuser).
3. **Suction pipe with foot valve and strainer.** Connects the sump to the eye. The foot valve is a non-return valve that keeps the pipe full; the strainer keeps out solids.
4. **Delivery pipe with delivery valve.** Carries the liquid to the required height; the valve regulates the flow.

Working:

1. The pump is **primed**, the delivery valve is closed and the motor is started.
2. The rotating impeller sets up a **forced vortex**; the pressure head rises from eye to rim by $\left(u_2^2 - u_1^2 \right) / \left(2g \right)$.
3. This lowers the pressure at the eye, and atmospheric pressure on the sump pushes water up the suction pipe into the eye.
4. When the impeller reaches its rated speed, the delivery valve is opened gradually. Water enters the eye axially, is flung outward along the vanes, and leaves the rim with high velocity and pressure; the work done on it is $V_{w2} u_2 / g$ per newton.
5. In the casing the velocity falls and the pressure rises further, and the water flows up the delivery pipe.

6. To stop, the delivery valve is closed first and then the motor is switched off, so that water does not run back through the pump.

Significance. Simple construction, high speed (direct motor drive) and smooth, large discharge make the centrifugal pump the most widely used pump for water supply, irrigation and industry.

QB V.6 AKTU 2025-26 | BME302 | Q2(e) | 7 Marks AKTU 2022-23 | KME302 | Q7(a)(i) | 10 Marks (part) AKTU 2021-22 | KME302 | Q7(a) | 10 Marks Derive the expression for the minimum starting speed of a centrifugal pump (and, for S4, the specific speed of a centrifugal pump).

SOLUTION

(a) Minimum starting speed

Idea. When a pump is started with the delivery valve closed, there is no flow; the liquid in the impeller only rotates and the impeller produces only the centrifugal (forced-vortex) head. The pump starts delivering only when this head equals the manometric head H_m. The speed at which this happens is the **minimum starting speed**.

Assumptions. No flow at starting, so the liquid turns as a solid body at angular velocity ω; D_1, D_2 are the inner and outer impeller diameters; N is in rpm.

1. **Forced-vortex pressure rise.** For an element of thickness dr and area dA at radius r, the net inward pressure force supplies the centripetal force:
$$dp\,dA = \rho\,dA\,dr\,\omega^2 r \quad \Rightarrow \quad dp = \rho\,\omega^2 r\,dr$$
2. **Integrate from r_1 to r_2.**
$$p_2 - p_1 = \frac{\rho\,\omega^2\,\left(r_2^2 - r_1^2\right)}{2} \quad \Rightarrow \quad \frac{p_2 - p_1}{\rho g} = \frac{u_2^2 - u_1^2}{2g}$$
3. **Starting condition.** Flow begins when this rise equals H_m:
$$\frac{u_2^2 - u_1^2}{2g} = H_m$$
4. **Put in blade velocities** $u_1 = \pi D_1 N/60$ and $u_2 = \pi D_2 N/60$:
$$\left(\frac{\pi N}{60}\right)^2 \frac{D_2^2 - D_1^2}{2g} = H_m$$
5. **Solve for N.**
$$N_{\min} = \frac{60}{\pi}\sqrt{\frac{2gH_m}{D_2^2 - D_1^2}}$$
6. **Second form.** With $H_m = \eta_{\text{mano}} V_{w2} u_2/g$, Step 3 gives $u_2^2 - u_1^2 = 2\eta_{\text{mano}} V_{w2} u_2$, and cancelling one factor $\pi N/60$:
$$N_{\min} = \frac{120\,\eta_{\text{mano}}\,V_{w2}\,D_2}{\pi\,\left(D_2^2 - D_1^2\right)}$$

Sketch: impeller with eye radius r_1 and rim radius r_2; parabolic pressure-head rise from eye to rim of height $\left(u_2^2 - u_1^2\right) / (2g)$, compared with the level H_m.

Significance. $N_{\min} \propto \sqrt{H_m}$; a larger rim diameter or smaller eye lowers it. The pump must always run above $N_{\min}$, otherwise it churns without delivering.

(b) Specific speed (S4)

N_s is the speed of a geometrically similar pump delivering $1\ \mathrm{m^3/s}$ against $1\ \mathrm{m}$.

1. $Q = \pi D B V_f$ with $B \propto D$ and $V_f \propto u \propto DN$, so $Q \propto D^3 N$.
2. $u \propto \sqrt{H_m}$ and $u \propto DN$, so $D \propto \sqrt{H_m}/N$.
3. Substitute: $Q \propto \dfrac{H_m^{3/2}}{N^3}\, N = \dfrac{H_m^{3/2}}{N^2}$, so $\dfrac{N^2 Q}{H_m^{3/2}} = \text{constant}$.
4. For $Q = 1$, $H_m = 1$, $N = N_s$: the constant is N_s^2, so

$$N_s = \frac{N\sqrt{Q}}{H_m^{3/4}}$$

N_s fixes the impeller type: radial (10–80), mixed (80–160), axial (160–500), with N in rpm, Q in $\mathrm{m^3/s}$, H_m in m.

V.2 Centrifugal pump numericals

QB V.7 **AKTU 2016-17 | ME401 | Q2(h) | 10 Marks** The outer diameter of a centrifugal pump impeller is 600 mm, twice the inner diameter. It runs at 1200 rpm against a total head of 75 m. The velocity of flow is constant at 3 m/s, the vanes are set back at 30° at outlet and the outlet width is 50 mm. Find (a) the vane angle at inlet, (b) the work done per second by the impeller and (c) the manometric efficiency.

SOLUTION

Given: $D_2 = 600\ \mathrm{mm} = 0.6\ \mathrm{m}$, $D_1 = 0.3\ \mathrm{m}$, $N = 1200\ \mathrm{rpm}$, $H_m = 75\ \mathrm{m}$, $V_{f1} = V_{f2} = 3\ \mathrm{m/s}$, $\phi = 30^\circ$, $B_2 = 50\ \mathrm{mm} = 0.05\ \mathrm{m}$. Radial entry ($V_{w1} = 0$).

To find: (a) θ, (b) work done per second, (c) η_{mano}.

1. Blade velocities: $u_1 = \dfrac{\pi D_1 N}{60} = \dfrac{\pi \times 0.3 \times 1200}{60} = 18.85\ \mathrm{m/s}$;
   $u_2 = \dfrac{\pi \times 0.6 \times 1200}{60} = 37.70\ \mathrm{m/s}$
2. a. Inlet vane angle: $\tan\theta = \dfrac{V_{f1}}{u_1} = \dfrac{3}{18.85} = 0.1592$, so $\theta = 9.04^\circ$
3. Outlet whirl: $V_{w2} = u_2 - \dfrac{V_{f2}}{\tan\phi} = 37.70 - \dfrac{3}{\tan 30^\circ} = 37.70 - 5.196 = 32.50\ \mathrm{m/s}$
4. Discharge: $Q = \pi D_2 B_2 V_{f2} = \pi \times 0.6 \times 0.05 \times 3 = 0.2827\ \mathrm{m^3/s}$
5. b. Work done per second:
   $\rho Q V_{w2} u_2 = 1000 \times 0.2827 \times 32.50 \times 37.70 = 346\,500\ \mathrm{W} = 346.5\ \mathrm{kW}$
6. Work per unit weight: $\dfrac{V_{w2} u_2}{g} = \dfrac{32.50 \times 37.70}{9.81} = 124.9\ \mathrm{m}$

7. c. $\eta_{\text{mano}} = \dfrac{gH_m}{V_{w2}u_2} = \dfrac{9.81 \times 75}{32.50 \times 37.70} = \dfrac{735.8}{1225.3} = 60.0\%$

Answer: $\theta = 9.04^\circ$; work done $= 346.5\ \text{kW}$; $\eta_{\text{mano}} = 60.0\%$.

QB V.8 AKTU 2018-19 | RME601 | Q5(b) | 7 Marks A centrifugal pump impeller of 0.5 m outer and 0.25 m inner diameter runs at 600 rpm and delivers 8000 litres per minute against a head of 8.5 m. The vanes are set back at 45° at outlet, there is no whirl and no shock at inlet, and the flow area is constant at 0.06 m². Find (i) the manometric efficiency, (ii) the vane angle at inlet and (iii) the least speed at which the pump will start delivering. *(The angle is printed unclearly; 45° is taken.)*

SOLUTION

The outlet vane angle is printed unclearly; as stated in the question, $\phi = 45^\circ$ is used.

Given: $D_2 = 0.5\ \text{m}$, $D_1 = 0.25\ \text{m}$, $N = 600\ \text{rpm}$, $Q = 8000\ \text{L/min} = \dfrac{8}{60} = 0.1333\ \text{m}^3/\text{s}$, $H_m = 8.5\ \text{m}$, $\phi = 45^\circ$, flow area $= 0.06\ \text{m}^2$ (same at inlet and outlet), $V_{w1} = 0$.

To find: (i) η_{mano}, (ii) θ, (iii) $N_{\min}$.

1. Flow velocity: $V_{f1} = V_{f2} = \dfrac{Q}{\text{area}} = \dfrac{0.1333}{0.06} = 2.222\ \text{m/s}$
2. Blade velocities: $u_1 = \dfrac{\pi \times 0.25 \times 600}{60} = 7.854\ \text{m/s}$; $u_2 = \dfrac{\pi \times 0.5 \times 600}{60} = 15.71\ \text{m/s}$
3. Outlet whirl: $V_{w2} = u_2 - \dfrac{V_{f2}}{\tan\phi} = 15.71 - \dfrac{2.222}{\tan 45^\circ} = 13.49\ \text{m/s}$
4. i. $\eta_{\text{mano}} = \dfrac{gH_m}{V_{w2}u_2} = \dfrac{9.81 \times 8.5}{13.49 \times 15.71} = \dfrac{83.39}{211.8} = 39.4\%$
5. ii. $\tan\theta = \dfrac{V_{f1}}{u_1} = \dfrac{2.222}{7.854} = 0.2829$, so $\theta = 15.8^\circ$
6. iii. Starting condition $\dfrac{u_2^2 - u_1^2}{2g} = H_m$:

$$\left(\frac{\pi N}{60}\right)^2 = \frac{2gH_m}{D_2^2 - D_1^2} = \frac{2 \times 9.81 \times 8.5}{0.5^2 - 0.25^2} = \frac{166.77}{0.1875} = 889.4$$

7. $\dfrac{\pi N}{60} = 29.82\ \text{rad/s}$, so $N_{\min} = \dfrac{60 \times 29.82}{\pi} = 569.6\ \text{rpm}$

Answer: $\eta_{\text{mano}} = 39.4\%$; $\theta = 15.8^\circ$; least starting speed $\approx 570\ \text{rpm}$ (below the running speed of 600 rpm, so the pump delivers).

Note. A manometric efficiency of about 39% is much lower than in a real pump (0.75–0.85), because the printed head of 8.5 m is small compared with the Euler head $V_{w2}u_2/g = 21.6\ \text{m}$. Solve as printed in the exam.

QB V.9 AKTU 2025-26 | BME302 | Q7(b) | 7 Marks A pump with a 0.30 m impeller delivers 3 m³/s against 30 m at 2000 rpm with an efficiency of 75%. A similar multistage pump is to deliver 5 m³/s against 200 m at 1500 rpm. Find the number of stages and the impeller diameter of each stage.

SOLUTION

Given: Pump 1: $D_1 = 0.30\,\text{m}$, $Q_1 = 3\,\text{m}^3/\text{s}$, $H_1 = 30\,\text{m}$, $N_1 = 2000\,\text{rpm}$. Multistage pump 2: $Q_2 = 5\,\text{m}^3/\text{s}$, total head $200\,\text{m}$, $N_2 = 1500\,\text{rpm}$. The pumps are similar, so their specific speeds are equal; the efficiency (75%) is the same for both and is not needed.

To find: number of stages n, impeller diameter D_2.

1. Specific speed of pump 1: $N_s = \dfrac{N_1\sqrt{Q_1}}{H_1^{3/4}} = \dfrac{2000\times\sqrt{3}}{30^{0.75}} = \dfrac{3464.1}{12.82} = 270.2$
2. Head per stage of pump 2 (same N_s, head of one stage):
   $H_2^{3/4} = \dfrac{N_2\sqrt{Q_2}}{N_s} = \dfrac{1500\times\sqrt{5}}{270.2} = 12.41$, so $H_2 = 12.41^{4/3} = 28.74\,\text{m}$
3. Number of stages: $n = \dfrac{200}{28.74} = 6.96$, round up: $n = 7$
4. Actual head per stage: $H_2 = \dfrac{200}{7} = 28.57\,\text{m}$
5. Diameter from the head coefficient $\dfrac{gH}{N^2D^2} = \text{const}$:
   $D_2 = D_1\dfrac{N_1}{N_2}\sqrt{\dfrac{H_2}{H_1}} = 0.30\times\dfrac{2000}{1500}\times\sqrt{\dfrac{28.57}{30}} = 0.30\times1.333\times0.9759 = 0.390\,\text{m}$
6. Check with the flow coefficient $\dfrac{Q}{ND^3} = \text{const}$:
   $D_2 = 0.30\times\left(\dfrac{5\times2000}{3\times1500}\right)^{1/3} = 0.30\times1.305 = 0.391\,\text{m}$, which agrees.

Answer: 7 stages, each impeller about $0.39\,\text{m}$ (390 mm) in diameter.

Note. A discharge of $3\,\text{m}^3/\text{s}$ through a 0.30 m impeller ($N_s \approx 270$, an axial-flow value) is not realistic; the original data were probably in smaller units (for example 0.03 and $0.05\,\text{m}^3/\text{s}$). The answers above depend only on the ratio Q_2/Q_1, so they are the same either way.

QB V.10 AKTU 2018-19 | RME601 | Q2(c) | 7 Marks Explain why centrifugal pumps are operated in series and in parallel. Describe common operating difficulties and their remedies.

SOLUTION

(i) Pumps in series and in parallel

One impeller can give only a limited head (about 40–100 m) and a limited discharge at a given speed. Two or more pumps (or impellers) are combined to meet a larger duty.

1. **Series (multistage) operation.** Impellers are mounted on one shaft, or pumps are connected so that the delivery of one is the suction of the next. The same discharge Q passes through each, and each adds its head:

$$\text{Total head} = nH_m, \qquad \text{discharge} = Q$$

   Used when a **high head** is needed with a moderate discharge: boiler feed, deep-well and high-rise building supply.
2. **Parallel operation.** Pumps draw from a common sump and deliver into a common main. Each works at the same head, and the discharges add:

$$\text{Total discharge} = nQ, \qquad \text{head} = H_m$$

   Used when a **large discharge** is needed at moderate head, or when demand varies (city water works, irrigation): pumps can be switched in or out, and one can be repaired while the others run.

Sketch: (a) two pumps in a line, pipe from pump 1 delivery to pump 2 suction; (b) two pumps side by side from one sump into one delivery main; with H–Q curves showing the head doubled (series) and the discharge doubled (parallel) at the same head.

(ii) Common operating difficulties and remedies

Difficulty	Likely causes	Remedies
Pump runs but delivers no water	Not primed; air leak in suction pipe or gland; foot valve leaking; speed too low ($N < N_{\min}$); suction lift too high	Prime fully; tighten joints and gland; repair foot valve; raise speed; lower the pump
Discharge or head less than rated	Impeller or strainer partly choked; worn wearing rings; wrong direction of rotation; air in water	Clean impeller and strainer; replace rings; correct motor connections; submerge suction pipe deeper
Cavitation (noise, vibration, pitting)	Low NPSH: high suction lift, long or narrow suction pipe, hot water	Reduce suction lift and suction losses; use larger suction pipe; flooded suction
Motor overloaded	Running at much larger discharge than design; speed too high; liquid heavier or more viscous	Throttle the delivery valve; reduce speed; choose correct motor
Excessive vibration and noise	Misalignment; bent shaft; unbalanced or damaged impeller; worn bearings	Realign coupling; balance or replace impeller; replace bearings
Pump loses prime after starting	Air leak on suction side; gland packing loose; suction pipe not submerged	Seal leaks; repack gland; extend suction pipe below water level

Significance. Correct combination of pumps meets the required head and discharge economically, and routine checks on priming, sealing and NPSH avoid most failures in service.

V.3 Cavitation in pumps

QB V.11 AKTU 2022-23 | KME302 | Q1(j) | 2 Marks Define cavitation.

SOLUTION

Cavitation is the formation of vapour bubbles in a flowing liquid where the local absolute pressure falls to the vapour pressure p_v, followed by the sudden collapse of these bubbles when they are carried into a region of higher pressure. The collapse produces very high local pressures that cause pitting of surfaces, noise, vibration and a fall in efficiency. In a centrifugal pump it occurs at the eye of the impeller; in a turbine, at the runner outlet and draft tube inlet.

QB V.12 AKTU 2018-19 | RME601 | Q1(e) | 2 Marks Define NPSH and explain why it must be calculated.

SOLUTION

Net positive suction head (NPSH) is the total head at the pump inlet above the vapour pressure head of the liquid:

$$\text{NPSH} = \frac{p_1}{\rho g} + \frac{V_1^2}{2g} - \frac{p_v}{\rho g} = H_{\text{atm}} - H_v - h_s - h_{fs}$$

It must be calculated to check that the pump will not cavitate: the NPSH available from the installation must exceed the NPSH required by the pump (given by the maker), $\text{NPSHA} > \text{NPSHR}$. It fixes the maximum suction lift at which the pump can be installed.

QB V.13 AKTU 2025-26 | BME302 | Q7(a) | 7 Marks Explain cavitation, its effects in centrifugal pumps and the precautions against it.

SOLUTION

Cavitation is the formation of vapour cavities in a liquid wherever the local absolute pressure falls to the vapour pressure, and their violent collapse when they reach a region of higher pressure.

Mechanism in a centrifugal pump

1. The pressure is lowest at the **eye of the impeller**, because the liquid has been lifted through the suction height h_s, has lost head h_{fs} in the suction pipe, and is accelerated into the eye. From the energy equation between sump and inlet:

$$\frac{p_1}{\rho g} = H_{\text{atm}} - h_s - h_{fs} - \frac{V_1^2}{2g}$$

2. If p_1 falls to the vapour pressure p_v, the liquid boils locally and vapour bubbles form; dissolved air is also released.

3. The bubbles are carried into the vane passages, where the pressure is higher, and collapse suddenly. The surrounding liquid rushes in and strikes the vane surface with local pressures reported to reach several thousand atmospheres, thousands of times per second.

Sketch: impeller with the eye marked as the lowest-pressure zone; bubbles forming at the eye and collapsing on the vane surfaces near the inlet; pitted region shaded.

Effects

1. **Pitting and erosion** of the vanes near the inlet, which can destroy the impeller.
2. **Noise and vibration**, like gravel passing through the pump.
3. **Fall in head, discharge and efficiency**, because vapour pockets block the flow passages; the H–Q curve drops sharply.
4. Fatigue and damage to bearings and shaft due to vibration.

Criterion. Cavitation is avoided when $\mathrm{NPSHA} > \mathrm{NPSHR}$, or when Thoma's factor $\sigma = \mathrm{NPSH}/H_m$ exceeds the critical value σ_c. The maximum suction lift is

$$h_{s,\,\max} = H_{\mathrm{atm}} - H_v - h_{fs} - \sigma_c H_m$$

Precautions

1. Keep the suction lift small, or place the pump below the sump level (flooded suction).
2. Use a short, large-diameter suction pipe with few bends and valves, to keep h_{fs} low.
3. Avoid running far above the design speed or discharge.
4. Use a double-suction impeller or an inducer to lower the eye velocity.
5. Make the impeller of cavitation-resistant material (stainless steel, bronze) with smooth surfaces.
6. Avoid pumping hot liquid at a high suction lift, since its vapour pressure is higher.

Significance. Cavitation fixes how high a pump may be set above the sump; checking NPSH at the design stage prevents costly damage.

V.4 Reciprocating pump

QB V.14 AKTU 2023-24 | BME302 | Q1(g) | 2 Marks AKTU 2022-23 | KME302 | Q1(i) | 2 Marks AKTU 2021-22 | KME302 | Q1(j) | 2 Marks Explain the slip of a reciprocating pump.

SOLUTION

Slip is the difference between the theoretical discharge Q_{th} (swept volume per second) and the actual discharge Q_{act}. It is caused by leakage past the piston and valves and by delayed closing of the valves.

$$\%\ \mathrm{slip} = \frac{Q_{\mathrm{th}} - Q_{\mathrm{act}}}{Q_{\mathrm{th}}} \times 100 = \left(1 - C_d\right) \times 100, \qquad C_d = \frac{Q_{\mathrm{act}}}{Q_{\mathrm{th}}}$$

with $Q_{\text{th}} = ALN/60$ (single-acting) or $2ALN/60$ (double-acting). Slip is usually 2–10%.

QB V.15 AKTU 2016-17 | ME401 | Q1(j) | 2 Marks Explain slip and negative slip.

SOLUTION

Slip is $Q_{\text{th}} - Q_{\text{act}}$, the shortfall of actual discharge below the swept volume, due to leakage past the piston and valves; $\%\ \text{slip} = \left(Q_{\text{th}} - Q_{\text{act}}\right)/Q_{\text{th}} \times 100$.

Negative slip occurs when $Q_{\text{act}} > Q_{\text{th}}$ ($C_d > 1$). It happens with a long suction pipe, a short delivery pipe and high speed: the momentum of the liquid column in the suction pipe keeps flowing after the suction stroke ends, opens the delivery valve before the delivery stroke begins, and pushes extra liquid out.

QB V.16 AKTU 2018-19 | RME601 | Q1(f) | 2 Marks Explain why a reciprocating pump is called a positive displacement pump.

SOLUTION

In a reciprocating pump the liquid is trapped in the cylinder and **actually pushed out (displaced)** by the piston. Each delivery stroke delivers a definite volume AL, whatever the delivery head, so $Q_{\text{th}} = ALN/60$ does not depend on H. Because the liquid is moved by positive (direct) displacement of a closed volume, and not by giving it velocity as in a centrifugal pump, it is called a positive displacement pump.

QB V.17 AKTU 2022-23 | KME302 | Q7(a)(ii) | 10 Marks (part) Derive the expression for the work done by a single-acting reciprocating pump.

SOLUTION

Assumptions. Friction in the pipes and acceleration of the liquid are neglected; no leakage; piston area $A = \pi D^2/4$, stroke L, speed N rpm; h_s = height of the cylinder centre above the sump (suction head), h_d = height of the delivery outlet above the cylinder centre (delivery head).

Sketch: sump, vertical suction pipe of height h_s to the cylinder, suction and delivery valves, delivery pipe rising h_d above the cylinder; piston driven by crank and connecting rod.

1. **Discharge.** One delivery stroke per revolution pushes out a volume AL; in one minute N strokes, so
$$Q = \frac{ALN}{60}$$
2. **Weight of liquid per second.**
$$W = \rho g Q = \frac{\rho g A L N}{60}$$

3. **Work on the suction stroke.** The pressure in the cylinder is h_s below atmospheric while the other face of the piston is at atmospheric pressure, so the piston must be moved against a net force $\rho g h_s A$ through the distance L:

$$\text{Work in suction stroke} = \rho g h_s A L$$

4. **Work on the delivery stroke.** The pressure in the cylinder is h_d above atmospheric, so the force is $\rho g h_d A$ through distance L:

$$\text{Work in delivery stroke} = \rho g h_d A L$$

5. **Work per revolution.** Add the two strokes:

$$\text{Work per revolution} = \rho g A L \left(h_s + h_d \right)$$

6. **Work per second.** Multiply by $N/60$ revolutions per second:

$$\text{Work done per second} = \frac{\rho g A L N \left(h_s + h_d \right)}{60} \text{ W} = W \left(h_s + h_d \right)$$

7. **Power.** $P = \dfrac{\rho g A L N \left(h_s + h_d \right)}{60 \times 1000}$ kW. For a double-acting pump the discharge, and so the power, is doubled: $\dfrac{2 \rho g A L N \left(h_s + h_d \right)}{60}$.

Significance. The work equals the weight of liquid lifted per second times the total static lift; it is proportional to the area of the ideal indicator diagram, a rectangle of height $h_s + h_d$ and width L. If pipe friction is included, $\frac{2}{3} h_{fs}$ and $\frac{2}{3} h_{fd}$ are added to the bracket.

QB V.18 AKTU 2023-24 | BME302 | Q7(b) | 7 Marks AKTU 2021-22 | KME302 | Q7(b) | 10 Marks Classify reciprocating pumps. Draw the indicator diagram and prove that the work done by the pump is proportional to the area of the indicator diagram.

SOLUTION

(i) Classification of reciprocating pumps

Basis	Types
Liquid in contact with piston	Single-acting (one side of the piston) · Double-acting (both sides)
Number of cylinders	Single-cylinder · Double-cylinder (duplex) · Triple-cylinder (triplex) · Multicylinder
Type of displacer	Piston pump · Plunger pump
Air vessels	With air vessels · Without air vessels
Drive	Crank (power) driven · Direct-acting (steam) · Hand operated

(ii) Indicator diagram

The indicator diagram is a graph of the **pressure head in the cylinder** (ordinate) against the **piston displacement** or stroke length (abscissa) over one revolution.

Sketch: horizontal axis = stroke L; a horizontal dashed line for atmospheric head H_{atm}. The suction line AB lies at a distance h_s below it (from A at the start to B at the end of the suction stroke); the delivery line CD lies at a distance h_d above it (from C back to D). Vertical lines BC and DA close the figure into a rectangle ABCD of height $h_s + h_d$ and width L.

- Suction stroke: cylinder head $= H_{\text{atm}} - h_s$, constant.
- Delivery stroke: cylinder head $= H_{\text{atm}} + h_d$, constant.

(iii) Proof: work done is proportional to the area of the indicator diagram

1. From the derivation of work done (no friction, no acceleration), for a single-acting pump
$$\text{Work done per second} = \frac{\rho g A L N}{60} \left(h_s + h_d \right)$$
2. For a given pump and liquid, ρ, g, A and N are constant. Let $K = \frac{\rho g A N}{60}$:
$$\text{Work done per second} = K \times L \left(h_s + h_d \right)$$
3. The indicator diagram is a rectangle of width L and height $\left(h_s + h_d \right)$, so
$$\text{Area of indicator diagram} = L \times \left(h_s + h_d \right)$$
4. Therefore
$$\text{Work done per second} \propto \text{Area of indicator diagram}$$

Extension (real diagram). Acceleration of the liquid tilts the suction and delivery lines (by $\pm h_{as}$ and $\pm h_{ad}$) but leaves the area unchanged, because the extra work at the start of each stroke is returned at the end. Friction adds parabolic bulges of mean height $\frac{2}{3} h_{f,\ \max}$, so the area becomes $L \left(h_s + h_d + \frac{2}{3} h_{fs} + \frac{2}{3} h_{fd} \right)$. In every case, work done = $K \times$ area.

Significance. The area of the indicator diagram, taken from a test, gives the work done directly; comparing it with the ideal rectangle shows the friction losses.

QB V.19 AKTU 2022-23 | KME302 | Q7(b) | 10 Marks AKTU 2021-22 | KME302 | Q2(e) | 10 Marks AKTU 2016-17 | ME401 | Q4 | 15 Marks Explain the ideal indicator diagram. Discuss the effect of acceleration and of friction in the suction and delivery pipes on the indicator diagram.

SOLUTION

Notation. Piston area A, crank radius r, stroke $L = 2r$, angular speed $\omega = 2\pi N/60$, crank angle θ from inner dead centre; suction pipe length L_s, area a_s, diameter d_s; delivery pipe L_d, a_d, d_d; static heads h_s, h_d; atmospheric head H_{atm}. Piston motion is simple harmonic.

(i) Ideal indicator diagram

The indicator diagram is the graph of cylinder pressure head against piston displacement for one revolution. In the ideal case friction and acceleration are neglected, so the pressure is constant during each stroke:

- Suction stroke: $H_{\text{atm}} - h_s$
- Delivery stroke: $H_{\text{atm}} + h_d$

Sketch: rectangle ABCD; AB (suction) at depth h_s below the atmospheric line, CD (delivery) at height h_d above it; width L, height $h_s + h_d$.

$$\text{Work done per second} = \frac{\rho g A L N}{60}\ (\ h_s + h_d\ )\ \propto\ \text{area of ABCD}$$

(ii) Effect of acceleration

1. **Piston motion.** $x = r\ (\ 1 -\ \cos\theta\ )$; differentiating, $v_{\text{piston}} = \omega r \sin\theta$ and $a_{\text{piston}} = \omega^2 r \cos\theta$.
2. **Continuity.** $A\, v_{\text{piston}} = a\, v_{\text{pipe}}$, so the liquid in the pipe accelerates at
$$a_{\text{pipe}} = \frac{A}{a}\,\omega^2 r \cos\theta$$
3. **Force needed.** Mass of the column $= \rho a L_p$; force $= \rho a L_p \times \frac{A}{a}\,\omega^2 r \cos\theta$.
4. **Acceleration head.** Pressure $=$ force$/a$; dividing by ρg:
$$h_a = \frac{L_p}{g}\,\frac{A}{a}\,\omega^2 r \cos\theta, \qquad h_{a,\ \max} = \frac{L_p}{g}\,\frac{A}{a}\,\omega^2 r\ \ (\ \theta = 0^\circ, 180^\circ\ )$$
5. **Pressures in the cylinder.**
   - Suction stroke: start ($\theta = 0$): $H_{\text{atm}} - h_s - h_{as}$; middle: $H_{\text{atm}} - h_s$; end ($\theta = 180^\circ$): $H_{\text{atm}} - h_s + h_{as}$.
   - Delivery stroke: start: $H_{\text{atm}} + h_d + h_{ad}$; middle: $H_{\text{atm}} + h_d$; end: $H_{\text{atm}} + h_d - h_{ad}$.
6. **Shape.** Since $h_a \propto\ \cos\theta$ varies linearly with x, the suction and delivery lines become sloping straight lines through the mid-points of the ideal lines: the rectangle becomes a **parallelogram** A'B'C'D'.
7. **Work.** The two triangles added and removed on each line are equal, so the **area, and the work done, are unchanged**. Proof:
$$W_a = \int_0^{\pi} \rho g A h_{a,\ \max}\ \cos\theta\, r \sin\theta\, d\theta = \rho g A r h_{a,\ \max} \left[\frac{\sin^2\theta}{2}\right]_0^{\pi} = 0.$$

Sketch: the ideal rectangle dashed; suction line tilted from h_{as} below AB at the start to h_{as} above it at the end; delivery line tilted from h_{ad} above CD at the start to h_{ad} below at the end.

Consequence. Acceleration lowers the pressure at the start of suction; if $H_{\text{atm}} - h_s - h_{as} < H_{\text{sep}}$ (about 2–2.5 m absolute), separation occurs. This limits the speed.

(iii) Effect of friction

1. **Pipe velocity.** $v = \frac{A}{a}\,\omega r \sin\theta$.

2. **Friction head (Darcy).**

$$h_f = \frac{4fL_p}{2g\,d}\left(\frac{A}{a}\omega r\,\sin\theta\right)^2, \qquad h_{f,\,\max} = \frac{4fL_p}{2g\,d}\left(\frac{A}{a}\omega r\right)^2 \quad (\,\theta = 90^\circ\,)$$

3. **Shape.** $h_f = h_{f,\,\max}\,\sin^2\theta$ is zero at both ends and greatest at mid-stroke, and it always opposes the flow. It adds a **parabola** below the suction line (pressure lowered) and above the delivery line (pressure raised).
4. **Mean friction head.** Work against friction on a stroke $= \rho g A r h_{f,\,\max}\int_0^{\pi}\sin^3\theta\,d\theta = \rho g A r h_{f,\,\max}\times\frac{4}{3}$. Dividing by $\rho g A\,(\,2r\,)$:

$$\bar{h}_f = \frac{2}{3}h_{f,\,\max}$$

5. **Work done including friction.** The area of the diagram increases:

$$\text{Work done per second} = \frac{\rho g A L N}{60}\left(\,h_s + h_d + \tfrac{2}{3}h_{fs} + \tfrac{2}{3}h_{fd}\,\right)$$

(iv) Combined effect

With both acceleration and friction, each stroke line is a sloping line (acceleration) with a parabola superposed (friction):

- Suction: start $H_{\text{atm}} - h_s - h_{as}$; middle $H_{\text{atm}} - h_s - h_{fs}$; end $H_{\text{atm}} - h_s + h_{as}$.
- Delivery: start $H_{\text{atm}} + h_d + h_{ad}$; middle $H_{\text{atm}} + h_d + h_{fd}$; end $H_{\text{atm}} + h_d - h_{ad}$.

Sketch: the parallelogram of (ii) with a curved suction line bulging downward by h_{fs} at mid-stroke and a curved delivery line bulging upward by h_{fd} at mid-stroke; ends of the lines at the parallelogram corners.

Effect	Change in diagram	Change in work
Acceleration	Rectangle becomes parallelogram	None
Friction	Parabolic bulges at mid-stroke	Increases by $\frac{2}{3}\,(\,h_{fs} + h_{fd}\,)$ head

Significance. Acceleration raises the peak forces and the risk of separation but costs no work; friction costs extra work. Air vessels reduce both by keeping the pipe velocity steady.

QB V.20 AKTU 2018-19 | RME601 | Q2(d) | 7 Marks Explain the effect of piston acceleration on the velocity and pressure in the suction and delivery pipes, and derive the pressure head due to acceleration.

SOLUTION

Explanation. The piston of a reciprocating pump starts each stroke from rest, reaches its highest velocity at mid-stroke and comes to rest again at the end. The liquid columns in the suction and delivery pipes must follow this motion, so they are

accelerated in the first half of each stroke and retarded in the second half. Accelerating a column needs an extra pressure difference, called the **acceleration head** h_a; during retardation the column gives this back.

Assumptions. Simple harmonic motion (long connecting rod); constant $\omega = 2\pi N/60$; incompressible liquid filling the pipe; $\theta = \omega t$ measured from inner dead centre. Pipe length L_p, area a; piston area A, crank radius r.

Sketch: crank of radius r at angle θ, connecting rod to the piston in the cylinder; piston displaced x from inner dead centre; suction pipe of length L_s below the cylinder.

1. **Piston displacement.** $x = r - r\cos\theta = r\,(1 - \cos\theta)$
2. **Piston velocity.**

$$v_{\text{piston}} = \frac{dx}{dt} = \frac{dx}{d\theta}\frac{d\theta}{dt} = \omega r \sin\theta$$

3. **Piston acceleration.**

$$a_{\text{piston}} = \frac{dv_{\text{piston}}}{dt} = \omega^2 r \cos\theta$$

4. **Velocity in the pipe (continuity).**

$$A\,v_{\text{piston}} = a\,v_{\text{pipe}} \quad \Rightarrow \quad v_{\text{pipe}} = \frac{A}{a}\,\omega r \sin\theta$$

5. **Acceleration in the pipe.**

$$a_{\text{pipe}} = \frac{A}{a}\,\omega^2 r \cos\theta$$

6. **Force to accelerate the column.** Mass $= \rho a L_p$:

$$F = \rho a L_p \times \frac{A}{a}\,\omega^2 r \cos\theta$$

7. **Pressure head.** Pressure $= F/a$; head $= F/(\rho g a)$:

$$h_a = \frac{L_p}{g}\,\frac{A}{a}\,\omega^2 r\cos\theta\,, \qquad h_{a,\,\max} = \frac{L_p}{g}\,\frac{A}{a}\,\omega^2 r \quad (\theta = 0^\circ, 180^\circ)$$

Effect on velocity and pressure

Position in stroke	θ	**Pipe velocity**	h_a	**Suction cylinder head**	**Delivery cylinder head**
Start	0°	0	$+h_{a,\,\max}$	$H_{\text{atm}} - h_s - h_{as}$ (lowest)	$H_{\text{atm}} + h_d + h_{ad}$ (highest)
Middle	90°	Maximum, $\frac{A}{a}\omega r$	0	$H_{\text{atm}} - h_s$	$H_{\text{atm}} + h_d$
End	180°	0	$-h_{a,\,\max}$	$H_{\text{atm}} - h_s + h_{as}$	$H_{\text{atm}} + h_d - h_{ad}$

Significance. h_a grows with pipe length, with A/a and with ω^2. It lowers the cylinder pressure at the start of suction; if this falls to the separation head (about 2–2.5 m absolute) the liquid column separates, which limits the speed and suction lift. It does not change the net work per cycle, and air vessels near the cylinder remove it from the long pipes.

QB V.21 AKTU 2018-19 | RME601 | Q6(b) | 7 Marks A double-acting reciprocating pump makes 85 double strokes per minute with simple harmonic motion. The piston diameter is 0.120 m and the stroke 0.220 m. The vertical suction pipe is 5 m long and 110 mm in diameter. Separation occurs at 2 m of water absolute, and the atmospheric head is 10.3 m of water. Find the maximum permissible suction lift.

SOLUTION

Given: double-acting, $N = 85\ \text{rpm}$ (85 double strokes per minute = 85 revolutions per minute), $D = 0.120\ \text{m}$, $L = 0.220\ \text{m}$ so $r = 0.110\ \text{m}$, suction pipe $L_s = 5\ \text{m}$, $d_s = 110\ \text{mm} = 0.110\ \text{m}$, $H_{\text{sep}} = 2\ \text{m}$ (abs), $H_{\text{atm}} = 10.3\ \text{m}$. No air vessel; suction friction neglected.

To find: maximum suction lift h_s.

Condition. The lowest pressure in the cylinder occurs at the start of the suction stroke and must not fall below the separation head:

$$H_{\text{atm}} - h_s - h_{as,\ \max} = H_{\text{sep}}$$

For a double-acting pump each side of the piston has its own suction stroke, and the acceleration head per stroke is the same as for a single-acting pump at the same speed.

1. $\omega = \dfrac{2\pi N}{60} = \dfrac{2\pi \times 85}{60} = 8.901\ \text{rad/s};\ \omega^2 = 79.23$
2. $\dfrac{A}{a_s} = \left(\dfrac{D}{d_s}\right)^2 = \left(\dfrac{0.120}{0.110}\right)^2 = 1.190$
3. $h_{as,\ \max} = \dfrac{L_s}{g}\dfrac{A}{a_s}\omega^2 r = \dfrac{5}{9.81} \times 1.190 \times 79.23 \times 0.110 = 5.287\ \text{m}$
4. $h_s = H_{\text{atm}} - H_{\text{sep}} - h_{as,\ \max} = 10.3 - 2 - 5.287 = 3.01\ \text{m}$

Answer: the maximum permissible suction lift is about $3.0\ \text{m}$. (The 5 m vertical suction pipe then dips about 2 m below the sump surface; the length used for h_{as} is the full pipe length, as given.)

V.5 Air vessels and comparison

QB V.22 AKTU 2023-24 | BME302 | Q2(e) | 7 Marks AKTU 2016-17 | ME401 | Q1(i) | 2 Marks Explain the working of an air vessel with a neat diagram and its importance in a reciprocating pump.

SOLUTION

Air vessel. An air vessel is a closed cast-iron chamber with an opening at its base through which liquid enters or leaves. The upper part contains compressed air and the lower part liquid. One vessel is fitted on the suction pipe close to the suction valve and one on the delivery pipe close to the delivery valve. *(For the 2-mark version, the first two lines of "Working" and the three functions below are enough.)*

Sketch: single-acting reciprocating pump; sump and long suction pipe rising to a suction air vessel just below the suction valve; delivery air vessel just after the delivery valve with the long delivery pipe leaving it; each vessel shows an air cushion above a liquid level; short pipe lengths between the vessels and the cylinder marked.

Working

1. **Delivery side.** At the beginning and end of the delivery stroke the piston moves slowly and delivers less than the mean flow; the compressed air expands and the vessel supplies the shortfall to the delivery pipe. Near mid-stroke the piston moves fast and delivers more than the mean; the excess liquid enters the vessel and compresses the air. During the suction stroke the vessel alone feeds the delivery pipe. Thus the long delivery pipe carries an almost steady flow.
2. **Suction side.** While the suction valve is closed (delivery stroke), the liquid in the long suction pipe keeps flowing into the suction vessel. During the suction stroke the vessel and the pipe together supply the cylinder. The long suction pipe again carries a steady flow.
3. Only the short pipes between the vessels and the cylinder carry the fluctuating flow. For a single-acting pump the flow into the delivery vessel is $q = A\omega r\left(\ \sin\theta - 1/\pi\ \right)$: positive (into the vessel) for $18.6^\circ < \theta < 161.4^\circ$ and negative otherwise.

Importance (functions)

1. Gives a **continuous, nearly uniform discharge** in the suction and delivery pipes.
2. **Saves work against friction**, because the pipe velocity becomes the steady mean instead of a fluctuating one: 84.8% saving for a single-acting and 39.2% for a double-acting pump.
3. **Reduces the acceleration head** in the long pipes, so the pump can run at a **higher speed** or with a **larger suction lift** without separation.
4. Reduces the peak forces on the piston and crank, giving smoother running and less vibration.

Significance. Air vessels make a reciprocating pump with long pipes practical: higher speed, less power and steadier flow.

QB V.23 AKTU 2018-19 | RME601 | Q6(a) | 7 Marks Show that the work saved against friction in the pipes by fitting air vessels to a double-acting reciprocating pump is 39.2%.

SOLUTION

Assumptions. Simple harmonic motion of the piston; friction head by Darcy, $h_f = \dfrac{4fL_p v^2}{2g\,d}$; with air vessels close to the cylinder the velocity in the long pipe is constant and equal to the mean velocity; the short pipe between vessel and cylinder is neglected. Notation: piston area A, pipe area a, crank radius r, $L = 2r$, $\omega = 2\pi N/60$.

Sketch: double-acting pump with air vessels on suction and delivery pipes; graph of pipe velocity against crank angle: without vessel a train of half-sine humps of peak $(A/a)\,\omega r$ (two per revolution), with vessel a horizontal line at $(2/\pi)\,(A/a)\,\omega r$.

(a) Without air vessel

1. Pipe velocity: $v = \dfrac{A}{a}\,\omega r \sin\theta$.
2. Friction head: $h_f = \dfrac{4fL_p}{2g\,d}\left(\dfrac{A}{a}\,\omega r\right)^2 \sin^2\theta = h_{f,\,\max}\,\sin^2\theta$.
3. Work against friction on one stroke ($dx = r\sin\theta\,d\theta$):

$$W_f = \int_0^{\pi} \rho g A\, h_{f,\,\max}\,\sin^2\theta\; r\sin\theta\,d\theta = \rho g A r\, h_{f,\,\max} \int_0^{\pi} \sin^3\theta\,d\theta$$

4. $\int_0^{\pi} \sin^3\theta\,d\theta = \left[-\cos\theta + \frac{\cos^3\theta}{3}\right]_0^{\pi} = \frac{2}{3} + \frac{2}{3} = \frac{4}{3}$
5. Mean friction head (work $\div\ \rho g A \times 2r$):

$$\bar{h}_f = \frac{\rho g A r\, h_{f,\,\max} \times \frac{4}{3}}{2\rho g A r} = \frac{2}{3}\,h_{f,\,\max}$$

6. For a double-acting pump this holds on each of the two strokes per revolution, so the work against friction per revolution is $\rho g\,(2AL)\,\frac{2}{3}h_{f,\,\max}$.

(b) With air vessel

7. Mean discharge of a double-acting pump: $Q = \dfrac{2AL}{T} = \dfrac{2A\times 2r}{2\pi/\omega} = \dfrac{2A\omega r}{\pi}$
8. Steady pipe velocity: $v_{\text{mean}} = \dfrac{Q}{a} = \dfrac{2}{\pi}\dfrac{A}{a}\,\omega r$
9. Friction head (constant):

$$h_f{}' = \frac{4fL_p}{2g\,d}\left(\frac{2}{\pi}\frac{A}{a}\,\omega r\right)^2 = \frac{4}{\pi^2}\,h_{f,\,\max}$$

10. Work against friction per revolution $= \rho g\,(2AL)\,\dfrac{4}{\pi^2}h_{f,\,\max}$ (same volume pumped).

(c) Work saved

11. Ratio of friction work, with to without:

$$\frac{\left(4/\pi^2\right) h_{f,\,\max}}{\left(2/3\right) h_{f,\,\max}} = \frac{6}{\pi^2} = \frac{6}{9.870} = 0.608$$

12. Percentage saved:

$$\text{Work saved} = \left(1 - 0.608\right) \times 100 = 39.2\%$$

Significance. For a single-acting pump the same method gives $v_{\text{mean}} = \left(1/\pi\right)\left(A/a\right)\omega r$ and a saving of $1 - 3/\left(2\pi^2\right) = 84.8\%$. The saving is smaller for a double-acting pump because its pipe flow is already smoother (mean velocity $2/\pi$ of the peak instead of $1/\pi$).

QB V.24 AKTU 2023-24 | BME302 | Q7(a) | 7 Marks State the differences between a centrifugal pump and a reciprocating pump.

SOLUTION

A centrifugal pump is a rotodynamic pump that raises the pressure of the liquid by whirling it in an impeller; a reciprocating pump is a positive displacement pump in which a piston pushes out a fixed volume each stroke.

S. No.	Feature	Centrifugal pump	Reciprocating pump
1	Principle	Rotodynamic: impeller gives kinetic energy, casing converts it to pressure	Positive displacement: piston pushes liquid out
2	Discharge	Large, continuous and uniform	Small, pulsating (smoothed by air vessels)
3	Head	Low to medium per stage; high only with multistage	High head possible with one cylinder
4	Discharge–head relation	Discharge falls as head rises	Discharge almost independent of head, $Q = ALN/60$
5	Speed	High; coupled directly to motor	Low; needs gear or belt drive
6	Priming	Required	Not required (self-priming)
7	Dirty, viscous liquids	Can handle sewage, slurry (open impeller)	Unsuitable; valves choke
8	Size, weight, cost	Compact, light, cheaper	Bulky, heavy, costlier for same discharge
9	Maintenance	Low; few wearing parts	Higher; valves, piston rings, glands wear
10	Efficiency	High near design point, falls away from it	High and nearly constant over a wide range
11	Delivery valve closed while running	Safe for a short time (shut-off head)	Dangerous; needs relief valve
12	Uses	Water supply, irrigation, process plants, boiler feed (multistage)	Hydraulic presses, metering, small high-pressure duties, oil wells

Selection rule: large discharge at moderate head and dirty liquids favour the centrifugal pump; small discharge at high pressure favours the reciprocating pump.

OUTSIDE THE CURRENT BME302 SYLLABUS

These questions appeared in the papers read for this bank, but their topics are not in the BME302 syllabus. They are included, with solutions, for completeness.

X.1 Notches, weirs and orifice flow

QB X.1 AKTU 2022-23 | KME302 | Q2(a)(iii) | 10 Marks (part) Differentiate between a notch and a weir.

SOLUTION

Notch	Weir
An opening cut in the side of a tank or small channel, with the liquid surface below the top edge	A concrete or masonry barrier built across a river or large open channel; water flows over its top
Small in size; usually a metal plate	Large structure
Used to measure discharge in laboratories and small channels	Used to measure discharge of rivers and large channels, and to raise the water level
Bottom edge over which water flows is called the **sill**	Top edge over which water flows is called the **crest**

Both are measured by the head H over the sill or crest; e.g. rectangular: $Q = \frac{2}{3} C_d L \sqrt{2g}\, H^{3/2}$.

QB X.2 AKTU 2021-22 | KME302 | Q3(b) | 10 Marks Explain the difference between a notch and a weir. In a laboratory test, 0.05 m³ of water flowing over a right-angled notch was collected in one minute. The head over the sill was 50 mm. Find the coefficient of discharge.

SOLUTION

(i) Notch and weir. A **notch** is a small opening (usually a sharp-edged metal plate) in the side of a tank or laboratory channel, used to measure discharge; the liquid surface is below its top edge, and its lower edge is the **sill**. A **weir** is a large masonry or concrete barrier built across a river or canal; water flows over its top edge, the **crest**. A weir measures the discharge of rivers and large channels and also raises the upstream water level. In both, the discharge depends only on the head H above the sill/crest.

(ii) Discharge over a triangular (V) notch. *Sketch:* a V-shaped notch of total angle θ with water surface a height H above the apex; a horizontal strip of thickness dh at depth h below the free surface.

1. Width of strip at depth h: $b = 2\,(\,H - h\,)\,\tan\,(\,\theta/2\,)$.
2. Velocity through the strip: $\sqrt{2gh}$; discharge $dQ = C_d \times 2\,(\,H - h\,)\,\tan\,(\,\theta/2\,)\,dh\sqrt{2gh}$.
3. Integrate from $h = 0$ to H:

$$Q = 2C_d \tan\frac{\theta}{2}\sqrt{2g}\int_0^H (\,H - h\,)\,h^{1/2}\,dh = 2C_d \tan\frac{\theta}{2}\sqrt{2g}\left(\frac{2}{3} - \frac{2}{5}\right)H^{5/2}$$

$$Q = \frac{8}{15} C_d \tan\frac{\theta}{2} \sqrt{2g}\, H^{5/2}$$

(iii) Numerical.

Given: $\theta = 90^\circ$ so $\tan(\theta/2) = 1$; volume $= 0.05\ \text{m}^3$ in $t = 60\ \text{s}$; $H = 50\ \text{mm} = 0.05\ \text{m}$.

To find: C_d.

1. Actual discharge: $Q = \dfrac{0.05}{60} = 8.333 \times 10^{-4}\ \text{m}^3/\text{s}$.
2. $H^{5/2} = 0.05^{2.5} = 5.590 \times 10^{-4}$; $\dfrac{8}{15}\sqrt{2 \times 9.81} = 2.362$.
3. $C_d = \dfrac{Q}{\frac{8}{15}\tan 45^\circ \sqrt{2g}\, H^{5/2}} = \dfrac{8.333 \times 10^{-4}}{2.362 \times 1 \times 5.590 \times 10^{-4}} = 0.631$

Answer: $C_d \approx 0.631$, a typical value for a sharp-edged V-notch.

QB X.3 AKTU 2022-23 | KCE303 | Q5(a) | 10 Marks Find the time needed to raise the water level in a tank of uniform cross-sectional area A, when an orifice of area a at the bottom discharges while a constant inflow Q enters the tank.

SOLUTION

A tank receives a steady inflow Q while an orifice at its bottom discharges at a rate that grows with the head. The level rises only while inflow exceeds outflow. We find the time to raise the level from H_1 to H_2.

Sketch: a vertical tank of plan area A; inflow pipe at the top delivering Q; orifice of area a at the bottom; water level at height h above the orifice, rising from H_1 to H_2.

Assumptions. A constant; coefficient of discharge C_d of the orifice constant; velocity of the water surface in the tank negligible.

Step 1. Outflow at head h. By Torricelli, $q = C_d a\sqrt{2gh} = k\sqrt{h}$, where $k = C_d a\sqrt{2g}$.

Step 2. Volume balance in time dt. Increase in stored volume = inflow − outflow:

$$A\,dh = \left(Q - k\sqrt{h}\right) dt \quad \Rightarrow \quad dt = \frac{A\,dh}{Q - k\sqrt{h}}$$

Step 3. Substitution. Put $\sqrt{h} = x$, so $h = x^2$ and $dh = 2x\,dx$:

$$dt = \frac{2A\,x\,dx}{Q - kx}$$

Step 4. Integrate. Write $\dfrac{x}{Q-kx} = -\dfrac{1}{k} + \dfrac{Q/k}{Q-kx}$. Then

$$\int \frac{x\,dx}{Q-kx} = -\frac{x}{k} - \frac{Q}{k^2}\ln(Q - kx)$$

Step 5. Apply limits $x_1 = \sqrt{H_1}$ to $x_2 = \sqrt{H_2}$:

$$T = \frac{2A}{k^2}\left[k\left(\sqrt{H_1} - \sqrt{H_2}\right) + Q\ln\frac{Q - k\sqrt{H_1}}{Q - k\sqrt{H_2}}\right]$$

Result.

$$T = \frac{2A}{\left(C_d a\sqrt{2g} \right)^2}\left[C_d a\sqrt{2g}\left(\sqrt{H_1} - \sqrt{H_2} \right) + Q \ln \frac{Q - C_d a\sqrt{2gH_1}}{Q - C_d a\sqrt{2gH_2}} \right]$$

The first term is negative and the logarithmic term positive; T is positive.

Significance. The level can rise only while $Q > C_d a\sqrt{2gh}$. It approaches (but never reaches in finite time) the equilibrium head $H_e = \left(Q/\left(C_d a\sqrt{2g} \right) \right)^2$, where inflow equals outflow; so H_2 must be less than H_e, otherwise the logarithm is undefined.

X.2 Hydrostatics, buoyancy and vortex motion

QB X.4 AKTU 2022-23 | KCE303 | Q2(a) | 10 Marks At sea level the pressure is 1 bar and the temperature 30 °C. Find the pressure at an altitude of 5 km assuming (i) constant air density, (ii) an isothermal atmosphere and (iii) a temperature falling linearly at 0.005 °C/m. Take $R = 287$ J/(kg·K).

SOLUTION

Basic equation. For a fluid at rest, with z measured upwards, $\dfrac{dp}{dz} = -\rho g$. For air, $p = \rho RT$, so $\rho = p/(RT)$.

Given: $p_0 = 1\ \text{bar} = 1.0 \times 10^5\ \text{N/m}^2$; $T_0 = 30 + 273 = 303\ \text{K}$; $z = 5000\ \text{m}$; $R = 287\ \text{J/(kg·K)}$; lapse rate $L = 0.005\ \text{K/m}$.

To find: p at $z = 5\ \text{km}$ for the three cases.

(i) Constant density (incompressible).

1. $\rho_0 = \dfrac{p_0}{RT_0} = \dfrac{1.0 \times 10^5}{287 \times 303} = 1.150\ \text{kg/m}^3$.
2. Integrating $dp = -\rho_0 g\, dz$: $p = p_0 - \rho_0 g z = 1.0 \times 10^5 - 1.150 \times 9.81 \times 5000$.
3. $p = 10^5 - 56\,405 = 43.6\ \text{kPa}$

(ii) Isothermal atmosphere ($T = T_0$).

1. $\dfrac{dp}{dz} = -\dfrac{pg}{RT_0} \Rightarrow \dfrac{dp}{p} = -\dfrac{g}{RT_0}\, dz$.
2. Integrate from p_0 at $z = 0$: $\ln \dfrac{p}{p_0} = -\dfrac{gz}{RT_0}$, so $p = p_0\, e^{-gz/(RT_0)}$.
3. $\dfrac{gz}{RT_0} = \dfrac{9.81 \times 5000}{287 \times 303} = 0.5640$.
4. $p = 1.0 \times 10^5 \times e^{-0.5640} = 56.89\ \text{kPa}$

(iii) Linear temperature fall ($T = T_0 - Lz$).

1. $\dfrac{dp}{p} = -\dfrac{g\, dz}{R(T_0 - Lz)}$.
2. Integrate: $\ln \dfrac{p}{p_0} = \dfrac{g}{RL} \ln \dfrac{T_0 - Lz}{T_0}$, so $p = p_0 \left(1 - \dfrac{Lz}{T_0} \right)^{g/(RL)}$.

3. $\dfrac{g}{RL} = \dfrac{9.81}{287 \times 0.005} = 6.836$; $1 - \dfrac{0.005 \times 5000}{303} = 0.9175$ (temperature at 5 km $= 278\,\text{K} = 5\,^\circ\text{C}$).
4. $p = 1.0 \times 10^5 \times 0.9175^{6.836} = \ 55.51\,\text{kPa}$

Answer: (i) $43.6\,\text{kPa}$, (ii) $56.89\,\text{kPa}$, (iii) $55.51\,\text{kPa}$. The constant-density result is far too low because air becomes lighter with height; the linear-lapse model is closest to the real atmosphere.

QB X.5 AKTU 2022-23 | KCE303 | Q3(a) | 10 Marks Find the range of b/h for stable equilibrium of a floating rectangular block of relative density S, width b and height h. A right circular cylinder of 0.3 m diameter and 0.6 m length, of specific weight 7500 N/m³, is to float vertically in kerosene of specific weight 8900 N/m³. Check its stability.

SOLUTION

Condition for stability. A floating body is stable when its metacentre M lies above its centre of gravity G, i.e. $GM = BM - BG > 0$, where $BM = I/V$ (I = second moment of the water-plane area about its longitudinal axis, V = volume displaced) and B is the centre of buoyancy.

(i) Rectangular block. *Sketch:* a block of width b, height h, length l (into the page), floating in water with depth of immersion d; G at mid-height, B at the centre of the immersed part, M above them.

1. **Depth of immersion.** Weight = buoyancy: $S\rho g\ (\ bhl\ ) = \rho g\ (\ bdl\ ) \Rightarrow d = Sh$.
2. **Positions above the base.** $OG = \dfrac{h}{2}$, $OB = \dfrac{d}{2} = \dfrac{Sh}{2}$, so $BG = \dfrac{h}{2}\ (\ 1 - S\ )$.
3. **Metacentric radius.** $I = \dfrac{lb^3}{12}$, $V = b\,Sh\,l$, so $BM = \dfrac{I}{V} = \dfrac{b^2}{12\,Sh}$.
4. **Stability:** $BM > BG \Rightarrow \dfrac{b^2}{12Sh} > \dfrac{h\ (\ 1 - S\ )}{2} \Rightarrow \dfrac{b^2}{h^2} > 6S\ (\ 1 - S\ )$.

$$\frac{b}{h} > \sqrt{6S\ (\ 1 - S\ )}$$

(Here S is the relative density with respect to the liquid in which the block floats.) A wide, shallow block is stable; a tall, narrow one tips over.

(ii) Cylinder in kerosene.

Given: $D = 0.3\,\text{m}$, $L = 0.6\,\text{m}$, $w_c = 7500\,\text{N/m}^3$, $w_k = 8900\,\text{N/m}^3$; axis vertical.

To find: GM and whether it is positive.

1. **Depth of immersion:** $w_c \dfrac{\pi D^2}{4} L = w_k \dfrac{\pi D^2}{4} d \Rightarrow d = \dfrac{7500 \times 0.6}{8900} = 0.5056\,\text{m}$.
2. $OG = \dfrac{0.6}{2} = 0.300\,\text{m}$; $OB = \dfrac{0.5056}{2} = 0.2528\,\text{m}$; $BG = 0.300 - 0.2528 = 0.0472\,\text{m}$.
3. $BM = \dfrac{I}{V} = \dfrac{\pi D^4/64}{\left(\pi D^2/4\right) d} = \dfrac{D^2}{16d} = \dfrac{0.3^2}{16 \times 0.5056} = 0.01113\,\text{m}$.
4. $GM = BM - BG = 0.01113 - 0.0472 = \ -0.0361\,\text{m}$

Answer: GM is negative (M lies below G), so the cylinder is **unstable** and cannot float with its axis vertical; it will tip over and float on its side. It would need a larger diameter or a shorter length to be stable.

QB X.6 AKTU 2022-23 | KCE303 | Q3(b) | 10 Marks Find the ratio of base diameter to height of a solid cone of relative density S floating (apex down) in a liquid, for stable equilibrium.

SOLUTION

Sketch: a solid cone of height H and base diameter D floating with apex O downward and axis vertical; immersed depth h; radius of the cone at the free surface r; G, B and M on the axis.

Step 1. Depth of immersion. The immersed part is a similar cone of height h. Volumes of similar cones vary as the cube of height. Weight = buoyancy:

$$S \times (\text{volume of cone}) = \text{volume immersed} \Rightarrow \left(\frac{h}{H}\right)^3 = S \Rightarrow h = HS^{1/3}$$

Step 2. G and B. The centroid of a cone is at $\frac{3}{4}$ of its height from the apex. So $OG = \frac{3H}{4}$, $OB = \frac{3h}{4}$, and

$$BG = \frac{3}{4}\,(\,H - h\,)$$

Step 3. Metacentric radius. Radius at the water line: $r = \frac{D}{2} \cdot \frac{h}{H}$. Then $I = \frac{\pi r^4}{4}$, $V = \frac{\pi r^2 h}{3}$:

$$BM = \frac{I}{V} = \frac{3r^2}{4h}$$

Step 4. Stability condition $BM > BG$:

$$\frac{3r^2}{4h} > \frac{3}{4}\,(\,H - h\,) \;\Rightarrow r^2 > h\,(\,H - h\,)$$

Step 5. Substitute r**.** $\frac{D^2 h^2}{4H^2} > h\,(\,H - h\,) \;\Rightarrow D^2 > \frac{4H^2\,(\,H - h\,)}{h} = 4H^2\,\frac{1 - S^{1/3}}{S^{1/3}}$.

Result.

$$\frac{D}{H} > 2\sqrt{\frac{1 - S^{1/3}}{S^{1/3}}}$$

Equivalently, in terms of the semi-vertical angle α ($\tan\alpha = D/\,(\,2H\,)$):
$\tan^2\alpha > \frac{1 - S^{1/3}}{S^{1/3}}$, i.e. $\cos^2\alpha < S^{1/3}$.

Significance. A flat (wide-based) cone is stable; a slender cone floating apex down is unstable. A heavier cone (larger S) sinks deeper and is stable at a smaller D/H.

QB X.7 AKTU 2022-23 | KCE303 | Q6(a) | 10 Marks Prove that in a forced vortex the rise of the liquid level at the ends equals the fall of the level at the axis of rotation.

SOLUTION

A **forced vortex** is flow in which the liquid rotates as a solid body at constant angular velocity ω (e.g. liquid in a cylinder rotated about its vertical axis), so the tangential velocity is $v = \omega r$.

Sketch: an open cylinder of radius R with liquid initially at height h_0 (dotted horizontal line). On rotation the free surface becomes a paraboloid: it falls by x at the axis and rises by y at the wall; the total height of the paraboloid is Z.

Step 1. Pressure variation. For a fluid element at radius r, the radial pressure gradient supplies the centripetal force, and the vertical gradient balances weight:

$$\frac{\partial p}{\partial r} = \rho\omega^2 r, \qquad \frac{\partial p}{\partial z} = -\rho g$$

$$dp = \rho\omega^2 r\, dr - \rho g\, dz$$

Step 2. Free surface. On the free surface p = atmospheric, so $dp = 0$: $g\, dz = \omega^2 r\, dr$. Integrating from the lowest point ($r = 0$, $z = 0$):

$$z = \frac{\omega^2 r^2}{2g}$$

The surface is a paraboloid of revolution. At the wall, its height is $Z = \dfrac{\omega^2 R^2}{2g}$.

Step 3. Volume of the paraboloid. Take a ring of radius r, thickness dr, under the curved surface:

$$V_p = \int_0^R 2\pi r\, z\, dr = \int_0^R 2\pi r \frac{\omega^2 r^2}{2g}\, dr = \frac{\pi\omega^2 R^4}{4g} = \frac{1}{2}\pi R^2 Z$$

So the paraboloid volume is half that of the circumscribing cylinder.

Step 4. Conservation of liquid volume. No liquid spills. After rotation the liquid fills a cylinder of height $(h_0 + y)$ minus the empty paraboloid:

$$\pi R^2 (h_0 + y) - \frac{1}{2}\pi R^2 Z = \pi R^2 h_0 \Rightarrow y = \frac{Z}{2}$$

Step 5. Since the surface at the axis is Z below the surface at the wall, $x + y = Z$, so $x = Z - \dfrac{Z}{2} = \dfrac{Z}{2}$.

Result.

$$x = y = \frac{Z}{2} = \frac{\omega^2 R^2}{4g}$$

The rise of the liquid at the wall equals the fall at the axis, each being half the paraboloid height. This is used to find the speed at which liquid just spills from, or just uncovers the bottom of, a rotating vessel.

X.3 Hydraulic devices

QB X.8 AKTU 2016-17 | ME401 | Q1(a)-(e),(h) | 2 Marks each Write short notes on (a) the torque converter, (b) the hydraulic ram, (c) the hydraulic accumulator, (d) the hydraulic intensifier, (e) the air lift pump and (h) the jet pump.

SOLUTION

(a) Torque converter. A hydraulic device that transmits power from a driving shaft to a driven shaft and changes (multiplies) the torque. It has three elements in one oil-filled casing: a pump impeller on the input shaft, a turbine runner on the output shaft and fixed guide (stator) blades between them. The stator gives the reaction that makes output torque $T_t = T_p + T_{\text{stator}}$, up to 2–3 times the input torque at low output speed. Used in automobile automatic transmissions.

(b) Hydraulic ram. A pump that uses the water-hammer pressure of a large quantity of water falling through a small height H to lift a small quantity q to a large height h, with no external power. A waste valve closes suddenly; the pressure rise opens the delivery valve to an air vessel. Efficiency (D'Aubuisson): $\eta = \dfrac{q\,(\,h - H\,)}{QH}$, where Q is the water wasted. Used to supply water to hilly villages.

(c) Hydraulic accumulator. A device that stores the energy of pressurised water when the demand is low and releases it when hydraulic machines (lifts, presses, cranes) need more. A loaded ram slides in a vertical cylinder; pressure $p = W/A$. Energy stored $= pAL = WL$, where L is the stroke. It lets a small pump meet large intermittent demands.

(d) Hydraulic intensifier. A device that raises the pressure of water using a large supply of low-pressure water. A fixed ram (small area a) enters a sliding cylinder that moves inside a larger fixed cylinder (area A). Force balance gives $p_{\text{high}} = p_{\text{low}}\,\dfrac{A}{a}$. Used in hydraulic presses, riveting and forging machines.

(e) Air lift pump. Compressed air is blown into the bottom of a delivery pipe submerged in a well. The air–water mixture has a lower density than the surrounding water, so the outside water pressure pushes the mixture up and out of the pipe. No moving parts below ground; low efficiency (about 20–40%). Used for deep wells, sandy or corrosive water.

(h) Jet pump. A pump with no moving parts in which a high-velocity jet of a driving fluid (water or steam) issues from a nozzle into a mixing chamber, entrains the surrounding fluid by momentum exchange, and the mixture's kinetic energy is converted to pressure in a diffuser. Often used with a centrifugal pump to lift water from deep wells; efficiency about 25–30%.

QB X.9 AKTU 2018-19 | RME601 | Q1(g) | 2 Marks Explain the use of an accumulator in a hydraulic system.

SOLUTION

A **hydraulic accumulator** stores the energy of pressurised liquid when machines need less than the pump delivers, and returns it when they need more. It is a vertical cylinder with a ram carrying a heavy load W; the stored liquid is at pressure $p = W/A$. Energy stored $= pAL = WL$ (L = stroke). It lets a small, steadily running pump supply large, intermittent demands of presses, lifts and cranes, and smooths pressure fluctuations.

QB X.10 AKTU 2018-19 | RME601 | Q2(e) | 7 Marks Describe the torque converter with a neat sketch: construction, working, characteristics, merits, limitations and applications.

SOLUTION

A **torque converter** is a hydraulic power-transmitting device that transmits power from one shaft to another and, unlike a fluid coupling, changes the torque: the output torque can be several times the input torque.

Sketch: an oil-filled closed casing with three bladed wheels in a ring: the pump impeller (keyed to the driving shaft), the turbine runner (keyed to the driven shaft) facing it, and a stationary row of guide blades (stator/reactor) fixed to the casing, placed between the turbine outlet and the pump inlet. Arrows show oil circulating pump → turbine → stator → pump.

1. Construction. - **Pump impeller** on the input (engine) shaft. - **Turbine runner** on the output shaft. - **Stator (fixed guide wheel)** between turbine exit and pump entry, held by the casing (in automobiles, on a one-way clutch). - The casing is filled with oil.

2. Working. 1. The engine drives the impeller; oil leaves it with high kinetic and pressure energy. 2. The oil strikes the turbine blades and drives the output shaft. 3. Leaving the turbine, the oil passes through the fixed stator blades, which change its direction so that it enters the pump in a favourable direction. 4. The stator receives a torque T_s from the oil; by the moment-of-momentum (angular-momentum) balance for the closed circuit,

$$T_t = T_p + T_s$$

so the turbine torque exceeds the pump torque. The torque ratio T_t/T_p is 2–3 at stall and falls as the output speed rises.

5. Efficiency: $\eta = \dfrac{T_t N_t}{T_p N_p}$.

3. Characteristics. Torque ratio is maximum when the output shaft is stalled ($N_t = 0$) and decreases with speed ratio N_t/N_p. Efficiency is zero at stall, rises to a peak of about 85–90% at an intermediate speed ratio, then falls. In automobiles the stator freewheels beyond the “coupling point”, and the unit then acts as a fluid coupling.

4. Merits. Smooth, shock-free and automatic variation of torque with load; no wear of contacting parts; damps vibration and shock loads; engine cannot stall under sudden load.

5. Limitations. Lower efficiency than a gearbox at most speeds; heating of the oil (needs cooling); more costly and heavier; some slip always exists.

6. Applications. Automatic transmissions of cars, buses and trucks; locomotives; earth-moving machines, cranes and hoists; marine drives.

QB X.11 AKTU 2018-19 | RME601 | Q7(a) | 7 Marks Describe the hydraulic press with a neat sketch: construction, working, characteristics, merits, limitations and applications.

SOLUTION

A **hydraulic press** is a device that uses a small force on a small plunger to produce a very large force on a large ram. It works on **Pascal's law**: pressure applied to a confined liquid is transmitted equally in all directions.

Sketch: two vertical cylinders connected at the bottom by a pipe and filled with water or oil: a small cylinder with a plunger of area a pushed down by force F (through a lever), and a large cylinder with a ram of area A carrying a platen that presses the job against a fixed upper head; non-return valves between a supply tank and the plunger cylinder.

1. Construction. Small plunger in a small cylinder (pump side); large ram in a large cylinder; connecting pipe; supply tank with suction and delivery valves; lever on the plunger; fixed top frame against which the job is pressed; release valve.

2. Working. 1. Force F on the plunger gives pressure $p = F/a$ in the liquid. 2. The same pressure acts on the ram: $W = pA$.

$$W = F\frac{A}{a}$$

3. With a lever of mechanical advantage L/l, $W = F\frac{L}{l}\frac{A}{a}$.
4. On each stroke the plunger pumps a small volume $a\,s$ into the ram cylinder; the ram rises by $a\,s/A$. On the return stroke the delivery valve closes and the suction valve refills the plunger cylinder from the tank.

3. Characteristics. Force gain $= A/a$; the ram moves a distance smaller by the same ratio, so work out ≤ work in (energy conservation); actual output is reduced by friction and leakage (mechanical efficiency typically 80–90%).

4. Merits. Very large forces from small effort; force can be controlled finely; simple construction; full force over the whole stroke; overload protection with a relief valve.

5. Limitations. Slow ram movement; leakage at seals; oil or water must be kept clean; fire risk with oil; needs maintenance of packing.

6. Applications. Forging, punching, deep drawing and metal forming; baling of cotton and paper; pressing of oilseeds; car lifts, jacks; testing machines (UTM); moulding of composites.

QB X.12 AKTU 2018-19 | RME601 | Q7(b) | 7 Marks A hydraulic accumulator is loaded with 400 kN. The ram is 300 mm in diameter and the stroke 6 m. Friction is 5% and the full stroke takes 2 minutes. The pump delivers 0.0075 m³/s while the accumulator descends. Find the total work supplied by the accumulator and the power delivered to the hydraulic appliance.

SOLUTION

Given: $W = 400\,\text{kN} = 400 \times 10^3\,\text{N}$; $D = 300\,\text{mm} = 0.3\,\text{m}$; stroke $L = 6\,\text{m}$; friction $= 5\%$ of W; time of descent $t = 2\,\text{min} = 120\,\text{s}$; pump discharge $Q_p = 0.0075\,\text{m}^3/\text{s}$.

To find: work supplied by the accumulator in one stroke; power delivered to the appliance.

Assumption. While the ram descends, friction acts upward and opposes the load, so the effective force on the liquid is $0.95W$. The pump delivers at the same pressure as the accumulator.

1. Ram area: $A = \dfrac{\pi}{4} \times 0.3^2 = 0.07069\,\text{m}^2$.
2. Effective load: $W_{\text{eff}} = W - 0.05W = 0.95 \times 400 = 380\,\text{kN}$.
3. Pressure of the liquid supplied: $p = \dfrac{W_{\text{eff}}}{A} = \dfrac{380 \times 10^3}{0.07069} = 5.376 \times 10^6\,\text{N/m}^2$ (5.376 MPa).
4. Work supplied by the accumulator in one full stroke: $W_{\text{eff}} \times L = 380 \times 10^3 \times 6 = 2280\,\text{kN·m}$ (2.28 MJ).
5. Volume supplied by the pump in 2 min: $0.0075 \times 120 = 0.9\,\text{m}^3$. Energy from the pump: $p \times 0.9 = 5.376 \times 10^6 \times 0.9 = 4.838 \times 10^6$ J.
6. Total energy to the appliance in 120 s: $2.280 \times 10^6 + 4.838 \times 10^6 = 7.118 \times 10^6$ J.
7. Power delivered: $P = \dfrac{7.118 \times 10^6}{120} = 59.32\,\text{kW}$

Answer: The accumulator supplies $2280\,\text{kN·m}$ of work per stroke, and the appliance receives about $59.3\,\text{kW}$ (accumulator about 19 kW plus pump about 40.3 kW).